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CHAPTER 1

QUADRATIC EQUATION

1. POLYNOMIAL

Algebraic expression containing many terms of the form cx^n, n being a non-negative integer is called a polynomial. i.e., $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_{n-1}x^{n-1} + a_nx^n$, where x is a variable, a_0, a_1, a_2 ..... a_n are constants $a_n \neq 0$

Example **:** $4x^4 + 3x^3 - 7x^2 + 5x + 3$, $3x^3 + x^2 - 3x + 5$.

1. **Real polynomial :** Let a_0, a_1, a_2 .... a_n be real numbers and x is a real variable.
Then $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$ is called real polynomial of real variable x with real coefficients.

   Example **:** $3x^3 - 4x^2 + 5x - 4$, $x^2 - 2x + 1$ etc. are real polynomials.

2. **Complex polynomial :** If a_0, a_1, a_2 ..... a_n be complex numbers and x is a varying complex number.
Then $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$ is called complex polynomial of complex variable x with complex coefficients.

   Example **:** $3x^2 - (2 + 4i)x + (5i - 4)$, $x^3 - 5ix^2 + (1 + 2i)x + 4$ etc. are complex polynomials.

3. **Degree of polynomial :** Highest power of variable x in a polynomial is called degree of polynomial.

   Example : $f(x) = a_0 + a_1x + a_2x^2 + ... + a_{n-1}x^{n-1}$ is a $n - 1$ degree polynomial.
   $f(x) = 4x^3 + 3x^2 - 7x + 5$ is a 3 degree polynomial.
   $f(x) = 3x - 4$ is single degree polynomial or linear polynomial.
   $f(x) = bx$ is an odd linear polynomial.

   A polynomial of second degree is generally called a quadratic polynomial. Polynomials of degree 3 and 4 are known as cubic and biquadratic polynomials respectively.

4. **Polynomial equation :** If $f(x)$ is a polynomial, real or complex, then $f(x) = 0$ is called a polynomial equation.

2. QUADRATIC EQUATION

Types of Quadratic Equation :

A quadratic polynomial $f(x)$ when equated to zero is called quadratic equation.

Example **:** $3x^2 + 7x + 5 = 0$, $-9x^2 + 7x + 5 = 0$, $x^2 + 2x = 0$, $2x^2 = 0$

or

An equation in which the highest power of the unknown quantity is two is called quadratic equation. Quadratic equations are of two types :

(1) **Purely quadratic equation :** A quadratic equation in which the term containing the first degree of the unknown quantity is absent is called a purely quadratic equation.

i.e. $ax^2 + c = 0$ where a, $c \in C$ and $a \neq 0$

(2) **Adfected quadratic equation :** A quadratic equation which contains terms of first as well as second degrees of the unknown quantity is called an adfected quadratic equation.

i.e. $ax^2 + bx + c = 0$ where $a, b, c \in C$ and $a \neq 0, b \neq 0$

(3) **Roots of a quadratic equation :** The values of variable x which satisfy the quadratic equation is called roots of quadratic equation.

Note :-

- *An equation of degree n has roots, real or imaginary.*
- *Surd and imaginary roots always occur in pairs in a polynomial equation with real coefficients i.e. If $2 - 3i$ is a root of an equation, then $2 + 3i$ is also its root. Similarly if $2 + \sqrt{3}$ is a root of given equation, then $2 - \sqrt{3}$ is also its root.*
- *An odd degree equation has at least one real root whose sign is opposite to that of its last term (constant term), provided that the coefficient of highest degree term is positive.*
- *Every equation of an even degree whose constant term is negative and the coefficient of highest degree term is positive has at least two real roots, one positive and one negative.*

Solution of Quadratic Equaiton :

(1) **Factorization method :** Let $ax^2 + bx + c = a(x-\alpha)(x-\beta) = 0$. Then $x = \alpha$ and $x = \beta$ will satisfy the given equation.

Hence, factorize the equation and equating each factor to zero gives roots of the equation.

Example **:** $3x^2 - 2x + 1 = 0 \Rightarrow (x-1)(3x+1) = 0$

$x = 1, -1/3$

(2) **Hindu method (Sri Dharacharya method) :** By completing the perfect square as

$$ax^2 + bx + c = 0 \Rightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Adding and subtracting $\left(\frac{b}{2a}\right)^2$, $\left[\left(x+\frac{b}{2a}\right)^2 - \frac{b^2-4ac}{4a^2}\right] = 0$ which gives, $\boxed{x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}}$

Hence the quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ has two roots, given by

$$\alpha = \frac{-b+\sqrt{b^2-4ac}}{2a}, \beta = \frac{-b-\sqrt{b^2-4ac}}{2a}$$

Note :- *Every quadratic equation has two and only two roots.*

Nature of Roots :

In quadratic equation $ax^2 + bx + c = 0$, the term $b^2 - 4ac$ is called discriminant of the equation, which plays an important role in finding the nature of the roots. It is denoted by Δ or D.

1. If $a, b, c \in R$ and $a \neq 0$, then :

(i) If $D < 0$, then equation $ax^2 + bx + c = 0$ has non-real complex roots.

(ii) If $D > 0$, the equaiton $ax^2 + bx + c = 0$ has real and distinct roots, namely $\alpha = \frac{-b+\sqrt{D}}{2a}$, $\beta = \frac{-b-\sqrt{D}}{2a}$

and then $ax^2 + bx + c = a(x-\alpha)(x-\beta)$ .... (i)

(iii) If $D = 0$, then equation $ax^2 + bx + c$ has has real and equal roots

$\alpha = \beta = \frac{-b}{2a}$ and then $ax^2 + bx + c = a(x-\alpha)^2$ .... (ii)

To represent the quadratic expression $ax^2 + bx + c$ in form (i) and (ii), transform it into linear factors.

(iv) If $D \geq 0$, then equation $ax^2 + bx + c = 0$ has real roots.

2. If $a, b, c \in Q \neq 0$, then :

(i) If $D > 0$ and D is a perfect square $\Rightarrow$ roots are unequal and rational.

(ii) If $D > 0$ and D is not a perfect square $\Rightarrow$ roots are irrational and unequal.

3. Conjugate roots :

The irrational and complex roots of a quadratic equation always occur in pairs. Therefore

(i) If one root be $\alpha + i\beta$ then other rooth will be $\alpha - i\beta$.

(ii) If one root be $\alpha + \sqrt{\beta}$ then other root will be $\alpha - \sqrt{\beta}$.

4. If D_1 and D_2 be the discriminants of two quadratic equations, then

(i) If $D_1 + D_2 \geq 0$, then

(a) At least one of D_1 and $D_2 \geq 0$.

(b) If $D_1 < 0$ then $D_2 > 0$.

(ii) If $D_1 + D_2 < 0$, then

(a) At least one of D_1 and $D_2 < 0$.

(b) If $D_1 > 0$ then $D_2 < 0$.

Illustration 1: If α is a root of $4x^2 + 2x - 1 = 0$, then other root of the equation is given by $4\alpha^3 - 3\alpha$.

Solution : Let β be other root of the given equation $4x^2 + 2x = 0$, of which α is a root.

Then sum of roots $= \alpha + \beta = \frac{-2}{4} = -\frac{1}{2}$

$\Rightarrow \quad \beta = -\frac{1}{2} - \alpha \qquad \because \ \alpha$ is a root of $4x^2 + 2x - 1 = 0$

$\Rightarrow \quad 4\alpha^2 + 2\alpha - 1 = 0 \quad \Rightarrow \quad 4\alpha^2 = 1 - 2\alpha \ ; \ 2\alpha^2 = \frac{1}{2} - \alpha$

$\Rightarrow \quad 4\alpha^3 = \alpha - 2\alpha^2 \ ; \quad \Rightarrow \quad 4\alpha^3 - 3\alpha = \alpha - 2\alpha^2 - 3\alpha$

$$= -2\alpha - \frac{1}{2} + \alpha \left(2\alpha^2 = \frac{1}{2} - \alpha \right) = -\frac{1}{2} - \alpha = \beta$$

Nature of Roots of Quadratic Equation $ax^2 + bx + c = 0$, $a \neq 0$, a, b, c are are real numbers.

(i) The roots of $ax^2 + bx + c = 0$ are real equal iff $D = b^2 - 4ac = 0$.

(ii) If $D > 0$, the roots are real and unequal.

(iii) If $D = a$ perfect square, then roots of $ax^2 + bx + c = 0$ are are rational provided a, b, c are rational. Otherwise roots are irrational

(iv) The above example suggests that if $a = 1$ and b, c are integers and the roots are rational, then roots of the quadratic equation must be integers.

(v) If $\alpha + \sqrt{\beta}$ be an irrational root of the quadratic equation, then $\alpha - \sqrt{\beta}$ is also a root of the quadratic equation provided all the coefficients are rational.

(vi) If $D < 0$, the roots are imaginary numbers. If $\alpha + i\,\beta$ is one root of the given equation, with real coefficients $\beta \neq 0$, then the other root must be its conjugate $\alpha - i\,\beta$ and vice-versa; where $\alpha, \beta \in R$ and $i = \sqrt{-1}$. In other words we can say that complex roots of an equation (if it exists) always occur in conjugate pairs.

(vii) If roots of the given quadratic equation $ax^2 + bx + c = 0$ are equal in magnitude but opposite in signs, then $b = 0$ = coefficient of x.

i.e. If $\alpha, -\alpha$ are roots of $ax^2 + bx + c = 0$, then sum of roots $= -\frac{b}{a} = \alpha + (-\alpha) = 0$

$\Rightarrow \ b = 0$.

Observation : The roots of $x^2 - 4 = 0$ are $x = +2, -2$ and that of $x^2 + 9 = 0$ are $3i$ and $-3i$.

(viii) If roots of $ax^2 + bx + c = 0$ are reciprocal, then coefficient of x^2 = constant term

i.e. $a = c$. (In this case product of roots $= 1 = \alpha . \ \frac{1}{\alpha} = \frac{c}{a}$)

Observation : The roots of $2x^2 + 5x + 2 = 0$ are -2 and $-1/2$ as roots are reciprocal as coefficient of x^2 = constant term = 2

(ix) If the sum of coefficients of $ax^2 + bx + c = 0$ is zero (i.e. if $a + b + c = 0$), then the roots of the equation and 1 and c/a.

Similarly if $a - b + c = 0$, then roots of $ax^2 + bx + c = 0$ are -1 and $-c/a$.

(x) If the quadratic equation is satisfied by more than two roots (real or complex), then it becomes an identity and in this case $a = b = c = 0$.

Proof :

Let x_1, x_2, x_3 be three distinct roots of $ax^2 + bx + c = 0$

then $\quad ax_1^2 + bx_1 + c = 0 \qquad \ldots$ (i)

$\quad ax_2^2 + bx_2 + c = 0 \qquad \ldots$ (ii)

$\quad ax_3^2 + bx_3 + c = 0 \qquad \ldots$ (iii)

By (i) – (ii), $a(x_1 + x_2) + b = 0 \qquad \ldots$ (iv) $\quad (x_1 \neq x_2)$

By (ii) – (iii), $a(x_2 + x_3) + b = 0 \qquad \ldots$ (v) $\quad (x_2 \neq x_3)$

Now (iv), (v), $a = 0$, $b = 0 \qquad (x_1 \neq x_3)$

In equation (i) putting $a = b = 0$, we get $c = 0$

(xi) A quadratic equation, whose roots are α and β can be written as

$(x-\alpha)(x-\beta) = 0 = x^2 - (\alpha+\beta)x + \alpha\beta$

i.e. $ax^2 + bx + c \equiv a(x-\alpha)(x-\beta)$.

Illustration 2 : Discuss the nature of roots of $x^2 - 2\sqrt{3}x - 1 = 0$.

Solution : Here we observe that

$D = b^2 - 4ac = (-2\sqrt{3})^2 - 4.1.(-1) = 12 + 4 = 16 = 4^2$

and roots of the given equation is

$$\frac{-b+\sqrt{D}}{2a} \text{ \& } \frac{-b-\sqrt{D}}{2a} \text{ i.e. } \frac{2\sqrt{3}+4}{2}, \frac{2\sqrt{3}-4}{2} \text{ i.e. } \sqrt{3}+2, \sqrt{3}-2.$$

Thus the roots of the given equation are irrational.

Illustration 3: Discuss the nature or roots of $x^2 - 2008x + 2007 = 0$.

Solution : Here, $D = (2008)^2 - 4.2007 = (2007+1)^2 - 4.1.2007$

$= (2007-1)^2 = (2006)^2$ = a perfect square.

Roots are $\frac{2008+2006}{2}, \frac{2008-2006}{2}$

i.e. 2007 and 1 are roots of the given equation.

3. PROPERTIES RELATED TO NATURE OF ROOTS OF QUADRATIC EQUATION

(i) Let D_1 and D_2 be the discriminants of two given quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ respectively and $D_1 + D_2 \geq 0$.

The at least one of D_1 & D_2 is ≥ 0

$\Rightarrow$ At **least one of the equations has real roots.**

Illustration 4: Discuss the nature of roots of equations $x^2 + ax + b = 0$ and $x^2 + cx + d = 0$, where a, b, c, d are real numbers and $ac = 2(b+d)$.

Solution : Let D_1 and D_2 be discriminants of the given equations $x^2 + ax + b = 0$ and $x^2 + cx + d = 0$ respectively.

Then $D_1 = a^2 - 4b$ and $D_2 = c^2 - 4d$.

Now $D_1 + D_2 = a^2 - 4b + c^2 - 4d$

$= a^2 + c^2 - 4(b+d)$

$= a^2 + c^2 - 2ac$ [$\because ac = 2(b+b)$]

$= (a-c)^2 \geq 0$

Consequently at least one of the given equatins has real roots.

(i) If $D_1 + D_2 < 0$, then at least one of D_1 and $D_2 < 0$

$\Rightarrow$ at least one of the given equations has imaginary roots.

For example, let us consider two quadratic equation $x^2 + 4x + 3 = 0$ and $x^2 + 2x + 4 = 0$.

Here $D_1 = 16 - 12 = 4$ & $D_2 = 4 - 16 = -12$

$D_1 + D_2 = 4 - 12 = -8 < 0$

$\Rightarrow$ at least one of the given equation has imaginary roots.

Here $x^2 + 2x + 4 = 0$ has imaginary roots.

(ii) If $D_1D_2 < 0$

Then the equation $(x_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2) = 0$ has two real roots.

In the above example we observe that $D_1D_2 = 4 \times 12 = -48 < 0$.

Hence the equation $(x_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2) = 0$ has two real and two complex roots.

(iii) $D_1D_2 > 0$

$\Rightarrow$ Either $D_1 > 0$ and $D_2 > 0$

$\Rightarrow$ $(a_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2) = 0$ has four real roots.

or $D_1 < 0$ and $D_2 < 0$

$\Rightarrow$ $(a_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2) = 0$ has four complex roots.

(iv) $D_1D_2 = 0$

$\Rightarrow$ $D_1 > 0$ and $D_2 = 0$ or $D_1 = 0$ and $D_2 > 0$

In this case $(a_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2) = 0$ has two equal real roots and two distinct real roots.

or $D_1 < 0$ and $D_2 = 0$ or $D_2 = 0$ or $D_1 = 0$ and $D_2 < 0$

In this case $(a_1x^2 + b_1x + c_1)(a_2X^2 + b_2x + c_2) = 0$ has two equal real roots and two imaginary roots.

4. CONDITION FOR COMMON ROOTS OF EQUATIONS

(i) Condition for One Common Root between two Quadratic Equations

Let α be a common root between given two quadratic equations

$ax^2 + bx + c = 0$ and $a_1x^2 + b_1x + c_1 = 0$

Then $a\alpha^2 + b\alpha + c = 0$

and $a_1\alpha^2 + b_1\alpha + c_1 = 0$.

By cross-multiplication, we have

$$\frac{\alpha^2}{bc_1 - b_1c} = \frac{\alpha}{a_1c - ac_1} = \frac{1}{ab_1 - a_1b}$$

$$\Rightarrow \quad \alpha = \frac{bc_1 - b_1c}{a_1c - ac_1} = \frac{a_1c - ac_1}{ab_1 = a_1b}$$

$\Rightarrow$ $(a_1c - ac_1)^2 = (ab_1 - a_1b)(bc_1 - b_1c)$, which is the required condition.

Remark : Let β and γ be other roots of the equations;

then $\alpha\beta = \frac{c}{a}$ & $\alpha\gamma = \frac{c_1}{a_1} \Rightarrow \beta = \frac{c}{a\alpha}$ & $\gamma = \frac{c_1}{a_1\alpha}$.

The quadratic equation with other roots of the equations is given by

$$x^2 - (\beta + \gamma)x + \beta\gamma = 0 \Rightarrow x^2 - \left(\frac{c}{a} + \frac{c_1}{a_1}\right)\frac{x}{\alpha} + \frac{cc_1}{aa_1\alpha^2} = 0$$

Illustration 5: If the quadratic equation $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ $(a \neq b)$ have a common root, then find the numerical value of $a + b$.

Solution : Let α be a common root between given quadratic equations.

Then $\alpha^2 + a\alpha + b = 0$

$\alpha^2 + b\alpha + a = 0$

On subtraction $(a - b)\alpha + b - a = 0 \Rightarrow \alpha = 1$.

$\therefore$ The common root is 1. Putting $\alpha = 1$, we get

$1 + a + b = 0 \Rightarrow a + b = -1 \Rightarrow |a + b| = 1$

$\Rightarrow$ Absolute value of $(a + b) = 1$

$\therefore$ Numerical value of $a + b$ is -1.

Illustration 6: If the equation $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a common root, then find the condition and the quadratic equation with other roots of the equation.

Solution : Let α, β be the roots of $x^2 + ax + bc = 0$ and α, γ be that of $x^2 + bx + ca = 0$ so that α is a common root.

Then $\alpha^2 + a\alpha + bc = 0$; $\alpha + \beta = -a$, $\alpha\beta = bc$

$\alpha^2 + b\alpha + ca = 0$; $\alpha + \gamma = -b$, $\alpha\gamma = ca$

Solving by cross-multiplication, we have

$$\frac{\alpha^2}{a^2c - b^2c} = \frac{\alpha}{bc - ca} = \frac{1}{b - a}$$

$$\Rightarrow \quad \frac{\alpha^2}{c(a-b)(a+b)} = \frac{\alpha}{c(b-a)} = \frac{1}{b-a}$$

$\Rightarrow$ $\alpha = -(a + b) = c \Rightarrow a + b + c = 0$

Which is the required condition for a common root between given equations.

also $\alpha\beta + bc = c\beta$ and $\alpha\gamma = ca = c\gamma$

$\Rightarrow \quad \beta = b \quad \Rightarrow \quad \gamma = a$

Thus the quadratic equation with roots β & γ is

$x^2 - (\beta + \gamma)x + \beta\gamma = 0$

$\Rightarrow \quad x^2 - (a + b)x + ab = 0$

$\Rightarrow \quad x^2 + cx + ab = 0$

(ii) Condition for two Quadratic Equations to have both Roots Common

Let α, β be roots of given quadratic equations $ax^2 + bx + c = 0$ and $a_1x^2 + b_1x + c_1 = 0$

$$\therefore \quad \alpha + \beta = \frac{-b}{a} = \frac{-b_1}{a_1} \quad \Rightarrow \quad \frac{a}{a_1} = \frac{b}{b}$$

$$\alpha\beta = \frac{c}{a} = \frac{c_1}{a_1} \qquad \frac{a}{a_1} = \frac{c}{c_1}$$

$\Rightarrow \quad \frac{a}{a_1} = \frac{b}{b_1} = \frac{c}{c_1}$. Which is the required condition.

Note :- *When both roots are common, then the two quadratic equations are identical.*

Illustration 1: Find a relation between a, b, c so that two quadratic equations $ax^2 + bx + c = 0$ and $1003x^2 + 1505x + 2007 = 0$ have a common root.

Solution : We observe that the discriminant of the quadratic equation $1003x^2 + 1505x + 2007 = 0$ is $(1505)^2 - 4 \times 1003 \times 2007 < 0$

$\Rightarrow$ Root of the equation are complex

We knwo that complex roots of an equation occur in conjugate pairs, and in this case whenever α be a common root between given equations, other root automatically becomes common.

Hence the given equations have both roots common and consequantly

$$\frac{a}{1003} = \frac{b}{1505} = \frac{c}{2007} = \frac{a+c}{3010}$$

$\Rightarrow \quad 2b = a + c \quad \Rightarrow$ a, b, c are in A.P.

which is the required relation.

5. RELATION BETWEEN ROOTS AND COEFFICIENTS OF A POLYNOMIAL EQUATION

$$\boldsymbol{a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + ... + a_{n-1}x + a_n = 0.}$$

Let $\alpha_1, \alpha_2, \alpha_3, ... \alpha_{n-1}, \alpha_n$ be the roots of the givne equation

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + ... + a_{n-1}x + a_n = 0$$

Then $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + ... + a_{n-1}x + a_n$

$$\equiv \alpha_0(x - \alpha_1)(x - \alpha_2)(x - \alpha_3) .... (x - \alpha_n) = 0$$

$$\equiv \alpha_0[x^n - (\alpha_1 + \alpha_2 + ... \alpha_n)x^{n-1} + (\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_1\alpha_4 + ... + \alpha_1\alpha_n + \alpha_2\alpha_3 + ... + \alpha_{n-1}\alpha_n)x^{n-2}$$
$$- (\alpha_1\alpha_2\alpha_3 + \alpha_1\alpha_2\alpha_4 + ... + \alpha_{n-2}\alpha_{n-1}\alpha_n)x^{n-3} + ... + (-1)^n \alpha_1\alpha_2\alpha_3\alpha_4 .... \alpha_n] = 0.$$

$$\Rightarrow \quad \frac{a_0}{a_0} = \frac{-a_0(\alpha_1 + \alpha_2 + ... + \alpha_n)}{a_3} = \frac{a_0(\alpha_1\alpha_2 + \alpha_1\alpha_3 + ... + \alpha_{n-1}\alpha_n)}{a_2}$$

$$\Rightarrow \quad \sum_{i=1}^{n} \alpha_i = \alpha_1 + \alpha_2 + \alpha_3 + ... + \alpha_n = \text{sum of all roots} = (-1)^1 \frac{a_1}{a_0} = (-1)^1 \frac{\text{coeff. of } x^{n-1}}{\text{coeff. of } x^n}$$

$$\sum_{1 \le i < j \le n} \alpha_i\alpha_j = \alpha_1\alpha_2 + \alpha_1\alpha_3 + ... + \alpha_{n-1}\alpha_n = \text{ sum of product of roots taken two at a time}$$

$$= (-1)^2 \frac{a_2}{a_0} = (-1)^2 \frac{\text{coeff. of } x^{n-2}}{\text{coeff. of } x^n}$$

$$\sum_{1=i<j<k\le n} \alpha_i\alpha_j\alpha_k = \alpha_1\alpha_2\alpha_3 + \alpha_1\alpha_2\alpha_4 + ... + \alpha_{n-2}\alpha_{n-1}\alpha_n$$

= sum of product of roots taken three at a time $= (-1)^3 \frac{a_3}{a_0} = (-1)^3 \frac{\text{coeff. of } x^{n-3}}{\text{coeff. of } x^n}$

Product of all roots = $a_1a_2a_3 .... a_n = (-1)^n \frac{a_n}{a_0} = (-1)^n \frac{\text{constant term}}{\text{coeff. of } x^n}$

Illustration 8: Find the condition in order that the equation $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_{n-1}x + a_n = 0$ has real roots ($a_0 \neq 0$).

Solution : Let $\alpha_1, \alpha_2, \alpha_3, ..., \alpha_n$ be the real roots of the given equation.

Then

$$\sum \alpha_i = \alpha_1 + \alpha_2 + \alpha_3 + ... + \alpha_n = -\frac{a_1}{a_0}$$

$$\sum_{1\le i<j\le n} \alpha_i\alpha_j = \alpha_1\alpha_2 + \alpha_1\alpha_3 + ... + \alpha_1\alpha_n + \alpha_2\alpha_3 + ... + \alpha_{n-1}\alpha_n = (-1)^2.\frac{a_2}{a_0} = \frac{a_2}{a_0}$$

From well-known inequality we know that

$$\frac{\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + ... + \alpha_n^2}{n} \ge \left(\frac{\alpha_1 + \alpha_2 + \alpha_3 + ... + \alpha_n}{n}\right)^2$$

$$\Rightarrow \quad \frac{(\Sigma\alpha_i)^2 - 2\Sigma\alpha_i\alpha_j}{n} \ge \frac{(\Sigma\alpha_i)^2}{n^2} \quad \Rightarrow \quad \frac{\frac{a_1^2}{a_0^2} - 2.\frac{a_2}{a_0}}{n} \ge \frac{\frac{a_1^2}{a_0^2}}{n^2}$$

$$\Rightarrow \quad na_1^2 - 2na_0a_2 \ge a_1^2 \quad \Rightarrow \quad (n-1)a_1^2 - 2na_0a_2 \ge 0$$

which is the required condition.

Note :- *The roots of the equation $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_2x^{n-2} + a_3x^{n-3} + ... + a_{n-1}x + a_n = 0$ can't be real if $(n-1)a_1^2 - 2na_2a_0 < 0$.*

6. RELATION BETWEEN ROOTS OF $f(x) = 0$ AND ITS DERIVATIVE $f'(x) = 0$

$f(x)$ be a continuous function and $f(x) = 0$ has n real roots, then $f'(x) = 0$ has $(n-1)$ roots, $f''(x) = 0$ has $(n-2)$ roots, $f'''(x) = 0$ has $(n-3)$ real roots and so on .....

Illustration 9: Find the number of real roots of the equation

$$(x - 7)^{2007} + (x + 7^2)^{2007} + (x - 7^3)^{2007} + ... + (x - 7^{2007})^{2007} = 0$$

Solution : Let $f(x) = (x-7)^{2007} + (x-7^2)^{2007} + (x-7^3)^{2007} + ... + (x-7^{2007})^{2007}$.

So that $f'(x) = 2007\,[(x-7)^{2006} + (x-7^2)^{2006} + (x-7^3)^{2006} + ... + (x-7^{2007})^{2006}]$

Clearly $f'(x) > 0$ for all $x \in R$ and hence

$\Rightarrow$ $f(x)$ is increasing

$\Rightarrow$ $f(x) = 0$ has one real and 2006 imaginary roots as every equation of an odd degree has at least one rest root.

Illustration 10: The equations $ax^2 + bx + a = 0$, $x^3 - 2x^2 + 2x - 1 = 0$ have two roots common, then find the value of $a + b$.

Solution : We have $x^3 - 2x^2 + 2x - 1 = 0$

$\Rightarrow$ $(x-1)(x^2 + x + 1) - 2x(x-1) = 0$

$\Rightarrow$ $(x-1)(x^2 + x - 2x + 1) = 0 \quad \Rightarrow \quad (x-1)(x^2 - x + 1) = 0$

$\Rightarrow$ $x = 1, -\omega, -\omega^2$, where ω being non-real cubic roots of unity.

Since the given equations $ax^2 + bx + c = 0$ and $x^3 - 2x^2 + 2x - 1 = 0$ have two common roots, hence $-\omega, -\omega^2$ are the common roots consequently

$a\omega^2 - b\omega + a = 0$ and $a\omega^4 - b\omega^4 - b\omega^2 + a = 0$

$\Rightarrow$ $a(1 + \omega^2) - b\omega = 0$ and $a(1 + \omega) - b\omega^2 = 0$

$\Rightarrow$ $-a\omega - b\omega = 0$ and $-a\omega^2 - b\omega^2 = 0$

$\Rightarrow$ $(a + b)\omega = 0$ and $(a + b)\omega^2 = 0$

$\Rightarrow$ $a + b = 0$

Note :- *When the given equations have a real root common, then $a = -2$.*

Illustration 11: Find the condition that $px^3 + qx^2 + rx + s = 0$ has exactly one real roots, where $p, q, r, s \in R$.

Solution : Let $f(x) = px^3 + qx^2 + rx + s$

so that $f'(x) = 3px^2 + 2qx + r$

Now $f(x) = 0$ has all real roots, if $f'(x) = 0$ has real roots.

i.e. $D \geq 0$

$\Rightarrow$ $4q^2 - 12pr \geq 0$ $\Rightarrow$ $q^2 - 3pr \geq 0$

But when $q^2 - 3pr < 0$, no root of $f'(x) = 0$ is real and hence $f(x) = 0$ has exactly one real root.

7. QUADRATIC EXPRESSION

An expression of the form $ax^2 + bx + c$, where $0 \neq a, b, c \in R$ is called a quadratic expression or quadratic polynomial in independent variable x.

Let $y = f(x) = ax^2 + bx + c = a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right] = a[x^2 - (\alpha + \beta)x + \alpha\beta] = [\alpha(x - \alpha)(x - \beta)$; α & β are roots of $ax^2 + bx + c = 0$.

Thus $ax^2 + bx + c = a(x - \alpha)(x - \beta)$,

where α & β are roots of the corresponding quadratic equation $ax^2 + bx + c = 0$.

Also $y = f(x) = a\left[x^2 + 2x.\frac{b}{2a} + \left(\frac{b}{2a}\right)^2\right] - \frac{b^2}{4a} + c = a\left[\left(x + \frac{b}{2a}\right)^2\right] - \frac{b^2 - 4ac}{4a}$

$\Rightarrow$ $\left(y + \frac{D}{4a}\right) = a\left[\left(x + \frac{b}{2a}\right)^2\right]; D = b^2 - 4ac$

which represents a parabola with vertex at $\left(\frac{-b}{2a}, \frac{-D}{4a}\right)$.

Student Task

Q.1 If the difference of the roots of the equation $x^2 + kx + 7 = 0$ is 6, then the possible values of k are

(a) 4 (b) -4 (c) ± 8 (d) ± 6

Q.2 If one root of equation $x^2 - ix - (1 + i) = 0$ is $1 + i$, then the other root is

(a) $1 - i$ (b) $-1 + i$ (c) -1 (d) $-1 - i$

Q.3 If the equation $(3x)^2 + (27 \times 3^{1/p} - 15)x + 4 = 0$ has equal roots then p =

(a) 0 (b) -2 (c) $-1/2$ (d) $-1/3$

Q.4 If $\alpha \neq \beta$ are $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$, then the equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

(a) $3x^2 + 19x + 3 = 0$ (b) $3x^2 - 19x + 3 = 0$

(c) $3x^2 - 19x - 3 = 0$ (d) $x^2 - 16x + 1 = 0$

Q.5 The set of values of p for which the roots of the equation $3x^2 + 2x + p(p - 1) = 0$ are of opposite sign is

(a) $(-\infty, 0)$ (b) $(0, 1)$ (c) $(1, \infty)$ (d) $(0, \infty)$

Answers :

1. (c) 2. (c) 3. (c) 4. (b) 5. (b)

8. SIGN OF QUADRATIC EXPRESSION

We have seen that

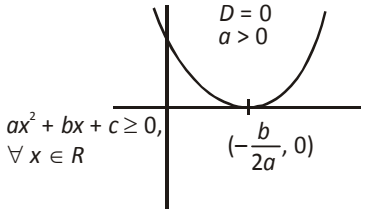
$ax^2 + bx + c = a(x - \alpha)(x - \beta)$; α & β being roots of corresponding equation $ax^2 + bx + c = 0$ which shows that the sign of $ax^2 + bx + c$ depends on 'a'.

Case-I : D - 0 ($\alpha + \beta$ are real & equal)

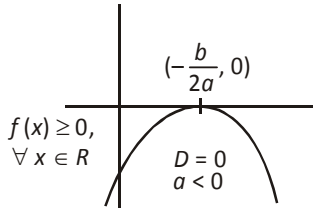
In this case $ax^2 + bx + c = a(x - \alpha)^2 > 0$, if $a > 0$

< 0, if $a < 0$

i.e. the quadratic expression $ax^2 + bx + c$ is positive or negative according as $a > 0$ or $a < 0$. It has value equal to zero at $x = \alpha$. Graphically it is shown as below :



min $(ax^2 + bx + c) = 0$, where $a > 0$



max $(ax^2 + bx + c) = 0$, if $a < 0$

Graph of $y = f(x)$ touches x-axis.

Case-II : $D < 0$ ($b^2 - 4ac < 0$),

In this case the roots of the corresponding equation are conjugate complex numbers say $p + iq$ & $p - iq$.

Let $\alpha = p + iq$, $\beta = p - iq$.

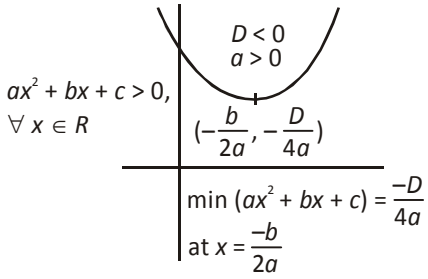
Now, $ax^2 + bx + c = a(x - \alpha)(x - \beta)$

$= a[(x - p) - iq][(x - p) + iq]$;

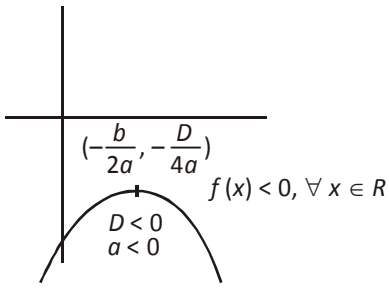
$= a[(x - p)^2 + q^2] = 0$ according as $a = 0$; because $(x - p)^2 + q^2 > 0$

which shows that the given expression $ax^2 + bx + c$ is always +ve or –ve according as $a > 0$ or $a < 0$.

Graphically it is shown as below.



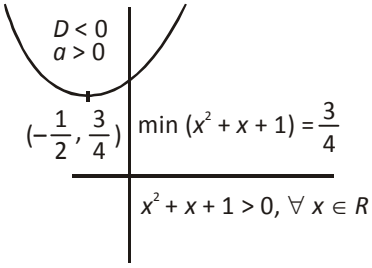
Graph $y = f(x)$ neither touches nor intersects x-axis.

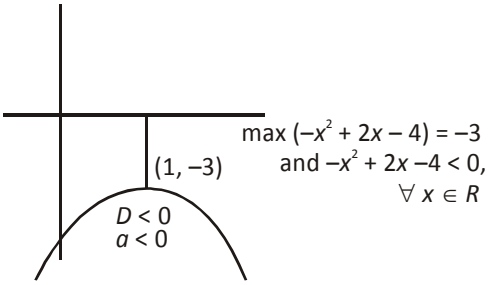


max $(ax^2 + bx + c) = \frac{-D}{4a}$ at $x = \frac{-b}{2a}$

For Example :

For the quadratic expressions $x^2 + x + 1$ and $-x^2 + 2x + 4$, their minimum and maximum values are 3/4 and –3 respectively, which is clear from the graph given below.





Case-III : $D > 0$ i.e. ($b^2 - 4ac > 0$),

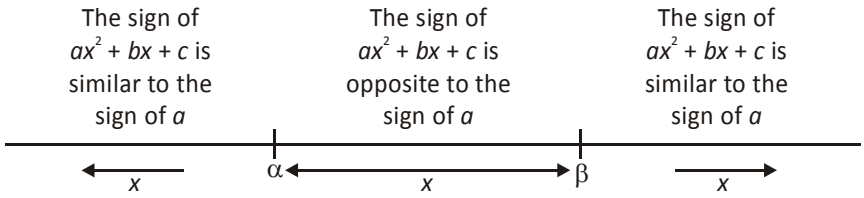
In this cases roots are real and distinct,

i.e. $\alpha, \beta \in R$ & $\alpha \neq \beta$.

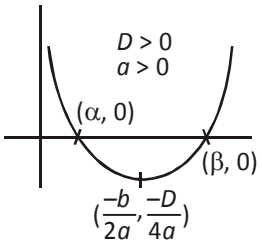
Let $\alpha < \beta$.

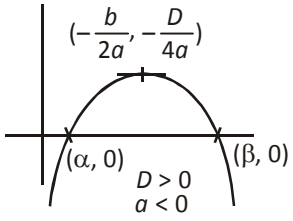
For $x < \alpha \Rightarrow x < \beta$	For $\alpha < x < \beta$	For $x > \beta \Rightarrow x > \alpha$
$\Rightarrow x - \alpha < 0$ & $x - \beta < 0$	$\Rightarrow x - \alpha > 0$ & $x - \beta < 0$	$\Rightarrow (x - \alpha)(x - \beta) > 0$
$\Rightarrow (x - \alpha)(x - \beta) > 0$	$\Rightarrow (x - \alpha)(x - \beta) < 0$	$\Rightarrow ax^2 + bx + c \gtrless 0$
$\therefore\ ax^2 + bx + c = a(x - \alpha)(x - \beta) \gtrless 0$	$\Rightarrow ax^2+bx+c = a(x-\alpha)(x-\beta) \gtrless 0$	according as $\gtrless 0$
if $a \gtrless 0$	according as $a \gtrless 0$	

The above cases can be interpreted on the number lines as



Clearly the graph $y = f(x)$ intersects x-axis at two points, which is apparent form the following graphs.





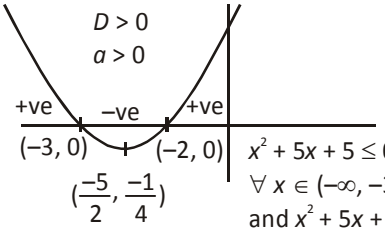
$\min (ax^2+bx+c) = \dfrac{-D}{4a}$ which occurs at $x = \dfrac{-b}{2a}$

$ax^2 + bx + c \geq 0,\ \forall\ x \in (-\infty, \alpha] \cup [B, \infty)$
and $ax^2 + bx + c \leq 0,\ \forall\ x \in [\alpha, \beta]$

$\max (ax^2 + bx + c) = \dfrac{-D}{4a}$ which occurs at $x = \dfrac{-b}{2a}$

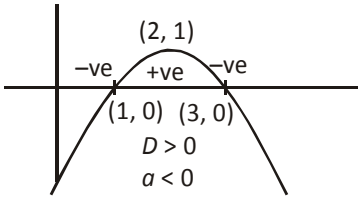
$ax^2 + bx + c \leq 0,\ \forall\ x \in (-\infty, \alpha] \cup [\beta, \infty)$
and $ax^2 + bx + c \leq 0,\ \forall\ x \in [\alpha, \beta]$

For Example : The quadratic expressions $y = x^2 + 5x + 6$ and $y = -x^2 + 4x - 3$ are plotted graphically as follows



$x^2 + 5x + 5 \leq 0,$
$\forall\ x \in (-\infty, -3] \cup [-2, \infty)$
and $x^2 + 5x + 6 \leq 0,$
$\forall\ x \in [-3, -2]$

$\min (x^2 + 5x + 6) = -1/4$
which occurs at $x = -5/2$



$\max (-x^2 + 4x - 3) = 1$ which occurs at $x = 2$
$-x^2 + 4x - 3 \leq 0$, in $(-\infty, 1] \cup [3, \infty)$
and $-x^2 + 4x - 3 \geq 0$ in $[1, 3)$

Clearly the graphs cross x-axis at two points.

9. SIGN SCHEME FOR RATIONAL EXPRESSIONS

Let $$f(x) = \frac{(x-a_1)^{k_1}(x-a_2)^{k_2}(x-a_3)^{k_3}.....(x-a_n)^{k_n}}{(x-b_1)^{n_1}(x-b_2)^{n_2}(x-b_3)^{n_3} - .....(x-b_m)^{n_m}}$$

where $k_1, k_2, k_3, ..., k_n, n_1, n_2, n_3, ..., n_m \in N$

and $a_1, a_2, a_3, ..., a_n, b_1, b_2, b_3, ..., b_m \in R$ such that

$a_i \neq b_j$ where $i = 1, 2, 3, ..., n$ and $j = 1, 2, 3, ..., m$.

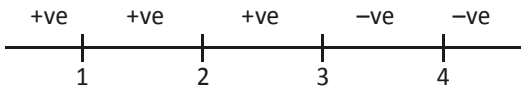
In order to discuss the sign scheme of $f(x)$ we proceed as follows :

Step I : We indicate all $a_j^{'s}$ & $b_j^{'s}$ on the number line with their respective position so that we have $n + m + 1$ section on the number line.

Step II : We write +ve, –ve alternatively from right most region of the number line keeping in view that if the index of critical points (a_j or b_j) is even, then we retain sign to the immediate left of the point. The concept will be clear from the following illustrated example :

Let us consider the sign scheme of $\frac{x^2-5x+6}{x^2-5x+4}$

$\Rightarrow$ 1, 2, 3, 4 are critical points where $x \neq 1, 4$.



Since all the indices are odd, so we write +ve, –ve alternatively from right most region of the number line.

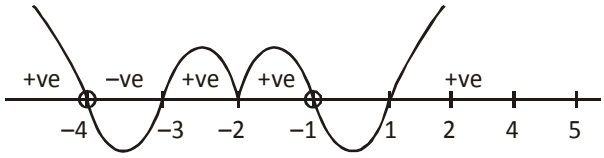
Thus $\frac{x^2-5x+6}{x^2-5x+4} \geq 0$, $\forall\ x \in (-\infty, 1) \cup [2, 3] \cup (4, \infty)$

and $\frac{x^2-5x+6}{x^2-5x+4} \leq 0$, $\forall\ x \in (1, 2] \cup [3, 4)$

Now let us take another example :

Let $f(x) = \frac{(x+2)^{2000}(x-1)^{199}(x-4)^{200}(x+3)^{2007}}{(x-2)^{1008}(x+1)^{2005}(x+4)^{305}(x-5)^{2004}}$

Here critical points are 1, 2, 4, 5, –1, –2, –3, –4. Out of which –1, –4, 2, 5 are points of discontinuties. Here we observe that index of $(x - 1)$, $(x + 3)$, $(x + 1)$, $(x + 4)$ are odd where as index of $(x + 2)$, $(x - 4)$, $(x - 2)$, $(x - 5)$ are even. The graph illustrates the sign scheme

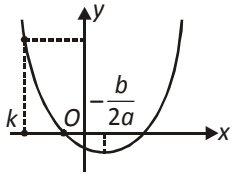


10. POSITION OF A NUMBER WITH RESPECT TO ROOTS OF AN EQUATION

Let α, β be real roots of $f(x) = ax^2 + bx + c = 0$ and $k = R$.

Case-I: k is less than both the roots $f(x) = 0$

(a) When $a > 0$



We see from $f(k) > 0$, $D \geq 0$ and $k < -\frac{b}{2a}$ .... (1)

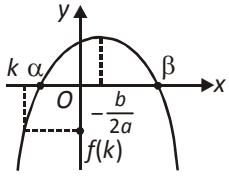
(b) When $a < 0$

We can see from graph $f(k) < 0$, $D \geq 0$ and $K <$ – b/2a ....(2)

From (1) and (2)

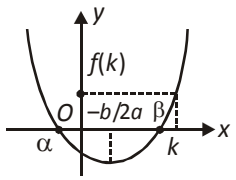
We conclude that when k is less thn both the roots of $f(x) = 0$ we must have $af(k) > 0$

$D \geq 0$ and $K <$ – $b/2a$



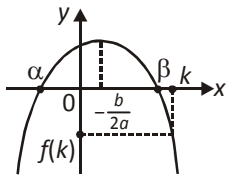
Case-II: k is greater than both the roots of $f(x) = 0$

(a) When $a > 0$



We can see from graph $f(k) > 0$, $D \geq 0$, $k > -\frac{b}{2a}$ .... (1)

(b) When $a < 0$

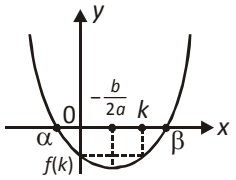


We can see from $f(k) > 0$, $D \geq 0$, k $> -\frac{b}{2a}$ .... (2)

From (1) and (2)

we conclude that when k is greater than both the roots of $f(x) = 0$ we must have $af(k) > 0$ and $k > -b/2a$.

Case-III: k lies between both the roots of $f(x) = 0$

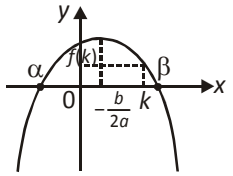


(a) When $a > 0$

But K can be smaller, greater or equal to $-\dfrac{b}{2a}$

$\therefore$ Only two conditions are thee $f(k) < 0$ and $D > 0$ .... (1)

(b) We can see from graph $f(k) > 0$ and $D > 0$



But k can be smaller, greater or equal to $-\dfrac{b}{2a}$

i.e. no conclusion can be made for $-\dfrac{b}{2a}$

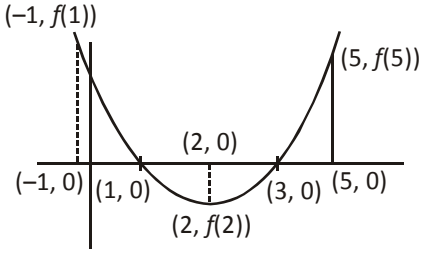
$\therefore$ Only two conditions are there $f(k) > 0$ and $D > 0$ .... (2)

From (1) and (2)

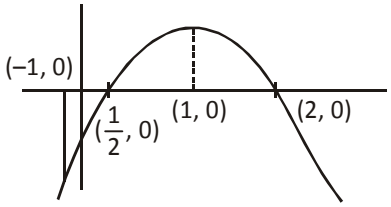
we conclude that when k lies between the roots $f(x) = 0$ we must have

$af(k) < 0$ and $D > 0$.

Illustration 12: Let us consider following two quadratic equations $x^2 - 4x + 3 = 0$ and $-2x^2 + 5x - 2 = 0$.



Solution : It is obvious from figure that – 1 is less than both roots and 5 is greater than both the roots of $f(x) = 0 = x^2 - 4x + 3$. $x = 2$ lies between the roots of $f(x) = 0$. Here $f(-1) = 8 > 0$, $f(5) = 8 > 0$, $f(2) = -1 < 0$.

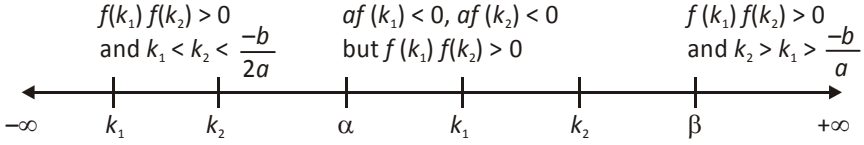


Also from the graph of $f(x) \equiv -2x^2 + 5x - 2$, it is clear that –1 and 5 are respectively less than and greater than both the roots of $f(x) = 0$ and 1 lies between the roots $f(-1) = -9 < 0$, $f(5) = -27 < 0$, $f(1) = 1 > 0$

(ii) Let k_1, k_2 $(k_1 < k_2)$ be two real numbers.

(a) The both k_1 & k_2 lie outside and less than both the roots of $f(x) = 0$,

If $af(k_1) > 0$, $af(k_2) > 0$

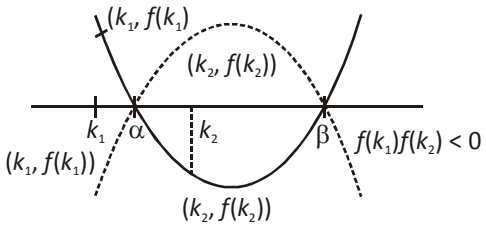


and $k_1 < k_2 < \dfrac{-b}{2a}$

(b) Both k_1 & k_2 lie outside and are greater than both the roots if

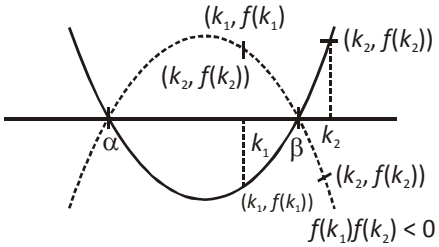
$af(k_1) > 0$, $af(k_2) > 0$ and $k_2 > k_1 > \dfrac{-b}{2a}$.

(c) Both k_1 & k_2 lie between both the roots
if $af(k_1) < 0$, $af(k_2) < 0$
The only smaller root α lies between k_1 and k_2
if $af(k_1) > 0$, $af(k_2) < 0$ combining $f(k_1) f(k_2) < 0$.



(d) The greater root b lies between k_1 & k_2

if $af(k_1) < 0$, $af(k_2) > 0$ and $f(k_1) f(k_2) < 0$ & $\frac{-b}{2a} < k_2$.



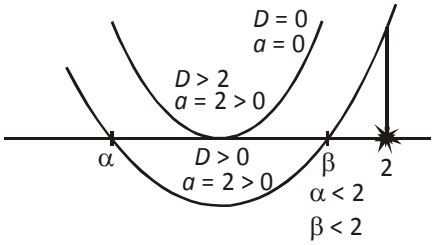
Both roots lie between k_1 & k_2
if $af(k_1) > 0$, $af(k_2) > 0$
i.e. $f(k_1) f(k_2) > 0$ and



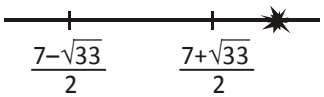
$\Rightarrow \quad k_1 < \frac{-b}{2a} < k_2$

Illustration 13: Let $f(x) = 2x^2 - 2(m + 1)x + m(m + 1)$, $m \in R$
Ror real roots of $f(x) = 0$,
$D \geq 0$
$\Rightarrow \quad 4(2m + 1)^2 - 4.2m(m + 1) \geq 0$
$\Rightarrow \quad 4m^2 + 4m - 2m^2 - 2m + 1 \geq 0$
$\Rightarrow \quad (m + 1)^2 + m^2 \geq 0$
Which is always non-negative for all real values of m.
Here coefficient of $x^2 = 2 > 0$.

(i) When both roots of $f(x) = 0$ are smaller than 2, then clearly $f(2) > 0$ and $\frac{-b}{2a} = \frac{2(2m+1)}{2\times 2} < 2$.



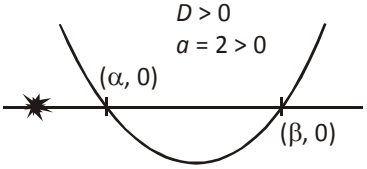
$\Rightarrow \quad 8 - 4(2m + 1) + m^2 + m > 0$ and $m < 3/2$
$\Rightarrow \quad m^2 - 7m + 4 > 0$ and m < 3/2



$\Rightarrow \quad \frac{7+\sqrt{33}}{2} > \frac{3}{2}$ $\quad \therefore$ Required interval is $m \in \left(-\infty, \frac{7-\sqrt{33}}{2}\right)$

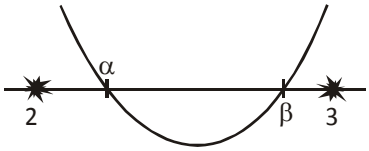
(ii) When both roots of $f(x) = 0$ are greater than 2, clearly $f(2) > 0$ and $\frac{-b}{2a} = \frac{2(2m+1)}{2\times 2} > 2$

$\Rightarrow \quad m \in \left(-\infty, \frac{7-\sqrt{33}}{2}\right) \cup \left(\frac{7+\sqrt{33}}{2}, \infty\right)$ and $m > \frac{3}{2}$



$\Rightarrow \quad m \in \left(\frac{7+\sqrt{33}}{2}, \infty\right)$

(iii) Both roots lie in (2, 3), we must have $f(2) > 0$, $f(3) > 0$ and $2 < -b/2a < 3$.



$\Rightarrow$ $m^2 - 7m + 4 > 0$ and $m^2 - 11m + 12 > 0$ and $2 < \frac{2m+1}{2} < 3$.

$\Rightarrow$ $m \in \left(-\infty, \frac{7-\sqrt{33}}{2}\right) \cup \left(\frac{7+\sqrt{33}}{2}, \infty\right), m \in \left(-\infty, \frac{11-\sqrt{73}}{2}\right) \cup \left(\frac{11+\sqrt{73}}{2}, \infty\right)$ and $\frac{3}{2} < m < \frac{5}{2}$

$\Rightarrow$ m does not exist satisfying above conditions.

(iv) In order that exactly on root lies in (2, 3). We must have $f(2)\, f(3) < 0$

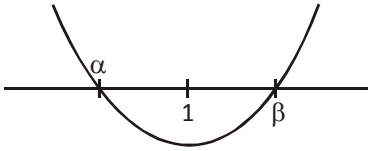
$\Rightarrow$ either $f(2) < 0$ and $f(3) > 0$ or $f(2) > 0$ & $f(3) < 0$.

$\Rightarrow$ $m \in \left(\frac{7-\sqrt{33}}{2}, \frac{7+\sqrt{33}}{2}\right)$ & $m \in \left(-\infty, \frac{11-\sqrt{73}}{2}\right) \cup \left(\frac{11+\sqrt{73}}{2}, \infty\right)$

$m \in \left(-\infty, \frac{7-\sqrt{33}}{2}\right) \cup \left(\frac{7+\sqrt{33}}{2}, \infty\right)$ and $m \in \left(\frac{11-\sqrt{73}}{2}, \frac{77+\sqrt{73}}{2}\right)$

$\Rightarrow$ $m \in \left(\frac{7-\sqrt{33}}{2}, \frac{11-\sqrt{73}}{2}\right) \cup \left(\frac{7+\sqrt{33}}{2}, \frac{11+\sqrt{73}}{2}\right)$

(v) Let $\alpha < 1$ and $\beta > 1$



$\Rightarrow$ $f(1) < 0$

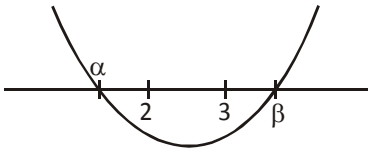
$\Rightarrow$ $2 - 2(2m + 1) + m(m + 1) < 0$

$\Rightarrow$ $m^2 - 3m < 0$ $\Rightarrow$ $m(m - 3) < 0$ $\Rightarrow$ $m \in (0, 3)$

(vi) Let $\alpha < 2$ and $\beta > 3$. In this case 2 and 3 lies between the roots and hence $f(2) < 0$ and $f(3) < 0$.

$\Rightarrow$ $m^2 - 7m + 4 < 0$ and $m^2 - 11m + 12 < 0$

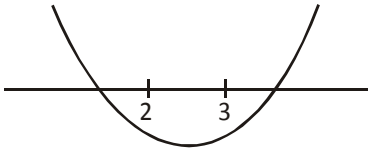
$\Rightarrow$ $m \in \left(\frac{7-\sqrt{33}}{2}, \frac{7+\sqrt{33}}{2}\right)$ and $m \in \left(\frac{11-\sqrt{73}}{2}, \frac{11+\sqrt{73}}{2}\right)$



$\Rightarrow$ $m \in \left(\frac{7-\sqrt{73}}{2}, \frac{7+\sqrt{33}}{2}\right)$

(vii) Both 2 and 3 lie betwen α & β, hence $f(2) < 0$, $f(3) < 0$

$\therefore$ $m \in \left(\frac{11-\sqrt{73}}{2}, \frac{7+\sqrt{33}}{2}\right)$



Some Useful Results :

(i) If $x = \alpha$ is a root repeated m times of $f(x) = 0$,
then $f(x)$ can be expressed as
$f(x) = (x - \alpha)^m g(x)$
where $g(x)$ is a polynomial of degree $(n - m)$.

(ii) **Rolle's Theorem :**
Let $f(x)$ be a real valued function defined on $[a, b]$ such that
$f(x)$ is continuous on $[a, b]$
$f(x)$ is differentiable on (a, b)
and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.
In otherwords, a polynomial equation $f(x) = 0$ has n real roots, then $f(x) = 0$ has $(n - 1)$ real roots.

More precisely, between two pair roots of $f(x) = 0$, there exists at least one root of $f'(x) = 0$

(iii) **Lagrange's Theorem :**

Let $f(x)$ be a function defined on $[a, b]$ such that it is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ satisfying $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

(iv) **Descarte's Rule of Sign :**

The maximum number of positive real roots of a polynomial equation $f(x) = 0$ is the number of changes of sign from positive to negative and negative to positive in $f(x)$.

The maximum number of negative real roots of a polynomial equation $f(x) = 0$ is the number of changes of sign from positive to negative and negative to positive in $f(-x)$.

Illustration 14: The maximum possible number of real roots of equation $x^5 - 6x^2 - 4x + 5 = 0$ is

(a) 0(b) 3 (c) 4 (d) 5

Solution : (b) $f(x) = x^5 - 6x^2 - 4x + 5 = 0$

\+ – + +

2 changes of sign $\Rightarrow$ maximum two positive roots.

$f(-x) = -x^5 - 6x^2 + 4x + 5$

– – + +

1 changes of sign Þ maximum one negative roots.

$\Rightarrow$ Total maximum possible number of real roots = 2 + 1 = 3.

Note :- *Descarte's Rule of sign simply gives the maximum number of positive or negative real roots. It doesnot give the exact number of positive or negative real root of f (x) = 0.*

Illustration 15: If $a, b, c \in R$ and $3a + 4b + 6c + 12d = 0$, then there exists at least one root of the cubic equation $ax^3 + bx^2 + cx + d = 0$ in (0, 1).

Solution : Here we have to show that one root of the equation $ax^3 + bx^2 + cx + d = 0$ lie in (0, 1)

Let $f'(x) = ax^3 + bx^2 + cx + d$

$\Rightarrow \quad f(x) = \dfrac{ax^4}{4} + \dfrac{bx^3}{3} + \dfrac{cx^2}{2} + dx + k$, where k is a constant of integration.

Here was observe that $f(x)$ is continuous in [0, 1] differentiable in (0, 1) and $f(0) = f(1) = k$. which shows that $f(x)$ satisfies all the conditions of Rolle's theorem and hence there exists at least one root of $f'(x) = 0$ in (0, 1)

i.e. $ax^3 + bx^2 + cx + d = 0$ has a root lying in (0, 1).

Student Task

Q.1 If α, β are the roots of $x^2 + bx - c = 0$, then the equation whose roots are b and c is

(a) $x^2 + \alpha x - \beta = 0$ (b) $x^2 - x(\alpha + \beta + \alpha\beta)x - \alpha\beta(\alpha + \beta) = 0$

(c) $x^2 - x(\alpha + \beta - \alpha\beta)x - \alpha\beta(\alpha + \beta) = 0$ (d) $x^2 + x(\alpha + \beta + \alpha\beta)x + \alpha\beta(\alpha + \beta) = 0$

Q.2 If the sum of the squares of the roots of the equation $x^2 - (\sin\alpha - 2)x - (1 + \sin\alpha) = 0$ is least, then α =

(a) $\dfrac{\pi}{4}$ (b) $\dfrac{\pi}{3}$ (c) $\dfrac{\pi}{2}$ (d) $\dfrac{\pi}{6}$

Q.3 Let $2\sin^2 x + 3\sin x - 2 > 0$ and $x^2 - x - 2 < 0$ (x is measured in radians). Then x lies in the interval.

(a) $\left(\dfrac{\pi}{6}, \dfrac{5\pi}{6}\right)$ (b) $\left(-1, \dfrac{5\pi}{6}\right)$ (c) (–1, 2) (d) $\left(\dfrac{\pi}{6}, 2\right)$

Q.4 If α, β are roots of the equation $ax^2 + 3x + 2 = 0$ ($a < 0$) then $\dfrac{\alpha^2}{\beta} + \dfrac{\beta^2}{\alpha}$ is

(a) 0 (b) 1 (c) 2 (d) Dependent of a

Q.5 If $\sin\alpha$ and $\cos\alpha$ are the roots of the equation $px^2 + qx + r = 0$, then the incorrect statement is

(a) $p^2 - q^2 + 2pr = 0$ (b) $\frac{q}{p} \in \left[-\sqrt{2}, \sqrt{2}\right]$

(c) $\frac{r}{p} \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ (d) $p^2 + q^2 - 2pr = 0$

Answers :

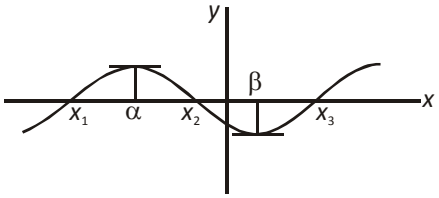
1. (d) 2. (c) 3. (d) 4. (d) 5. (d)

11. ANALYSIS OF CUBIC EQUATION

Let $f(x) = ax^3 + bx^2 + cx + d = 0$, $a \neq 0$, with real coefficient is a cubic equation.

(i) The equation may have three real roots or one real and two non-real complex roots. Hence this cubic equation always have one root.

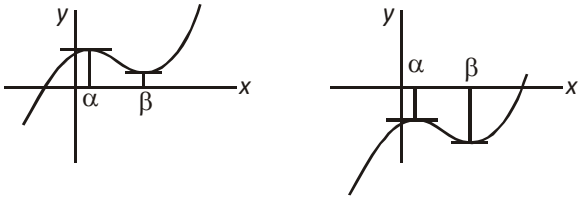
(ii) If the equation $f(x) = 0$ has three real distinct roots then its graph will be as following for $a < 0$.



Clearly the tangent is parallel to x-axis at two points $x = \alpha$, $x = \beta$.

Hence $f'(\alpha) = f'(\beta) = 0$ or $f'(x) = 0$ has two real distinct roots for which $f(\alpha)\, f(\beta) < 0$

(iii) If $f'(x) = 0$ has no may or may not have three real distinct roots. As in following cases



Here $f'(\alpha) = f'(\beta) = 0$. But $f(x) = 0$ has only one real root.

(iv) If x_1, x_2 and x_2 are the roots of $f(x) = 0$ then

$$x_1 + x_2 + x_3 = -\frac{b}{a} \; ; \; x_1x_2 + x_2x_3 + x_3x_1 = \frac{c}{a} \; ; \; x_1x_2x_3 = -\frac{d}{a}$$

12. EQUATION AND INEQUATION CONTAINING ABSOLUTE VALUE

1. Equations containing absolute values

By definition : $|x| = \begin{cases} x, \text{ if } x \geq 0 \\ -x, \text{ if } x < 0 \end{cases}$

Important forms containing absolute value :

Form I : The equation of the form $|f(x) + g(x)| = |f(x)| + |g(x)|$ is equivalent of the system $f(x).g(x) \geq 0$.

Form II : The equation of the form $|f_1(x)| + |f_2(x)| + |f_3(x)| + ... + |f_n(x)) = g(x)$ .... (i)

Where $f_1(x), f_2(x), f_3(x) ..... f_n(x)$, $g(x)$ are functions of x and $g(x)$ may be a constant.

Equations of this form can be solved by the method of interval. We first find all critical points of $f_1(x), f_2(x) ..... f_n(x)$. If coefficient of x is +ve, then graph starts with +ve sign and if it is negative, then graph starts with negative sign. Then using the definition of the absolute value, we pass form equation (i) to a collection of system which do not contain the absolute value symbols.

2. Inequations containing absolute value

By definition, $|x| < a \Rightarrow -a < x < a$ $(a > 0)$, $|x| \leq a \Rightarrow -a \leq x \leq a$,

$|x| > a \Rightarrow x < -a$ or $x > a$ and $|x| \geq a \Rightarrow x \leq -a$ or $x \geq a$

Illustration 16: The roots of $|x-2|^2 + |x-2| - 6 = 0$ are

(a) 0, 4 (b) –1, 3 (c) 4, 2 (d) 5, 1

Solution : (a) We have $|x-2|^2 + |x-2| - 6 = 0$

Let $|x-2| = X$

$X^2 + X - 6 = 0$

$\Rightarrow \quad X = \dfrac{-1 \pm \sqrt{1+24}}{2} = 2, -3 \Rightarrow X = 2$ and $X = -3$

$\therefore \quad x = 4$ or $x = 0$

Illustration 17 The set of all real numbers x for which $x^2 - |x+2| + x > 0$, is

(a) $-(\infty, -2) \cup (2, \infty)$ (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

(c) $(-\infty, -1) \cup (1, \infty)$ (d) $(\sqrt{2}, \infty)$

Solution : (b) **Case-I :** If $x + 2 \geq 0$ i.e. $x \geq -2$, we get

$x^2 - x - 2 + x > 0 \Rightarrow x^2 - 2 > 0 \Rightarrow (x - \sqrt{2})(x + \sqrt{2}) > 0$



$\Rightarrow \quad x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

But $x \geq -2$

$\therefore \quad x \in [-2, -\sqrt{2}) \cup (\sqrt{2}, \infty)$ ..... (i)

Case-II : $x + 2 < 0$ i.e. $x < -2$, then

$x^2 + x + 2 + x > 0 \Rightarrow x^2 + 2x + 2 > 0$

$\Rightarrow \quad (x+1)^2 + 1 > 0$. Which is true for all x

$\therefore \quad x \in (-\infty, -2)$ ..... (ii)

From (i) and (ii), we get, $x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

13. FORMATION OF NEW EQUATIONS WITH THE HELP OF GIVEN EQUATIONS

Let us consider a polynomial equation of n degree

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_n = 0 \quad .... (i)$$

The general root of this equation is x. To find a new equation having general root y, where $y = f(x)$, replace x by $g(x)$ in (i) where $x = g(y)$ obtained from $y = f(x)$. Under condition, let us have an illustration.

Illustration 18: If α, β, γ are the roots of $x^2 + x^2 + x + 9 = 0$. Then find the equation whose roots are $\dfrac{\alpha+1}{\alpha-1}, \dfrac{\beta+1}{\beta-1}, \dfrac{\gamma+1}{\gamma-1}$.

Solution : Let $y = \dfrac{x+1}{x-1} \Rightarrow x = \dfrac{y+1}{y-1}$

Hence to obtained new equation replace x by $\dfrac{x+1}{x-1}$ which is given by

$$\left(\frac{x+1}{x-1}\right)^3 + \left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x+1}{x-1}\right) + 9 = 0$$

$(x+1)^3 + (x+1)(x-1) + (x+1)(x-1)^2 + 9(x-1)^3 = 0$

$\Rightarrow \quad 12x^3 - 24x^2 + 29x + 10 = 0$

14. QUADRATIC EXPRESSION IN TWO VARIABLES

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, is a general quadratic expression in two variables x and y. The expression can be resolved into two linear rational factors if

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

Illustration 19: Find the value of k if $x^2 + 4xy + y^2 + 4x + 2y + k = 0$ can be resolved into two linear factors.

Solution : Comparing with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$,

we have, $a = 1, b = 1, h = 2, g = 2, f = 1, c = k$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Rightarrow \quad 1 \times 1 \times k + 2 \times 1 \times 2 \times 2 - 1 \times 1 - 1 \times 4 - k \times 4 = 0$

$\Rightarrow \quad 3k = 3 \Rightarrow k = 1.$

MISCELLANEOUS EXAMPLES

Example 1 : Solve the equation $\sqrt{3x^2+6x+7}+\sqrt{5x^2+10x+14}=4-2x-x^2$.

Solution : $\sqrt{3x^2+6x+7}+\sqrt{5x^2+10x+14}=4-2x-x^2$

i.e., $\sqrt{3(x+1)^2+4}+\sqrt{5(x+1)^2+9}=5-(x+1)^2$

LHS $\geq \sqrt{4}+\sqrt{9}=5$, since $(x + 1)^2$ is always positive or zero.

RHS ≤ 5. Hence LHS = RHS only when RHS = 5 i.e., when $x = 1$

Example 2 : If the roots of the equation $ax^2 + bx + c = 0$ be $\frac{k+1}{k}$ and $\frac{k+2}{k+1}$, prove that $(a + b + c)^2 = b^2 - 4ac$.

Solution : Given equation is $ax^2 + bx + c = 0$ ... (1)

Roots of equation (1) are $\frac{k+1}{k}$ and $\frac{k+2}{k+1}$

$\therefore \quad \frac{k+1}{k}+\frac{k+2}{k+1}=-\frac{b}{a}$ ... (2)

and $\frac{k+2}{k}=\frac{c}{a}$ ... (3)

From (3), $k = \frac{2a}{c-a}$. Putting the value of k in (2), we get

$$\frac{c+a}{2a}+\frac{2c}{c+a}=-\frac{b}{a} \text{ or } \frac{(c+a)^2+4ac}{2a(c+a)}=-\frac{b}{a}$$

or, $a(c + a)^2 + 4a^2c = -2abc - 2a^2b$ or $(c + a)^2 + 4ac = -2bc - 2ab$

or $(c + a)^2 + 2b(c + a) = -4ac$ or $(c + a)(c + a + 2b) = -4ac$

or $(a + b + c - b)(a + b + c + b) = -4ac$ or $(a + b + c)^2 - b^2 = -4ac$

or $(a + b + c)^2 = b^2 - 4ac$.

Example 3 : Solve the equation $x(x + 1)(x + 2)(x + 3) = 0.5625$

Solution : Multiplying separately $x(x + 3)$ and $(x + 1)(x + 2)$ we obtain

$(x^2 + 3x)(x^2 + 3x + 2) = 0.5625$

Introducing an auxiliary unknown $y = x^2 + 3x$, we get, after simple transformations, a quadratic equation

$y^2 + 2y - 0.5625 = 0$,

whose roots are $y_1 = 0.25$ and $y_2 = -2.25$. Returning to the initial unknown, we infer that (*) is equivalent to two equations.

$x^2 + 3x - 0.25 = 0$ and $x^2 + 3x + 2.25 = 0$.

The first equation has two different roots, $x = \frac{-3+\sqrt{10}}{2}$ and $x = \frac{-3-\sqrt{10}}{2}$, and the second equation has one root of multiplicity 2, $x = -3/2$.

Example 4 : Prove that the roots of the equation $\lambda(x - c) + (x - c) + (x - b)(x - d) = 0$ are real for all values of λ if $a < b < c < d$

Solution : $x^2(\lambda + 1) - x\{\lambda(a + c) + (b + d)\} + \lambda ac + bd = 0$

Roots are real, if the discriminant is greater than zero

$\{\lambda (a + c) + (b + d)\}^2 - 4 (\lambda + 1) (\lambda ac + bd) > 0$

$\lambda^2 (a - c)^2 + 2\lambda \{(a + c) (b + d) - 2ac - 2bd\} + (b-d)^2 > 0$ for all λ.

The two conditions required are (i) coefficient of λ^2 should be positive and (ii) discriminant should be negative.

Discriminant = $4 \{(a + c) - (b + d) - 2ac - 2bd\}^2 - 4 \{(a - c) (b - d)\}^2 < 0$

i.e, $\{(a + c) - (b + d) - 2ac - 2bd - (a - c) (b - d)\} \{(a - c) (b - d) - 2ac - 2bd\}$

ie. $- 4 (a - d) (b - c) (a - b) (c - d) < 0$

Which is true if $a < b < c < d$

Example 5 : The number of solutions of the equation $4x (x - 3) - 5|2x - 3| + 13 = 0$ is

(a) 1 (b) 2 (c) 3 (d) 4

Solution : **(d)** $4x(x - 3) - 5|2x - 3| + 13 = (2x - 3)^2 - 5|2x - 3| + 4 = 0$

As $(2x - 3)^2 \equiv |2x - 3|^2$, this gives $|2x - 3| = 1$ or 4 giving 4 solutions.

Example 6 : The equation $\log_5 x + \left(\log_{(x^2+3)} 25\right)^{-1} = \log_{25} 10$ has

(a) No solution (b) One solution (c) Two solutions (d) Four solutions

Solution : **(b)** The equation can be written as

$$\log x + \frac{1}{2}\log_5 (x^2 + 3) = \frac{1}{2}\log_5 10$$

leading to $x\sqrt{x^2+3} = \sqrt{10}$ i.e., $(x^2 + 5) (x^2 - 2) = 0$

Of the two values $x = \pm\sqrt{2}$ log x exists only when $x = \sqrt{2}$.

Example 7 : For what value of m does the sum of the squares of the roots of the equation $(3 + m)x^2 + (2m + 5)x - (m + 4) = 0$ assumes the least value ?

Solution : $(3 + m)x^2 + (2m + 5)x - (m + 4) = 0$

$$\alpha + \beta = -\left(\frac{2m+5}{3+m}\right) \text{ and } \alpha\beta = -\frac{(m+4)}{3+m}$$

$$s = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \frac{(2m+5)^2 + 2(m+4)(m+3)}{(m+3)^2} = \frac{6m^2+34m+49}{m^2+6m+9}$$

$\therefore\ m^2(6 - s) + 2m(17 - 3s) + (49 - 9s) = 0$

For m to be real, discriminant must be ≥ 0

$(17 - 3s)^2 - (6 - s) (49 - 9s) \geq 0$

$s - 5 \geq 0\ \ \therefore\ s \geq 5$

Least value of $s = 5$, and this corresponds to $\frac{6m^2+34m+49}{m^2+6m+9} = 5$ (i.e.) $m^2 + 4m + 4 = 0$ or $m = -2$.

Example 8 : The roots of $x^2 - 8|x| + 12 = 0$

(a) Do not form a progression (b) Form an A..P. with Zero sum

(c) Form an A.P. with non-zero sum (d) Form a G.P.

Solution : **(b)** The equation is $x^2 - 8|x| + 12 = (|x| - 6) (|x| - 2) = 0$

or $x = -6, -2, 2, 6$ which form an A.P. with zero sum

Example 9 : Without actually solving the equation $x^2 - 6x + 4 = 0$. Show that both the solution lie outside [1, 5]. Prove that the solution lie one each of the intervals $\left(\frac{1}{2}, 1\right)$ and $\left(5, 5\frac{1}{2}\right)$.

Solution : $f(x) = x^2 - 6x + 4 = (x - 1) (x - 5) - 1$ and $(x - 1) (x - 5) < 0$ in (1, 5) so that $f(x) < 0$ in [1, 5] and the two solutions are outside [1, 5] Also.

$f(x) = \left(x-\frac{1}{2}\right)\left(x-\frac{11}{2}\right) + \frac{5}{4}$ and $f\left(\frac{1}{2}\right) = f\left(\frac{11}{2}\right) > 0$ giving $\left(\frac{1}{2}\right) f(1) < 0$ and

$f(5).f\left(\frac{11}{2}\right) < 0$. $f(x)$, being a polynomial is continuous for all x and it has to cross the x-axis once at least in $\left(\frac{1}{2},1\right)$ and $\left(5,5\frac{1}{2}\right)$. If follows that there is one root each in $\left(\frac{1}{2},1\right)$ and $\left(5,5\frac{1}{2}\right)$.

Example 10 : If x and y are real satisfy the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$. Show that $3 \le x \le 6$ and $1 \le y \le 10$.

Solution : Arranged as a quadratic equation in x, the given equation is

$9x^2 + 2(y - 46)\,x + (y^2 - 20y + 244) = 0$.

As x is real $4(y - 46)^2 - 4.9\,(y^2 - 20y + 244) \ge 0$,

i.e., $(y - 1)(y - 10) \le 0$ leading to $1 \le y \le 10$.

As a quadratic in y, the equation is $y^2 + 2(x - 10)y + (9x^2 - 92x + 244) = 0$.

As y is real, $4(x - 10)^2 - 4(9x^2 - 92x + 244) \ge 0$,

i.e., $(x - 3)(x - 6) \le 0$ leading to $3 \le x \le 6$.

$$\frac{x^3+y^3+z^3-3xyz}{x^3+y^3+z^3+3xyz} = \frac{77}{317}.$$

As $77.4 = 308 < 317$ and $77.5 = 385 > 317$, the result follows.

UNSOLVED EXERCISE

SECTION-A
(Single Correct Option)

1. If $\alpha = \cos 2\pi/7 + i \sin 2\pi/7$, $p = \alpha + \alpha^2 + \alpha^4$ and $q = \alpha^3 + \alpha^5 + \alpha^6$, then the equation whose roots are p, q is
(a) $x^2 - x + 2 = 0$ (b) $x^2 + x - 2 = 0$
(c) $x^2 - x - 2 = 0$ (d) $x^2 + x + 2 = 0$

2. The solution of $\left|\frac{x}{x-1}\right| + |x| = \frac{x^2}{|x-1|}$ is
(a) $x \geq 0$ (b) $x > 0$
(c) $x \in (1, \infty)$ (d) None of these

3. The number of real roots of the equation $1 + a_1x + a_2x^2 + ... + a_nx^n = 0$ where $|x| < 1/3$ and $|a_n| < 2$ is
(a) n, if n is even
(b) 1 if n is odd
(c) 0 for any natural number n
(d) None of these

4. Let α and β be the roots of the equation $x^2 + x + 1 = 0$. The equation whose roots are α^{19}, β^7 is
(a) $x^2 - x - 1 = 0$ (b) $x^2 - x + 1 = 0$
(c) $x^2 + x - 1 = 0$ (d) $x^2 + x + 1 = 0$

5. The number of solutions of the equation $1! + 2! + 3! + ... + (x-1)! + x! = k^2$ and $k \in I$ are
(a) Two (b) Four
(c) Three (d) No solution

6. For the equation $|x^2 - 2x - 3| = b$ which statement or statements are true
(a) For $b < 0$ there are no solutions
(b) For $b = 0$ there are three solutions
(c) For $0 < b < 1$ there are four solutions
(d) For $b = 1$ there are two solutions

7. The solution set of $(3/5)^x = x - x^2 - 9$ is
(a) ϕ (b) All real
(c) All $x \in N$ (d) None

8. If x is real, then the maximum value of $y = 2(a - x)(x + \sqrt{(x^2 + b^2)})$
(a) $a^2 + b^2$ (b) $a^2 - b^2$
(c) a^2 (d) None of these

9. If $0 \leq p \leq \pi$, then the quadratic equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$
(a) Real roots
(b) Imaginary roots
(c) Nothing can be said
(d) None of these

10. If x is real, the minimum value of $\frac{(a+x)(b+x)}{c+x}$ for $a > c$, $b > c$ & $x > -c$ is
(a) $\sqrt{(b + a)}$
(b) $c^2 + a^2$
(c) $(\sqrt{(a - c)} + \sqrt{(b - c)})^2$
(d) None of these

11. If $ax^2 + 2bx + c = 0$ and $a_1x^2 + 2b_1x + c_1 = 0$ have a common root and a/a_1, b/b_1, c/c_1 are in A.P., then a_1, b_1, c_1 are in
(a) A.P. (b) G.P.
(c) H.P. (d) None of these

12. The range of values of a for which all the roots of the equation $(a-1)(1 + x + x^2)^2 = (a + 1)(1 + x^2 + x^4)$ are imaginary is
(a) $(-\infty, -2]$ (b) $(2, \infty)$
(c) $-2 < a < 2$ (d) None of these

13. The number of positive integral solutions of $\frac{x^2(3x-4)^3(x-2)^4}{(x-5)^5(2x-7)^6} \leq 0$ is
(a) Four (b) Three
(c) Two (d) Only one

14. If $y = 2[x] + 3 = 3[x - 2] + 5$, then $[x + y]$ is ($[x]$ denotes the integral part of x)
(a) 10 (b) 15
(c) 12 (d) None of these

15. The solution set of the following equations of
$$\begin{cases} 4(\log_2 x)^2 + 1 = 2\log_2 y \\ \log_2 x^2 \geq \log_2 y \end{cases}$$
(a) $(\sqrt{2}, 2)$ (b) $(2, 2)$
(c) $(\sqrt{2}, 1)$ (d) None of these

16. The number of positive integral solution of the equation $7x^2 - 2xy + 3y^2 = 27$ is
(a) One (b) Two
(c) Three (d) No solution

17. Values of a for which exactly one root of $5x^2 + (a + 1)x + a = 0$ lies in the interval $1 < x < 3$ is
(a) $a > 2$
(b) $-12 < a < -3$
(c) $a > 0$
(d) None

18. The number of real roots of $(6 - x)^4 + (8 - x)^4 = 16$ is
(a) 0 (b) 2
(c) 4 (d) None of these

19. The number of real solutions of the equation $\left(\frac{9}{10}\right)^x = -3 + x - x^2$ is

(a) 0 (b) 1
(c) 2 (d) None of these

20. The equation $(x-a)^3 + (x-b)^3 + (x-c)^3 = 0$, has

(a) All the roots real
(b) One real and two imaginary roots
(c) Three real roots namely $x = a, x = b, x = c$
(d) None of these

21. If $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$ then $f(-5 + 2\sqrt{-4})$ is

(a) –160 (b) 160
(c) 0 (d) None of these

22. The value of p for which both the roots of the equation $4x^2 - 20px + (25p^2 + 15p - 66) = 0$ are less than 2, lies in

(a) (4/5, 2) (b) $(2, \infty)$
(c) (–1, –4/5) (d) $(-\infty, -1)$

23. If $5\{x\} = x + [x]$ and $[x] - \{x\} = 1/2$, where $\{x\}$ and $[x]$ are fractional and integral part of x then the number of solutions of the equation is

(a) One (b) Two
(c) Three (d) No solution

24. The solution set of the equation $|x + 1| - |x| + 3|x - 1| - 2|x - 2| = x + 2$ is

(a) $[2, \infty) \cup \{-2\}$ (b) $(-\infty, 0] \cup [1, \infty)$
(c) $(-\infty, 0] \cup [2, \infty)$ (d) None of these

25. If the roots of the equation $x^2 + 2ax + b = 0$, are real and distinct and they differ by at most 2m, then b lies in the interval

(a) $(a^2 - m^2, a^2)$ (b) $[a^2 - m^2, a^2)$
(c) $(a^2, a^2 + m^2)$ (d) None of these

26. Let $F(x)$ be a function defined by $F(x) = x - [x]$, where $[x]$ is the greatest integer less than or equal to x. Then the number of solutions of $F(x) + F(1/x) = 1$ is/are

(a) 0 (b) Infinite
(c) 1 (d) 2

27. If the roots of the equations $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then

(a) $a < 2$ (b) $2 \le a \le 3$
(c) $3 < a \le 4$ (d) $a > 4$

28. The harmonic mean of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ is

(a) 2 (b) 4
(c) 6 (d) 8

29. The root of the equation $2(1 + i)x^2 - 4(2 - i)x - 5 - 3i = 0$ which has greater modulus is

(a) $(3 - 5i)/2$ (b) $(5 - 3i)/2$
(c) $(3 - i)/2$ (d) None of these

30. The equation whose roots are the nth power of the roots of the equation $x^2 - 2x \cos\theta + 1 = 0$ is given by

(a) $x^2 + 2x \cos n\theta + 1 = 0$
(b) $x^2 - 2x \cos n\theta + 1 = 0$
(c) $x^2 - 2x \sin n\theta + 1 = 0$
(d) $x^2 + 2x \sin n\theta + 1 = 0$

31. The greatest and the least value of $\frac{x^2 + 2x + 1}{x^2 + 2x + 7}$ where x is real, are

(a) 1 and 0 (b) –1 and 0
(c) 1 and 2 (d) None of these

32. The equation $|x+1|^{\log_{(x+1)}(3+2x-x^2)} = (x-3)|x|$ has

(a) Unique (b) Two solution
(c) No solution (d) More than two

33. The system of equation $|x - 1| + 3y = 4, x - |y - 1| = 2$ has

(a) No solution
(b) A unique solution
(c) Two solution
(d) More than two solution

34. The number of ordered 4-tuple (x, y, z, w) (x, y, z, $w \in [0, 10]$) which satisfies the inequality $2^{\sin^2 x} 3^{\cos^2 y} 4^{\sin^2 z} 5^{\cos^2 w} \ge 120$ is

(a) 0 (b) 144
(c) 81 (d) Infinite

35. The number of real solutions of the system of equation $x = \frac{2z^2}{1+z^2}, y = \frac{2x^2}{1+x^2}, z = \frac{2y^2}{1+y^2}$ is

(a) 1 (b) 2
(c) 3 (d) 4

36. If α, β, γ are the roots of the equation $x^3 + a_0x^2 + a_1x + a_2 = 0$, then $(1 - \alpha^2)(1 - \beta^2)(1 - \gamma^2)$ is equal to

(a) $(1 + a_1)^2 - (a_0 + a_2)^2$
(b) $(1 + a_1)^2 + (a_0 + a_2)^2$
(c) $(1 - a_1)^2 + (a_0 - a_2)^2$
(d) None of these

SECTION-B
(Assertion-Reason)

1. **Statement-1 :** The equation whose roots are reciprocal of the roots of the equation $10x^2 - x - 5 = 0$ is $5x^2 + x - 10 = 0$.

Statement-2 : To obtain a quadratic equation whose roots are reciprocal of the roots of the given equation $ax^2 + bx + c = 0$ change the coefficients a, b, c to c, b, a. ($c \ne 0$).

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

2. **Statement-1:** If a and b are integers and roots of $x^2 + ax + b = 0$ are rational then they must be integers.

Statement-2: If the co-efficient of x^2 in a quadratic equation is unity then its roots must be integers.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

3. **Statements-1:** The equation $(x-a)(x-c) + \lambda(x-b)(x-d) = 0$ where $a < b < c < d$ has non real roots if $\lambda > 0$.

Statements-2: The equation $(a, b, c \in R)$ $ax^2 + bx + c = 0$ has non-real roots if $b^2 - 4ac < 0$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

4. **Statement-1 :** There is just one quadratic equation with real coefficients, one of whose roots is $\frac{1}{3+\sqrt{7}}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

5. **Statement-1 :** The roots of $x^2 + 2\sqrt{2008}\,x + 501 = 0$ are irrational.

Statement-2 : If the discriminant of the equation $ax^2 + bx + c = 0$, $a \neq 0$ $(a, b, c \in R)$ is a perfect square then the roots are rational.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

6. **Statement-1 :** If $ax^2 + bx + c = 0$ and $1000x^2 + 1504x + 2008 = 0$ have a common root, then a, b, c are in A.P. (where a, b, c are real numbers).

Statement-2 : If a, b, c, a_1, b_1, c_1, all are non-zero and both roots of equations $ax^2 + bx + c = 0$ and $a_1x^2 + b_1x + c_1 = 0$ are common then $\frac{a}{a_1} = \frac{b}{b_1} = \frac{c}{c_1}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

SECTION-C
(Previous Years Questions)

1. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is

(a) 6 (b) -6
(3) 3 (d) -3

2. Let α and β be the roots of equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q and r are in AP and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is

(a) $\frac{\sqrt{61}}{9}$ (b) $\frac{2\sqrt{17}}{9}$
(c) $\frac{\sqrt{34}}{9}$ (d) $\frac{2\sqrt{13}}{9}$

3. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_0}$ is

(a) 1
(b) 2
(c) 3
(d) 4

4. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

(a) $(p^3 + q)x^2 - (p^2 + 2q)x + (p^3 + q) = 0$

(b) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

(c) $(p^3 - q)x^2 - (5p^2 - 2q)x + (p^3 - q) = 0$

(d) $(p^3 - q)x^2 - (5p^2 + 2q)x + (p^3 - q) = 0$

5. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\alpha/2$, 2β be the roots of the equation $x^2 - qx + r = 0$. Then, the value of r is

(a) 2/9 (p – q)(2q – p) (b) 2/9 (q – p)(2p – q)

(c) 2/9 (q – 2p)(2q – p) (d) 2/9 (2p – q)(2q – p)

6. If a, b, c are the sides of a triangle ABC such that $x^2 - 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ has real roots, then

(a) $\lambda < \frac{4}{3}$ (b) $\lambda > \frac{5}{3}$

(c) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$ (d) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$

7. If one root is square of the other root of the equation $x^2 + px + q = 0$, then the relation between p and q is

(a) $p^3 - q(3p - 1) + q^2 = 0$

(b) $p^3 - q(3p + 1) + q^2 = 0$

(c) $p^3 + q(3p - 1) + q^2 = 0$

(d) $p^3 + q(3p + 1) + q^2 = 0$

8. The set of all real number x for which $x^2 - |x + 2| + x > 0$ is

(a) $(-\infty, -2) \cup (2, \infty)$

(b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

(c) $(-\infty, -1) \cup (1, \infty)$

(d) $(\sqrt{2}, \infty)$

9. The number of solutions of $\log_4(x - 1) = \log_2(x - 3)$ is

(a) 3 (b) 1

(c) 2 (d) 0

10. For the equation $3x^2 + px + 3 = 0$, $p > 0$, if one of the root is square of the other, then p is equal to

(a) 1/3 (b) 1

(c) 3 (d) 2/3

11. If α and β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then

(a) $0 < \alpha < \beta$ (b) $\alpha < 0 < \beta < |\alpha|$

(c) $\alpha < \beta < 0$ (d) $\alpha < 0 < |\alpha| < \beta$

12. The equation $\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$ has

(a) no solution (b) one solution

(c) two solutions

(d) more than two solutions

13. The equation $x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$ has

(a) atleast one real solution

(b) exactly three real solutions

(c) exactly one irrational solution

(d) complex roots

14. If α and β are the roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always

(a) two real roots

(b) two positive roots

(c) two negative roots

(d) one positive and one negative root

15. The equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ has

(a) no root (b) one root

(c) two equal roots

(d) infinitely many roots

16. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is

(a) 4 (b) 1

(c) 3 (d) 2

17. Both the roots of the equation $(x - b)(x - c) + (x - a)(x - c)(x - b) = 0$ are always

(a) positive (b) negative

(c) real (d) none of these

18. Let $a > 0$, $b > 0$ and $c > 0$. Then, both the roots of the equation $ax^2 + bx + c = 0$

(a) are real and negative

(b) have negative real parts

(c) have positive real parts

(d) none of above

19. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, a, b, $c \in R$ have a common root, then a : b : c is:

(a) 1 : 2 : 3 (b) 3 : 2 : 1

(c) 1 : 3 : 2 (d) 3 : 1 : 2

20. A value of *b* for which the equations $x^2 + bx - 1 = 0$, $x^2 + x + b = 0$ have one root in common in

(a) $-\sqrt{2}$ (b) $-i\sqrt{3}$

(c) $i\sqrt{5}$ (d) $\sqrt{2}$

21. Let a, b be the roots of the equation, $(x - a)(x - b) = c$, $c \neq 0$. Then the roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are

(a) a, c (b) b, c

(c) a, b (d) a + c, b + c

22. If $a \in R$ and the equation $-3(x-[x]^2)+2(x-[x])+a^2=0$ (where, [x] denotes the greatest integer $\le x$) has no integral solution, then all possible values of *a* lie in the interval
(a) $(-1, 0)\cup(0, 1)$ (b) (1, 2)
(c) (– 2, – 1) (d) $(-\infty, -2)\cup(2, \infty)$

23. For all 'x', $x^2+2ax+(10-3a)>0$, then the interval in which '*a*' lies is
(a) $a < -5$ (b) $-5 < a < 2$
(c) $a > 5$ (d) $2 < a < 5$

24. If the roots of the equation $x^2-2ax+a^2+a-3=0$ are real and less than 3, then
(a) $a < 2$ (b) $2 \le a \le 3$
(c) $3 < a \le 4$ (d) $a > 4$

25. The real number *k* for which the equation, $2x^3+3x+k=0$ has two distinct real roots in [0, 1]
(a) lies between 1 and 2
(b) lies between 2 and 3
(c) lies between – 1 and 0
(d) does not exist

26. Let a, b, c be real numbers, $a \neq 0$. If *a* is a root of $a^2x^2+bx+c=0$, β is the root of $a^2x^2+2bx+2c=0$ has a root γ that always satisfies
(a) $\gamma=\frac{\alpha+\beta}{2}$ (b) $\gamma=\alpha+\frac{\beta}{2}$
(c) $\gamma=\alpha$ (d) $\alpha<\gamma<\beta$

27. The largest interval for which $x^{12}-x^9+x^4-x+1>0$ is
(a) $-4 < x \le 0$ (b) $0 < x < 1$
(c) $-100 < x < 100$ (d) $-\infty < x < \infty$

SECTION-D
(School / Board pattern)

1. The number of roots of the equation $\sqrt{x-2}\,(x^2-4x+3)=0$ is
(a) 1 (b) 2
(c) 3 (d) 4

2. The number of integral roots of the equation $\sqrt{(x+3)-4\sqrt{x-1}}+\sqrt{(x+8)-6\sqrt{x-1}}=1$ is
(a) 1 (b) 4
(c) 5 (d) 6

3. The value of '*a*' for which the equation $x^3+ax+1=0$ and $x^4+ax^2+1=0$, have a common root is
(a) $a = 2$ (b) $a = -2$
(c) $a = 0$ (d) None of these

4. If x^2+px+1 is a factor of the expression ax^3+bx+c, then
(a) $a^2+c^2=-ab$
(b) $a^2-c^2=-ab$
(c) $a^2-c^2=ab$
(d) None of these

5. If one root of a quadratic equation be $p+\sqrt{q}$, the equation is
(a) $x^2-2px+2pq=0$
(b) $x^2-2px+(p^2-q^2)=0$
(c) $x^2+2px+(p^2+q^2)=0$
(d) $x^2-2px+(p^2-q)=0$

6. If *a*, *b*, *c* are in A.P. and $ax^2-2bx+c=0$ and $fgx^2-2egx+ef=0$ have a common root, then *e*, *f*, *g* are
(a) In A.P. (b) In G.P.
(c) In H.P. (d) None of these

7. If α, β are the roots of $ax^2+bx+c=0$ and the equation has $1/\alpha$, $1/\beta$ also for its roots, then
(a) $a = c$ (b) $b = a$
(c) $a = -c$ (d) $b = a = c$

8. The number of solutions of $\sqrt{3x^2+6x+7}+\sqrt{5x^2+10x+14}=4-2x-x^2$ is
(a) One (b) Two
(c) Three (d) Four

9. The real roots of the equation $x^2+5|x|+4=0$ are
(a) {–1, –4} (b) {1, 4}
(c) {–4, 4} (d) None of these

10. Let α, β be the roots of $x^2+x+1=0$. Then the equation whose roots are α^{229} and β^{1004}
(a) $x^2-x-1=0$ (b) $x^2-x+1=0$
(c) $x^2+x-1=0$ (d) $x^2+x+1=0$

11. If the expression $a^2(b^2-c^2)x^2+b^2(c^2-a^2)x+c^2(a^2-b^2)$ is a perfect square, then
(a) *a*, *b*, *c* are in A.P.
(b) a^2, b^2, c^2 are in A.P.
(c) a^2, b^2, c^2 are in H.P.
(d) a^2, b^2, c^2 are in G.P.

12. The necessary and sufficient condition for the equation $(1-a^2)x^2+2ax-1=0$ to have roots lying in the interval (0, 1) is
(a) $a > 0$ (b) $a < 0$
(c) $a > 2$ (d) None of these

13. If *x* be real and $2x^2+x-3=18/2x^2+x$, then *x* is equal to
(a) $\left\{-2, \frac{3}{2}\right\}$ (b) $\left\{2, -\frac{3}{2}\right\}$
(c) {–2, 3, 1 – 1} (d) {3, –2, 1, – 1}

14. Given that tan *A* and tan *B* are the roots of $x^2-ax+b=0$. The value of $\sin^2(A+B)$ is
(a) $\frac{a^2}{a^2+(1-b)^2}$ (b) $\frac{a^2}{a^2+b^2}$
(c) $\frac{a^2}{(b+a)^2}$ (d) $\frac{a^2}{b^2+(1-a)^2}$

15. The value of α, for which the equation $x^2 - (\sin\alpha - 2)x - (1 + \sin\alpha) = 0$ has roots whose sum of square is least is

(a) $\pi/4$ (b) $\pi/3$

(c) $\pi/2$ (d) $\pi/6$

16. If $(ax^2 + c)y + (a'x^2 + c') = 0$ and x is a rational function of y and ac is negative, then

(a) $ac' + a'c = 0$ (b) $a/a' = c/c'$

(c) $a^2 + c^2 = a'^2 + c'^2$ (d) $aa' + cc' = 1$

17. The values of the parameter a for which the quadratic equation $(1-2a)x^2 - 6ax - 1 = 0$ and $ax^2 - x + 1 = 0$ have at least one root in common are

(a) $a = 0, a = 1/2$

(b) $a = 1/2, a = 2/9$

(c) $a = 2/9$

(d) $a = 0, a = 1/2, a = 2/9$

18. If the roots of the quadratic equation $x^2 - 4x - \log_3 a = 0$ are real, then the least value of a is

(a) 81 (b) 1/81

(c) 1/64 (d) None of these

19. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the value of the determinant

$$\begin{vmatrix} 1 & \cos(\beta-\alpha) & \cos\alpha \\ \cos(\alpha-\beta) & 1 & \cos\beta \\ \cos\alpha & \cos\beta & 1 \end{vmatrix}$$ is

(a) $\sin(\alpha + \beta)$ (b) $\sin\alpha \sin\beta$

(c) $1 + \cos(\alpha + \beta)$ (d) None of these

20. The real values of a for which the quadratic equation $2x^2 - (a^3 + 8a - 1)x + a^2 - 4a = 0$ possesses roots of opposite signs are given by

(a) $a > 5$ (b) $0 < a < 4$

(c) $a > 0$ (d) $a > 7$

CHAPTER 2

COMPLEX NUMBER

1. INTRODUCTION

You may recall that the equation

$$x^2 + 4 = 0 \quad \text{.... (1)}$$

has no solution in the set $\mathbb{R}$ of real numbers because the square of every real number is either positive or zero. Therefore, we feel the necessity to extend the system of real numbers to a new kind of system, namely, complex number system that allows the square root of negative numbers. The Greeks were the first to recognize the fact that square root of a negative number does not exist in the real number system. But the indian mathematician Mahavira (A.D. 850) was the first to state this difficulty clearly as he mentioned in his work Ganitasara Sangraha that "in the nature of things a negative (quantity) is not a square (quantity), it has, therefore no square root ...."

Now, the solutions of equation (1) are

$$x = \pm 2\sqrt{-1}$$

We assume that square root of –1 is denoted by the symbol i (called imaginary unit and pronounced as iota), where is a number such that $i^2 = -1$. So, the solution of (1) can be written as $x = \pm 2i$.

The symbol i was first introduced in mathematics by the famous Swiss Mathematician, Leonhard Euler (1707–1783) in 1748, possibly because i is the first letter of the Latin word 'imaginarius'.

Note :- *If $a < 0$, then $\sqrt{a} = \sqrt{|a|}\,i$.*

Integral Powers of *i*

We have, $i = \sqrt{-1}$, $i^2 = -1$. Therefore,

$i^3 = i^2 \times i = (-1) \times i = -i$,

$i^4 = i^2 \times i^2 = (-1) \times (-1) = 1$.

TRICKS (S) FOR PROBLEM SOLVING

- For any $n \in \mathbb{N}$

1. $$i^{2n} = (i^2)^n = (-1)^n = \begin{bmatrix} 1, \text{when } n \text{ is even} \\ -1, \text{when } n \text{ is odd} \end{bmatrix}$$

2. $$i^{2n+1} = (i^{2n})i = (-1)^n i = \begin{bmatrix} i, \text{when } n \text{ is even} \\ -i, \text{when } n \text{ is odd} \end{bmatrix}$$

- The sum of four consecutive powers of i is zero. For example,

  $i^{10} + i^{11} + i^{12} + i^{13} = 0$

- Also, for any $n \in \mathbb{N}$, the value of i^{-n} is found out by writing this as $\frac{1}{i^n}$ and solving i^n.
- Thus, any integral power of i can be expressed in terms of ±1 or ± i.
- For any two real numbers a and b, $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ is true only when at least one of a and b is either zero or positive. If both a and b are positive real numbers, then the calculation

  $\sqrt{-a} \times \sqrt{-b} = \sqrt{(-a)(-b)} = \sqrt{ab}$ is wrong.

  The correct calculation is

  $\sqrt{-a} \times \sqrt{-b} = (\sqrt{-1}\sqrt{a})(\sqrt{-1}\sqrt{b})$

$= (i\sqrt{a}) \times (i\sqrt{b}) \ = i^2(\sqrt{a})(i\sqrt{b})$

$= i^2(\sqrt{a} \times \sqrt{b}) = (-1)(\sqrt{ab}) \ = -\sqrt{ab}$.

Thus, the calculation $\sqrt{-2} \times \sqrt{-3} = \sqrt{(-2) \times (-3)} = \sqrt{6}$ is wrong.

The correct result is $\sqrt{-2} \times \sqrt{-3} = (i\sqrt{2})(i\sqrt{3})$

$= i^2(\sqrt{2} \times \sqrt{3}) \ = -\sqrt{6}$.

2. COMPLEX NUMBERS

An expression of the form $x + iy$, where x and y are real numbers and $i = \sqrt{-1}$, is called a complex nums not hold good in case of complex numbers having non-zero imaginary parts. For example, the statement $8 + 5i > 4 + 2i$ makes no sense.

3. ALGEBRA OF COMPLEX NUMBERS

1. **Addition :** For two complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, the subtraction of z_2 from z_1 is defined as

$z = z_1 - z_2 = (a_1 + a_2) + i\,(b_1 + b_2)$

2. **Subtraction :** For two complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, the subtraction of z_2 from z_1 is defined as

$z_1 - z_2 = z_1 + (-z_2) = (a_1 - a_2) + i\,(b_1 - b_2)$.

3. **Multiplication :** Multiplication of two complex numbers $z_1 = a + ib$ and $z_2 = c + id$ is defined as

$z_1 \cdot z_2 = (a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc)$

The product of complex numbers can be easily computed if we actually carry out the multiplication as given below :

$(a + ib)\,(c + id) = ac + iad + ibc + i^2bd$

$= ac + i(ad + bc) - bd \quad (\because\ i^2 = -1)$

$= (ac - bd) + i(ad + bc)$.

4. **Division :** Division of two complex numbers

$z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$,

where $x_2 + iy_2 \neq 0$, is defined as

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)}$$

$$= \frac{x_1x_2 + y_1y_2 + i(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + \frac{i(x_1y_1 - x_1y_2)}{x_2^2 + y_2^2}$$

4. MULTIPLICATIVE INVERSE AND CUNJUGATE OF A COMPLEX NUMBER

1. **Multiplicative Inverse of a Non-zero Complex Number :** Multiplicative inverse of a non-zero complex number $z = a + ib$ is defined as

$$z^{-1} = \frac{1}{z} = \frac{1}{a + ib} = \frac{1}{a + ib} \times \frac{a - ib}{a - ib} = \frac{a - ib}{a^2 + b^2} = \frac{a}{a^2 + b^2} - i\frac{b}{a^2 + b^2}$$

i.e., $z^{-1} = \dfrac{\mathrm{Re}(z)}{|z|^2} + i\dfrac{[-\mathrm{Im}(z)]}{|z|^2}$

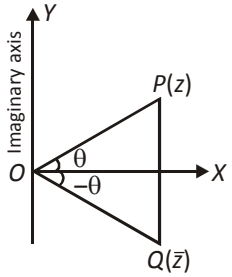
Conjugate of a Complex Number

Conjugate of a complex number $z = a + ib$ is defined as $\overline{z} = a - ib$.

For example, $\overline{4 + 5i} = 4 - 5i$ and $\overline{4 - 5i} = 4 + 5i$. io

IMPORTANT POINT(S) TO NOTE

Geometrically, the conjugate of z is the reflection or point image of z in the real axis.



Properties of Conjugate

1. $\overline{(\overline{z})} = z$
2. $z = \overline{z}$ if and only if z is purely real
3. $z = -\overline{z}$ if and only if z is purely imaginary
4. $z + \overline{z} = 2\text{Re}(z)$ and $z - \overline{z} = 2i\,\text{Im}(z)$
5. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
6. $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$
7. $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$
8. $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}, z_2 \neq 0$
9. If $z = f(z_1)$, then $\overline{z} = f(\overline{z_1})$
10. $\left(\overline{z^n}\right) = (\overline{z})^n$
11. $z_1\overline{z_2} + \overline{z_1}z_2 = 2\text{Re}(\overline{z_1}z_2) = 2\text{Re}(z_1\overline{z_2})$

Note :-

- If z lies in the first quadrant, then $\overline{z}$ lies in the fourth quadrant and vice-versa.
- If z lies in the second quadrant, then $\overline{z}$ lies in the third quadrant and vice-versa.
- $z + \overline{z}$ is purely real whereas $z - \overline{z}$ is purely imaginary.

Method of writing the Complex Number $\frac{a+ib}{c+id}$ in the form $A + iB$

We have, $\frac{a+ib}{c+id} = \frac{(a+ib)(c-id)}{(c+id)(c-id)}$

[Multiplying the numerator and the denominator by the conjugate of the denominator]

$$= \frac{(ac+bd)+(bc-ad)}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + i\frac{bc-ad}{c^2+d^2}$$

$= A + iB$, where $A = \frac{ac+bd}{c^2+d^2}$ and $B = \frac{bc-ad}{c^2+d^2}$.

TRICKS(S) FOR PROBLEM SOLVING

To put the complex number $\frac{a+ib}{c+id}$ in the form $A + iB$, multiply the numerator and the denominator by the conjugate of the denominator.

5. MODULUS OF A COMPLEX NUMBER

Modulus of a complex number $z = a + ib$, denoted as mod (z) or $|z|$, is defined as

$|z| = \sqrt{a^2 + b^2}$, where $a = Re(z)$, $b = Im(z)$.

Sometimes, $|z|$ is called absolute value of z, Note that $|z| \geq 0$.

For example, if $z = 3 + 2i$, then $|z| = \sqrt{3^2 + 2^2} = \sqrt{13}$.

CAUTION

Then $|z|$ symbol used here is different from $|x|$ symbol used for real numbers.

Properties of Modulus

1. $|z| \geq 0$ and $|z| = 0$ if and only if $z = 0$, i.e., $x = 0$, $y = 0$
2. $|z| = |\bar{z}| = |-z| = |-\bar{z}|$.
3. $z\bar{z} = |z|^2$

Explanation

Let $z = a + ib, |z| = \sqrt{a^2 + b^2}$. Then

$$z\bar{z} = (a+ib)(a-ib) = a^2 + b^2 = \left(\sqrt{a^2+b^2}\right)^2 = |z|^2 .$$

4. $-|z| \leq Re(z) \leq |z|$ and $-|z| \leq Im\,(z) \leq |z|$

Explanation

Let $z = a + ib$, then $|z|^2 = a^2 + b^2$.

But $a^2 \leq a^2 + b^2$ $(\because\ b^2 \geq 0)$

Thus, $a^2 \leq |z|^2 \Rightarrow |a|^2 \leq |z|^2 \Rightarrow |a \leq |z|$

$\Rightarrow -|z| \leq a \leq |z|$. $(\because\ |x| \leq k \Rightarrow -k \leq x \leq k)$

$\Rightarrow -|z| \leq Re(z) \leq |z|$.

Similarly, $-|z| \leq Im(z) \leq |z|$.

5. $|z^n| = |z|^n$
6. $|z_1 z_2| = |z_1||z_2|$
7. $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$
8. $|z_1 \pm z_2| \leq |z_1| + |z_2|$

Exaplanation

$$|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2})$$

$$= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = z_1\bar{z}_1 + z_2\bar{z}_2 + z_1\bar{z}_2 + z_2\bar{z}_1$$

$$= |z_1|^2 + |z_2|^2 + (z_1\bar{z}_2 + \overline{z_1\bar{z}_2})$$

$$= |z_1|^2 + |z_2|^2 + 2\mathrm{Re}(z_1\bar{z}_2) \leq |z_1|^2 + |z_2|^2 + 2|z_1\bar{z}_2| \quad (\because\ |\bar{z}| = |z|)$$

$$= |z_1|^2 + |z_2|^2 + 2|z_1||\bar{z}_2| = |z_1|^2|z_2|^2 + 2|z_1||z_2| \qquad (\because |\bar{z}| = |z|)$$

$$= (|z_1| + |z_2|)^2$$

$\Rightarrow$ $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$.

Taking positive square roots, we get

$|z_1 + z_2| \leq |z_1| + |z_2|$.

9. $|z_1 - z_2| \geq |z_1| - |z_2|$
10. $|z_2 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$

Explanation :

Let $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, then

$z_1 + z_2 = (a_1 + a_2) + i(b_1 + b_2)$ and $z_1 - z_2 = (a_1 - a_2) + i(b_1 - b_2)$

Thus, $|z_1 + z_2|^2 + |z_1 - z_2|^2$

$= (a_1 + a_2)^2 + (b_1 + b_2)^2 + (a_1 - a_2)^2 + (b_1 - b_2)^2$

$= 2(a_1^2 + a_2^2) + 2(b_1^2 + b_2^2)$ [$\because (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$]

$= 2(a_1^2 + b_1^2) + 2(a_2^2 + b_2^2) = 2|z_1|^2 + 2|z_2|^2$.

11. $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2Re(z_1\bar{z}_2)$.

Explanation

$$|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2}) = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2)$$
$$= z_1\bar{z}_1 + z_2\bar{z}_2 + z_1\bar{z}_2 + z_2\bar{z}_1 = |z_1|^2 + |z_2|^2 + (z_1\overline{z_2})(\overline{z_1\bar{z}_2})$$
$$= |z_1|^2 + |z_2|^2 + 2\text{Re}(z_1\bar{z}_2) \quad [\because z + \bar{z} = 2\text{Re}(z)]$$

12. $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2\text{Re}(z_1\overline{z_2})$

Explanation

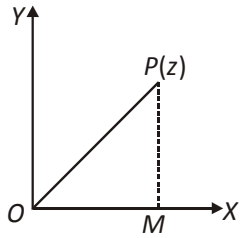
$$|z_1 - z_2|^2 = (z_1 - z_2)(\overline{z_1 - z_2}) = (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$$
$$= z_1\bar{z}_1 + z_2\bar{z}_2 - z_1\bar{z}_2 - z_2\bar{z}_1 = |z_1|^2 + |z_2|^2 - z_1\bar{z}_2 - (\overline{z_1\bar{z}_2})$$
$$= |z_1|^2 + |z_2|^2 - 2\text{Re}(z_1\bar{z}_2)$$

13. $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Rightarrow \frac{z_1}{z_2}$ is purely imaginary

or, $\text{Re}\left(\frac{z_1}{z_2}\right) = 0$

IMPORTANT POINT(S) TO NOTE

Gemetrically, $|z|$ represents the distance of point *P* fromthe origin, i.e., $|z| = OP$



Most of the complex equations are solve using the property $z\bar{z} = |z|^2$.

Square Roots of a Complex Number

Let $z = a + ib$ and let the square root of z be the complex number $x + iy$. Then

$\sqrt{a + ib} = x + iy$ or $(a + ib) = (x + iy)^2 = (x^2 - y^2) + (2xy)i$

Equating real and imaginary part, we get

$a = x^2 - y^2$ ...(1)

and, $b = 2xy$ ...(2)

Now, $x^2 + y^2 = \sqrt{\left(x^2 - y^2\right)^2 + 4x^2y^2} = \sqrt{a^2 + b^2}$ ...(3)

Solving the equation (1) and (3), we get

$$x = \pm\sqrt{\left(\frac{\sqrt{a^2 + b^2} + a}{2}\right)} \text{ and } y = \pm\sqrt{\left(\frac{\sqrt{a^2 + b^2} - a}{2}\right)}$$

From (2), we can determine the sign of xy. If $xy > 0$, then x and y will have same sign. Thus

$$\sqrt{a + ib} = \pm\left[\sqrt{\left(\frac{\sqrt{a^2 + b^2} + a}{2}\right)} + i\sqrt{\left(\frac{\sqrt{a^2 + b^2} - a}{2}\right)}\right]$$

If $xy < 0$, then

$$\sqrt{a+ib} = \pm\left[\sqrt{\left(\frac{\sqrt{a^2+b^2}+a}{2}\right)} - i\sqrt{\left(\frac{\sqrt{a^2+b^2}-a}{2}\right)}\right]$$

Illustration 1: Find the locus of complex number z if $\left|\frac{1-iz}{z-i}\right| = 1$.

Solution : Given $\left|\frac{1-iz}{z-i}\right| = 1$

$$\Rightarrow \left|\frac{i^4+i^3z}{z-i}\right| = 1 \Rightarrow |i^3|\left|\frac{z+i}{z-i}\right| = 1 \Rightarrow |z+i| = |z-i|$$

Which represents the equation of perpendicular bisector of i and $-i$ and that will be the x-axis.

Illustration 2: Find the least and greatest value of $|z|$ which satisfies $\left|z+\frac{1}{z}\right| = a$, where $a \in R$.

Solution : Let $z = r(\cos\theta + i\sin\theta)$ then $\frac{1}{z} = \frac{1}{r}(\cos\theta - i\sin\theta)$

$$a = \left|z+\frac{1}{z}\right| = \left|z-\left(-\frac{1}{z}\right)\right| \geq \left||z| - \frac{1}{|z|}\right| \quad -a \leq r - \frac{1}{r} \leq a$$

$r^2 + ar - 1 \geq 0$ and $r^2 - ar - 1 \leq 0$

$$\left(r \geq \frac{-a+\sqrt{a^2+4}}{2} \text{ or } r \leq \frac{-a-\sqrt{a^2+4}}{2}\right) \text{ and } \frac{a-\sqrt{a^2+4}}{2} \leq r \leq \frac{a+\sqrt{a^2+4}}{2}$$

$$\therefore \quad \frac{-a+\sqrt{a^2+4}}{2} \leq r \leq \frac{a+\sqrt{a^2+4}}{2}, \quad r_{max} = \frac{a+\sqrt{a^2+4}}{2}; \; r_{min} = \frac{-a+\sqrt{a^2+4}}{2}$$

Illustration 3: If $n \in N$, prove that $\left(\frac{1+\sin\phi+i\cos\phi}{1+\sin\phi-i\cos\phi}\right)^n = \cos\left(\frac{n\pi}{2}-n\phi\right) + i\sin\left(\frac{n\pi}{2}-n\phi\right)$

Solution : We have $1 + \sin\theta + i\cos\theta = 1 + \cos\left(\frac{\pi}{2}-\phi\right) + i\sin\left(\frac{\pi}{2}-\theta\right)$

$$= 2\cos^2\left(\frac{\pi}{4}-\frac{\phi}{2}\right) + 2i\sin\left(\frac{\pi}{4}-\frac{\phi}{2}\right)\cos\left(\frac{\pi}{4}-\frac{\phi}{2}\right)$$

$$= 2\cos\left(\frac{\pi}{4}-\frac{\phi}{2}\right)\left(\cos\left(\frac{\pi}{4}-\frac{\phi}{2}\right) + i\sin\left(\frac{\pi}{4}-\frac{\phi}{2}\right)\right) = 2\cos\left(\frac{\pi}{2}-\frac{\phi}{2}\right)e^{i(\pi/4-\phi/2)}$$

Similarly, $1+\sin\phi-i\cos\phi = 2\cos^2\left(\frac{\pi}{2}-\frac{\phi}{2}\right) - 2i\sin\left(\frac{\pi}{4}-\frac{\phi}{2}\right)\cos\left(\frac{\pi}{4}-\frac{\phi}{2}\right)$

$$= 2\cos\left(\frac{\pi}{4}-\frac{\phi}{2}\right)e^{-i(\pi/4-\phi/2)}$$

$$\therefore \quad \left(\frac{1+\sin\phi+i\cos\phi}{1+\sin\phi-i\cos\phi}\right)^n = \left(\frac{e^{i\left(\frac{\pi}{4}-\frac{\phi}{2}\right)}}{e^{-i\left(\frac{\pi}{4}-\frac{\phi}{2}\right)}}\right)^n$$

$$= e^{in\left(\frac{\pi}{2}-\phi\right)} = \cos\left(\frac{n\pi}{2}-n\phi\right) + i\sin\left(\frac{n\pi}{2}-n\phi\right)$$

Student Task

Q.1 If $\sum_{k=0}^{200} i^k + \prod_{p=1}^{50} i^p = x + iy$, then (x, y) is equal to

(a) (0, 1) (b) (1, –1) (c) (2, 3) (d) (4, 8)

Q.2 The smallest integer for which $\left(\frac{1+i}{1-i}\right)^n = 1$ is

(a) $n = 8$ (b) $n = 12$ (c) $n = 16$ (d) $n = 4$

Q.3 If $z = x - iy$ and $z^{1/3} = p + iq$ then $\left(\frac{x}{p} + \frac{y}{q}\right)/(p^2 + q^2)$ is equal to

(a) 1 (b) –1 (c) 2 (d) –2

Q.4 If $x = \frac{1+i}{2}$ then the expression $2x^4 - 2x^2 + x + 3$ equals

(a) $3 - \frac{i}{2}$ (b) $3 + \frac{i}{2}$ (c) $\frac{3+i}{2}$ (d) $\frac{3-i}{2}$

Q.5 The complex numbers $\sin x + i \cos 2x$ and $\cos x - i \sin 2x$ are conjugate to each other, for

(a) $x = n\pi$ (b) $x = \left(\frac{n+1}{2}\right)\pi$ (c) $x = 0$ (d) No value of x

Answers :

1. (b) 2. (d) 3. (d) 4. (a) 5. (d)

6. ARGAND PLANE AND GEOMETRICAL REPRESENTATION OF COMPLEX NUMBERS

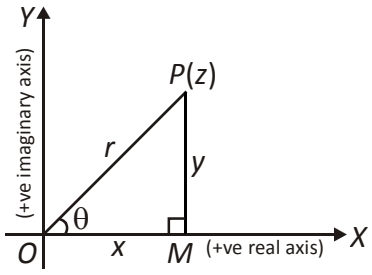
Let O be the origin and OX and OY be the x-axis and y-axis, respectively. Then, any complex number $z = x + iy = (x, y)$ may be represented by a unique point P whose coordinates are (x, y).

The representation of complex numbers as points in a plane forms an **Argand diagram.**

The plane on which complex numbers are represented is known as the **complex plane** or **Argand's plane** or **Gaussian plane.** The x-axis is called the **real axis** and y-axis the **imaginary axis.**

The complex number $z = x + iy$ is known as the **affix** of the point (x, y) which it represents.

Polar Form of a Complex Number



Let O be the origin and OX and OY be the x-axis and y-axis, respectively. Let $z = x + iy$ be a complex number represented by the point $P(x, y)$.

Draw $PM \perp OX$. Then,

$OM = x$ and $PM = y$. Join OP. Let $OP = r$ and $\angle XOP = \theta$. Then,

$z = x + iy = r(\cos\theta + i \sin\theta)$

This form of z is called **polar or trigonometric form**.

Comparing real and imaginary parts, we get

$x = r \cos\theta$ .... (1)

and, $y = r \sin\theta$ .... (2)

Squaring (1) and (2) and adding, we get

$r^2 = x^2 + y^2$ or $r = \sqrt{x^2 + y^2} = |z|$

Thus, r is known and is equal to the modulus of the complex number z.

Substituting the value of r in (1) and (2), we get

$$\cos\theta = \frac{x}{\sqrt{x^2+y^2}} \text{ and } \sin\theta = \frac{y}{\sqrt{x^2+y^2}} \quad \text{.... (3)}$$

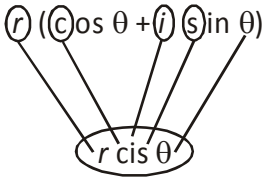
Dividing (2) by (1), we get $\tan\theta = \frac{y}{x}$.

The form $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ of the complex number z is called exponential form.

Any value of θ satisfying (3) is known as **ampmlitude or argument** of z and written as $\theta = \arg(z)$ or $\theta = \text{amp } z$.

IMPORTANT POINT(S) TO NOTE

- The unique value of θ such that $-\pi < \theta \leq \pi$ for which $x = r\cos\theta$ and $y = r\sin\theta$, is known as the **principal value of the argument.**
- The general value of argument is $(2n\pi + \theta)$, where n is an integer and θ is the principal value or arg (z).
- While reducing a complex number to polar form, we always take the principal value.
- The complex number $z = r(\cos\theta + i\sin\theta)$ can also be written as r cis θ



- Argument is taken as positive or negative according as it is measured in the anticlockwise direction or clockwise direction.

TRICK(S) FOR PROBLEM SOLVING

- If $x > 0$, $y > 0$ (i.e., z is in first quadrant), then

$$\arg z = \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

- If $x < 0$, $y > 0$ (i.e., z is in second quadrant), then

$$\arg z = \theta = \pi - \tan^{-1}\left(\frac{y}{|x|}\right)$$

- If $x < 0$, $y < 0$ (i.e., z is in third quadrant), then

$$\arg z = \theta = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$$

- If $x > 0$, $y < 0$ (i.e., z is in fourth quadrant), then

$$\arg z = \theta = -\tan^{-1}\left(\frac{|y|}{x}\right)$$

- Argument of the complex number 0 is not defined.
- $\arg(x+i0) = \begin{cases} 0, \text{ if } x > 0 \\ \pi, \text{ if } x < 0 \end{cases}$
- $\arg(0+iy) = \begin{cases} \pi/2, & \text{if } y > 0 \\ 3\pi/2, & \text{if } y < 0 \end{cases}$
- If $\arg z = \frac{\pi}{2}$ or $\frac{-\pi}{2}$, then z is purely imaginary.
- If $\arg z = 0$ or π, then z is purely real.

Properties of Argument

1. $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
2. $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$
3. $\arg\left(\frac{z}{\bar{z}}\right) = 2\arg z$
4. $\arg(z^n) = n \arg z$
5. If $\arg\left(\frac{z_2}{z_1}\right) = \theta$, then $\arg\left(\frac{z_1}{z_2}\right) = 2k\pi - \theta,\ k \in I.$
6. $\arg \bar{z} = -\arg z = \arg\left(\frac{1}{z}\right)$
7. $\arg(z\bar{z}) = \arg\left(|z|^2\right) = \arg$ (positive real number) = 0.
8. $\arg(iz) = \arg(i) + \arg z = \frac{\pi}{2} + \arg z$

Particular Cases of Polar Form

1. $1 = 1 + i0 = \cos 0 + i \sin 0$
2. $-1 = -1 + i\,0 = \cos\pi + i \sin\pi$
3. $i = 0 + i1 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$
4. $-i = 0 + i(-1) = \cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)$
5. $1 - i = \sqrt{2}\left[\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right]$
6. $-1 - i = \sqrt{2}\left[\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right]$

7. EULERIAN REPRESENTATION OF A COMPLEX NUMBER

Since $e^{i\theta} = \cos\theta + i\sin\theta$, thus any non-zero complex number $z = x + iy = r(\cos\theta + \sin\theta)$ can be represented in Eulerian form as

$z = re^{i\theta} = r(\cos\theta + i\sin\theta)$,

where $|z| = r$ and $\theta = \arg(z)$.

KEY POINT(S) TO REMEMBER

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \text{ and } \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Logarithm of a Complex Number

$$\log(x + iy) = \log_e(re^{i\theta}) = \log_e r + i\theta = \log_e\sqrt{(x^2 + y^2)} + i\tan^{-1}\left(\frac{y}{x}\right)$$

$\log_e(z) = \log_e|z| + i\arg(z)$

CAUTION

Since the argument of a complex number is not unique, the log of a complex number cannot be unique. In general,

$\log_e(z) = \log_e|z| + i[2k\pi + \arg(z)],\ k \in I$

KEY POINT(S) TO REMEMBER

$$\log i = \log_e e^{\frac{i\pi}{2}} = \frac{i\pi}{2}, \log(\log i) = \log\left(\frac{i\pi}{2}\right) = \log i + \log\left(\frac{\pi}{2}\right) = \frac{i\pi}{2} + \log(\pi/2)$$

Vector Representation of a Complex Number

If P is the point (a, b) on the argand plane corresponding to the complex number $z = a + ib$.

Then, $\overrightarrow{OP} = a\hat{i} + b\hat{j}$,

$\therefore \quad \left|\overrightarrow{OP}\right| = \sqrt{a^2 + b^2} = |z|$

and, arg (z) = direction or the vector $\overrightarrow{OP} = \tan^{-1}\left(\frac{b}{a}\right)$

8. DE'MOIVRE'S THEOREM

If n is any integer, then

$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

TRICK(S) FOR PROBLEM SOLVING

- If n is any rational number, then $\cos n\theta + i\sin n\theta$ is one of the values of $(\cos\theta + i\sin\theta)^n$.
- $(\cos\theta + i\sin\theta)^{-n} = \cos(-n)\theta + i\sin(-n)\theta = \cos n\theta - i\sin(n)\theta$.
- $(\cos\theta - i\sin\theta)^n = [\cos(-\theta) + i\sin(-\theta)]^n = \cos(-n\theta) + i\sin(-n\theta) = \cos n\theta - i\sin n\theta$.
- $\frac{1}{\cos\theta + i\sin\theta} = (\cos\theta + i\sin\theta)^{-1} = \cos\theta - i\sin\theta$.
- The theorem cannot be applied to $(\cos\theta + i\sin\theta)^n$
  i.e., θ must be same with cos and sin both.
- The theorem is not directly applicable to $(\sin\theta + i\cos\theta)^n$, rather

$$(\sin\theta + i\cos\theta)^n = \left[\cos\left(\frac{\pi}{2} - \theta\right) + i\sin\left(\frac{\pi}{2} - \theta\right)\right]^2 = \cos n\left(\frac{\pi}{2} - \theta\right) + i\sin\left(\frac{\pi}{2} - \theta\right)$$

- $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \ldots (\cos\theta_n + i\sin\theta_n)$
  $= \cos(\theta_1 + \theta_2 + \ldots + \theta_n) + i\sin(\theta_1 + \theta_2 + \ldots + \theta_n)$.

Roots of a Complex Number

If $z = r(\cos\theta + i\cos\theta)$ and n is a positive integer, then

$$z^{\frac{1}{n}} = r^{\frac{1}{n}}\left[\cos\left(\frac{2k\pi + \theta}{n}\right) + i\sin\left(\frac{2k\pi + \theta}{n}\right)\right]$$

where $k = 0, 1, 2, 3, \ldots (n - 1)$.

9. CUBE ROOTS OF UNITY

Cube roots of unity

Let $z = 1^{1/3}$ or $z^3 - 1 = 0 \Rightarrow (z - 1)(x^2 + z + 1) = 0$.

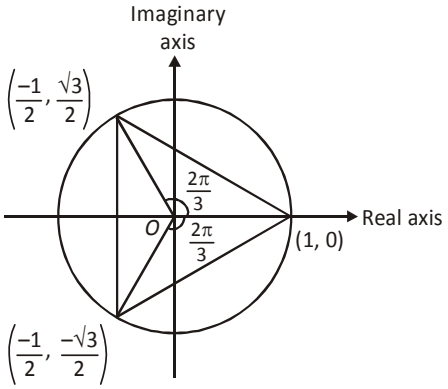
i.e., $z = 1, \frac{-1 + i\sqrt{3}}{2}, \frac{-1 - i\sqrt{3}}{2}$

Put, $\omega = \frac{-1 + i\sqrt{3}}{2}$, then $\omega^2 = \frac{-1 - i\sqrt{3}}{2}$. Thus, cube roots of unity are 1, ω, ω^2.

Properties of Cube Roots of Unity

1. $1 + \omega + \omega^2 = 0$
2. $\omega^3 = 1$
3. $\omega^{3n} = 1$, $\omega^{3n+1} = \omega$, $\omega^{3n+2} = \omega^2$

4. $\bar{\omega} = \omega^2$ and $(\bar{\omega})^2 = \omega, \omega\bar{\omega} = \omega^3, \omega = e^{\frac{2\pi i}{3}}, \omega^2 = e^{\frac{2\pi i}{3}}$



5. If $a + b\omega + c\omega^2 = 0$, then $a = b = c$ provided a, b, c are real.
6. If these roots are marked on the argand plane, then these are vertices of an equilateral triangle with circumcentre at origin.

TRICK(S) FOR PROBLEM SOLVING

Let n be an integer. Then, there exists integer k such that either $n = 3k + 1$ or $3k + 2$.

$n = 3k \Rightarrow \omega^n = \omega^{3k} = (\omega^3)^k = 1^k = 1.$

$n = 3k + 1 \Rightarrow \omega^n = \omega^{3k+1} = (\omega^3)^k \cdot \omega^2 = 1 \cdot \omega = \omega.$

$n = 3k + 2 \Rightarrow \omega^n = \omega^{3k+2} = (\omega^3)^k \cdot \omega^2 = 1 \cdot \omega^2 = \omega^2$

Thus, any integral power of ω is either 1 or ω or ω^2.

Fourth Roots of Unity

The four, fourth roots of unity are given by the solution set of the equation $x^4 - 1 = 0$

$\Rightarrow \quad (x^2 - 1)(x^2 + 1) = 0 \Rightarrow x = \pm 1, \pm i$

Fourth roots of unity are vertices of a square which lie on coordinate axes.

Some Useful Relations

1. $x^2 + y^2 = (x + iy)(x - iy)$
2. $x^3 + y^3 = (x + y)(x + y\omega)(x + y\omega^2)$
3. $x^3 - y^3 = (x - y)(x - y\omega)(x - y\omega^2)$
4. $x^2 + xy + y^2 = (x - y\omega)(x - y\omega^2)$, in particular, $x^2 + x + 1 = (x - \omega)(x - \omega^2)$
5. $x^2 - xy + y^2 = (x + y\omega)(x + y\omega^2)$, in particular, $x^2 - x + 1 = (x + \omega)(x + \omega^2)$
6. $x^2 + y^2 + z^2 - xy - xz - yz = (x + y\omega + z\omega^2)(x + y\omega^2 + z\omega)$
7. $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z)$

***n*th roots of Unity**

Since $1 = \cos 0 + i \sin 0$, therefore,

$(1)^{1/n} = (\cos 0 + i \sin 0)^{1/n}$

$= \cos\frac{2\pi r + 0}{n} + i\sin\frac{2\pi r + 0}{n}$; $r = 0, 1, 2, ..., (n-1)$ $= \cos\frac{2\pi r}{n} + i\sin\frac{2\pi r}{n}$; $r = 0, 1, 2, ..., (n-1$

$= e^{i\frac{2r\pi}{n}}$; $r = 0, 1, 2, ..., (n-1) = 1, e^{(i2\pi/n)}, e^{(i4\pi/n)}, ..., e^{[i2(n-1)\pi/n]}$

$= 1, \alpha, \alpha^2, \alpha^3, ..., \alpha^{n-1}$, where $\alpha = e^{(i2\pi/n)}$

Properties of *n*th Roots of Unity

1. $1 + \alpha + \alpha^2 + ... + \alpha^{n-1} = 0$
2. $1 \cdot \alpha \cdot \alpha .... \alpha^{n-1} = (-1)^{n-1}$
3. The n, nth roots of unity lie on the unit circle $|z| = 1$ and form the vertices of a regular polygon of n sides.

Explanation

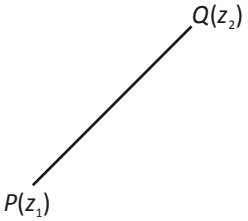
In the argand plane, the n roots are equally inclined at angle $2\pi/n$ to each other and they have the same magnitude. Hence, the roots form the vertices of a regular n-sided polygon.

4. The n, nth roots of unity form a G.P. with common ratio $e^{(i2\pi/n)}$

10. GEOMETRY OF COMPLEX NUMBERS

1. **Distance formula :** The distance between two points $P(z_1)$ and $Q(z_2)$ is given by

$PQ = |z_2 - z_1| = |$ affix of Q – affix of $P|$



2. **Section formula :** If $R(z)$ divides the line segment joining $P(z_1)$ and $Q(z_2)$ in the ratio $m_1 : m_2$ $(m_1, m_2 > 0)$, then

(i) For internal division $z = \dfrac{m_1 z_2 + m_2 z_1}{m_1 + m_2}$

(ii) For external division $z = \dfrac{m_1 z_2 - m_2 z_1}{m_1 - m_2}$

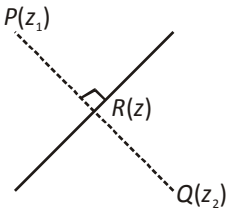
3. **Equation of the perpendicular bisector :** If $P(z_1)$ and $Q(z_2)$ are two fixed points and $R(z)$ is moving point such that it is always at equal distance from $P(z_1)$ and $Q(z_2)$ then locus of $R(z)$ is perpendicular bisector of PQ

i.e., $PR = QR$ or $|z - z_1| = |z - z_2|$

$\Rightarrow \quad |z - z_1|^2 = |z - z_2|^2$

After solving,

$$z(\bar{z}_1 - \bar{z}_2) + \bar{z}(z_1 - z_2) = |z_1|^2 - |z_2|^2$$



4. **Equation of a straight line :**

(i) **Parametric form :** Equation of a straight joining the points having affixes z_1 and z_2 is $z = tz_1 + (1 - t)z_2$, when $t \in \mathbb{R}$

(ii) **Non-parametric form :** Equation of a straight line joining the points having affixes z_1 and z_2 is

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0 \quad \Rightarrow \quad z(\bar{z}_1 - \bar{z}_2) - \bar{z}(z_1 - z_2) + z_1\bar{z}_2 - z_2\bar{z}_1 = 0$$

TRICK(S) FOR PROBLEM SOLVING

➤ Three points z_1, z_2 and z_3 are collinear if,

$$\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} = 0$$

➤ If three points $A(z_1)$, $B(z_2)$, $C(z_3)$ are collinear, then slope of AB = slope of BC = slope of AC

$$\Rightarrow \quad \frac{z_1 - z_2}{z_1 - \bar{z}_2} = \frac{z_2 - z_3}{\bar{z}_2 - \bar{z}_3} = \frac{z_1 - z_3}{\bar{z}_1 - \bar{z}_3}$$

➤ If z_1, z_2, z_3 are collinear, then $\arg\left(\dfrac{z_3 - z_1}{z_2 - z_1}\right) = 0$ i.e. $\dfrac{z_3 - z_1}{z_2 - z_1}$ is purely real.

(iii) **General equation of a straight line :** The general equation of a straight line is of the form $\bar{a}z + a\bar{z} + b = 0$, where a is complex number and b is real number.

(iv) **Slope of a line :** The complex slope of the line

$\bar{a}z + a\bar{z} + b = 0$ is $-\frac{a}{\bar{a}} = -\frac{\text{coeff. of } \bar{z}}{\text{coeff. of } z}$ and real slope of the line is $-\frac{\text{Re}(a)}{\text{Im}(a)} = -i\frac{(a+\bar{a})}{(a-\bar{a})}$.

Explanation

Put $z = x + iy$ in the equation $\bar{a}z + a\bar{z} + b = 0$, where $b \in R$

we get $(\bar{a} + a)x + i(\bar{a} - a)y + b = 0$

whose slope $= \frac{a+\bar{a}}{i(a-\bar{a})}$, which is a real quantity.

(v) **Length of perpendicular :** The length of perpendicular from a point z_1 to the line $\bar{a}z + a\bar{z} + b = 0$ is given by $\frac{|\bar{a}z_1 + a\bar{z}_1 + b|}{|a| + |\bar{a}|}$ or $\frac{|\bar{a}z_1 + a\bar{z}_1 + b|}{2|a|}$.

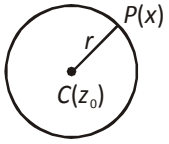
(vi) No heading the pair of lines joining the points z_1, z_2 and z_3, z_4 are perpendicular if

- $\arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = \pm\frac{\pi}{2}$
- $\left(\frac{z_1 - z_2}{z_3 - z_4}\right)$ is purely imaginary

5. **Equation of circle :**

(i) The equation of a circle whose centre is at point having affix z_0 and radius r is $|z - z_0| = r$

(ii) If the centre of the circle is at origin and radius r, then its equations is $|z| = r$.



(iii) General equation of circle

The equation of a circle in the argand plane is of the form

$$z\bar{z} + \bar{a}z + a\bar{z} + b = 0 \qquad \text{.... (1)}$$

where b is purely real

The circle represented by (1) has centre at $-a$ and radius = $\sqrt{|a|^2 - b}$.

(iv) Equation of circle in diameter form

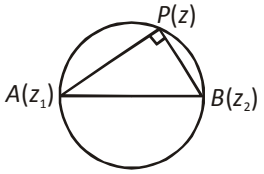
The equation of a circle described on $A(z_1)$ and $B(z_2)$ as its diameter is

$$(z - z_2)(\bar{z} - \bar{z}_1) + (\bar{z} - \bar{z}_1)(z - \bar{z}_1) = 0$$

Explanation

Let a circle be described on $A(z_1)$ and $B(z_2)$ as its diameter, then for any point $P(z)$ on the circle

$AP \perp BP \Rightarrow \arg\left(\frac{z - z_2}{z - z_1}\right) = \pm\frac{\pi}{2}$



$\Rightarrow \quad \frac{z - z_2}{z - z_1} + \frac{\bar{z} - \bar{z}_2}{\bar{z} - \bar{z}_1} = 0$

$\Rightarrow \quad (z - z_2)(\bar{z} - \bar{z}_1) + (\bar{z} - \bar{z}_2)(z - z_1) = 0$

(v) $|z - z_0| < r$ represens interior of the circle $|z - z_0| = r$ and $|z - z_0| > r$ represents exterior for the circle $|z - z_0| = r$.

(vi) If A, B and C are three points in argand plane such that $AC = AB$ and $\angle CAB = \theta$ then use the rotation about A to find $e^{i\theta}$, but if $AC \neq AB$ use coni method.

$\overrightarrow{AC} = \overrightarrow{AB}\, e^{i\theta}$ or $(z_3 - z_1) = (z_2 - z_1)e^{i\theta}$ or $\dfrac{z_3 - z_1}{z_2 - z_1} = e^{i\theta}$.

(vii) If four points z_1, z_2, z_3 and z_4 are con-cyclic then

$\dfrac{(z_4 - z_1)(z_2 - z_3)}{(z_4 - z_2)(z_1 - z_3)}$ is = real or, $\arg\left(\dfrac{(z_2 - z_3)(z_4 - z_1)}{(z_1 - z_3)(z_4 - z_2)}\right) = \pm\pi, 0$

Student Task

Q.1 If $|z - 4 + 3i| \leq 2$, then

(a) $3 \leq |z| \leq 4$ (b) $3 \leq |z| \leq 6$ (c) $3 \leq |z| \leq 7$ (d) $1 \leq |z| \leq 2$

Q.2 If $|z_1 - 1| < 1$, $|z_2 - 2| < 2$, $|z_3 - 3| < 3$, then $|z_1 + z_2 + z_3|$ is less then

(a) 7 (b) 12 (c) 14 (d) 6

Q.3 Let 3 – i and 2 + i be the affixes of two points A and B in the argand plane and P represents the complex number $z = x + iy$, then the locus of P if $|z - 3 + i| = |z - 2 - i|$ is

(a) Circle on AB as diameter (b) The line AB

(c) The perpendicular bisector of AB (d) An ellipse

Q.4 For a complex number z, the minimum value of $|z| + |z - 2|$ is equal to

(a) –2 (b) 0 (c) 4 (d) 2

Q.5 The inequality $|z - 4| < |z - 2|$ represents the region given by

(a) $Re(z) > 0$ (b) $Re(z) < 0$ (c) $Re(z) > 3$ (d) $Re(z) < 3$

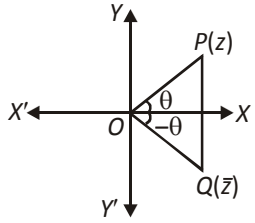
Answers :

1. (d) 2. (b) 3. (c) 4. (a) 5. (d)

11. GEOMETRICAL REPRESENTATION OF ALGEBRAIC OPERATIONS ON COMPLEX NUMBERS

I. Geometrical Representation of Conjugate of

(a) Complex Number : Let $z = x + iy$ be any complex number. It is represented by a point $P(x, y)$ in the argand plane.



Now, $\overline{z} = x - iy$, $\overline{z}$ will be represented by the point $Q(x, -y)$.

Clearly, Q is the image of the point P in the real axis.

Note :- Let $z = x + iy$ and $|z| = r$, $\arg z = \theta$, then

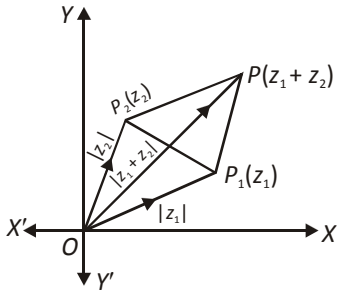
$z = x + iy = r(\cos\theta + i\sin\theta)$,

$\overline{z} = x - iy = r(\cos\theta - i\sin\theta) = r[\cos(-\theta) + i\sin(-\theta)]$

Hence, $\arg(\overline{z}) = -\theta$.

II. Geometrical Representation of $z_1 + z_2$

Let P_1, P_2 represent two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, respectively in the complex plane, then $P_1 \equiv (x_1, y_1)$ and $P_2 \equiv (x_2, y_2)$. Join the origin O with the points P_1 and P_2. Complete the parallelogram OP_1PP_2 with OP_1 and OP_2 as two adjacent sides. Let the coordinates of P be (x, y).



Since the diagonals of a parallelogram bisect each other, OP and P_1P_2 have the same mid-point.

$\Rightarrow \quad \left(\frac{x+0}{2}, \frac{y+0}{2}\right) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$\Rightarrow \quad \frac{x}{2} = \frac{x_1+x_2}{2}$ and $\frac{y}{2} = \frac{y_1+y_2}{2}$

$\Rightarrow \quad x = x_1$ and $y = y_1 + y_2$.

$\therefore$ The coordinates of P are $(x_1 + x_2, y_1 + y_2)$. Hence, the point P represents the complex number $(x_1 + x_2) + i(y_1 + y_2) = (x_1 + iy_1) + (x_2 + iy_2) = z_1 + z_2$.

Cor. We know that sum of two sides of triangle is greater than the third side.

$\therefore$ In ΔOPP_1, $OP \le OP_1 + P_1P$ .... (1)

(The equality holds when the points O, P_1 and P are collinear)

But $OP = |z_1 + z_2|$, $OP_1 = |z_1|$, $P_1P = OP_2 = |z_2|$

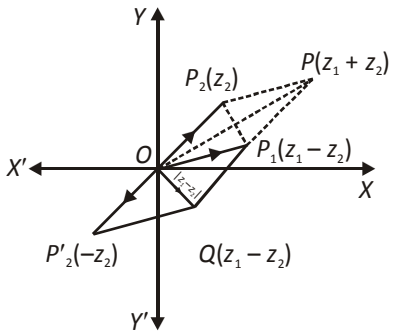
$\therefore$ From (1), we have $|z_1 + z_2| \le |z_1| + |z_2|$, which is called the triangle inequality.

III. Geometrical Representation of $z_1 - z_2$

Let P_1, P_2 represent two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, respectively in the complex plane, then $P_1 \equiv (x_1, y_1)$ and $P_2 \equiv (x_2, y_2)$. Join the origin O with the points P_1 and P_2. Produce P_2O to P'_2 such that $OP'_2 = OP_2$. Since $P_2P'_2$ is bisected at $O(0, 0)$, the coordinates of P'_2 are $(-x_2, -y_2)$.

$\therefore$ P'_2 represents the complex number $-x_2 - iy_2 = -z_2$.

Complete the parallelogram $OP_1QP'_2$ with OP_1 and OP'_2 as two adjacent sides. Then, Q represents the sum of the complex numbers z_1 and $-z_2$, i.e., $z_1 - z_2$ and $OQ = |z_1 - z_2|$.



Cor. 1. Complete the parallelogram OP_1PP_2, then P represents the complex number $z_1 + z_2$.

We know that the sum of the squares of the diagonals of a parallelogram is equal to the sum of the squares of the four sides.

$\therefore$ In parallelogram OP_1PP_2, $(OP)^2 + (P_2P_1)^2 = (OP_1)^2 + (P_1P)^2 + (OP_2)^2 + (P_2P_1)^2$

But $P_1P = OP_2$ and $P_2P = OP_1$

$\therefore$ $(OP)^2 + (P_2P_1)^2 = 2(OP_2)^2 + 2(OP_2)^2$

Also, $OP = |z_1 + z_2|$, $P_2P_1 = OQ = |z_1 - z_2|$,

$OP_1 = |z_1|$, $OP_2 = |z_2|$

$\therefore$ $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$.

Cor. 2. Since the absolute difference of two sides of a triangle is less than the third side, we have in ΔOP_2P_2

$|OP_1 - OP_2| \le P_2P_1$

i.e., $||z_1| - |z_2|| \le |z_1 - z_2|$.

This inequality is also called triangle inequality and it is deducible from the triangle inequality, as givne below :

$|z_1| = |(z_1 - Z_2) + z_2| \le |z_1 - z_2| + z_2|$

$\Rightarrow \quad |z_1| - |z_2| \le |z_1 - z_2|$ .... (1)

Interchanging z_1 and z_2 in (1), we have

$|z_2| - |z_1| \le |z_2 - z_1|$

$\Rightarrow \quad -(|z_1| - |z_1|) \le |z_1 - z_2|$ .... (2) [$\because |z_2 - z_1| = |-(z_1 - z_2)| = |z_1 - z_2|$, since $|-z| = |z|$]

Combining (1) and (2), we have

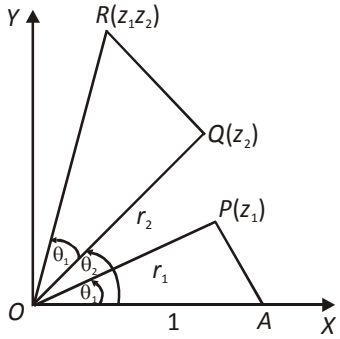
$||z_1| - |z_2|| \le |z_1 - z_2|$.

IV. Geometrical Representation of the Product z_1z_2

Let the complex numbers

$z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

be represented by the points P and Q, respectively, in the Argand plane. Join OP and OQ. Then,



$OP = r_1, \angle XOP = \theta_1$

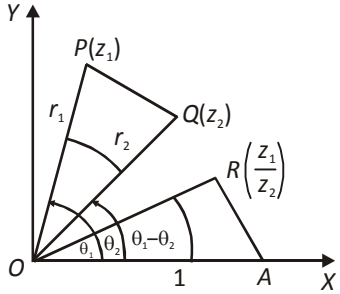
$OQ = r_2, \angle XOQ = \theta_2$.

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \left(\frac{r_1}{r_2}\right) e^{i(\theta_i - \theta_2)}$$

V. Geometrical Representation of the Quotient $\frac{z_1}{z_2}, z_2 \neq 0$

Let the complex numbers

$z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$



be represented by the points P and Q, respectively in the Argand plane. Join OP and OQ. Then,

$OP = r, \angle XOP = \theta_1$

$OQ = r_2, \angle XOQ = \theta_2$.

Let A represents the number 1 on the real axis. On side OA, construct a triangle OAR similar to the triangle OQP

$$\therefore \quad \frac{OR}{OP} = \frac{OA}{OQ} \text{ or } \frac{OR}{r_1} = \frac{1}{r_2} \text{ or } \frac{r_1}{r_2}$$

$\angle XOR = \angle QOP = \angle XOP - \angle XOP - \angle XOQ = \theta_1 - \theta_2$.

Thus, the point R has polar coordinates $\left(\frac{r_1}{r_2}, \theta_1 - \theta_2\right)$.

Hence, R represents the complex number

$$\frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)] = \frac{r_1}{r_2} \cdot \frac{\cos\theta_1 + i\sin\theta_1}{\cos\theta_2 + i\sin\theta_2} = \frac{z_1}{z_2}$$

Note :- If $P(z_1)$ and $Q(z_2)$ are two points in the Argand plane then arg $(z_2 - z_1)$ = the angle which PQ makes with real axis.

TRICK(S) FOR PROBLEM SOLVING

- If z is a variable point and z_1, z_2 are two fixed points in the argand plane, then
  - (i) $|z - z_1| = |z - z_2| \Rightarrow$ Locus of z is the perpendicular bisector of the line segment joining z_1 and z_2.
  - (ii) $|z - z_1| + |z - z_2|$ = constant ($\neq |z_1 - z_2|$) $\Rightarrow$ Locus z is an ellipse.
  - (iii) $|z - z_1| + |z - z_2| = |z_1 - z_2| \Rightarrow$ Locus of z is the line segment joining z_1 and z_2
  - (iv) $|z - z_1| - |z - z_2| = |z_1 - z_2| \Rightarrow$ Locus of z is a straight line joining z_1 and z_2 but z does not lie between z_1 and z_2.
  - (v) $|z - z_1| - |z - z_2|$ = constant ($\neq |z - z_2|$) $\Rightarrow$ Locus of z is a hyperbola.
  - (vi) $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2 \Rightarrow$ Locus of z is a circle with z_1 and z_2 as the extremities of diameter.
  - (vii) $|z - z_1| = k\,|z - z_2|$, $(k \neq 1) \Rightarrow$ Locus of z is a circle.
  - (viii) $\arg\left(\frac{z - z_1}{z - z_2}\right) = a$ (fixed) $\Rightarrow$ Locus of z is a segment of circle.
  - (ix) $\arg\left(\frac{z - z_1}{z - z_2}\right) = \pm\frac{\pi}{2} \Rightarrow$ Locus of z is a circle with z_1 and z_2 as the vertices of diameter.
  - (x) $\arg\left(\frac{z - z_1}{z - z_2}\right) = 0$ or $\pi \Rightarrow$ Locus of z is a straight line passing through z_1 and z_2.
- $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$

  Thus, $|z_1| + |z_2|$ is the greatest possible value of $|z_1 + z_2|$ and $||z_1| - |z_2||$ is the least possible value of $|z_1 + z_2|$.
- If $\left|z + \frac{1}{z}\right| = a$, the greatest and least values of $|z|$ are respectively, $\frac{a + \sqrt{a^2 + 4}}{2}$ and $\frac{-a + \sqrt{a^2 + 4}}{2}$.
- The area of the triangle whose vertices are z, iz and $z + iz$ is $\frac{1}{2}|z|^2$.
- The area of the triangle with vertices z, wz and $z + wz$ is $\frac{\sqrt{3}}{4}|z|^2$.
- If z_1, z_2, z_3 be the vertices of an equilateral triangle and z_0 be the circumcentre, then $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$.
- If z_1, z_2, z_3 be the vertices of a triangl, then the triangle is equilateral iff $(z_1 - z_2)^2 + (z_3 - z_1)^2 = 0$

  or, $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

  or, $\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$.

- The equation $|z-z_1|^2+|z-z_2|^2=k$ (where k is a real number) will represent in circle with centre at $\frac{1}{2}(z_1-z_2)$ and radius $\frac{1}{2}\sqrt{2k-|z_1-z_2|^2}$, provided $k \geq \frac{1}{2}|z_1-z_2|^2$.
- The one and only one case in which $|z_1|+|z_2|+....+|z_n|=|z_1+z_2+....+z_n|$ is that the numbers $z_1, z_2, ... z_n$ have the same amplitude.
- If three points z_1, z_2, z_3 connected by relation $az_1+bz_2+cz_3=0$ where $a+b+c=0$, then the three points are collinear.
- If z is a complex number, then e^z is periodic.
- If three complex numbers are in A.P., then they lie on a straight line in the complex plane.

MISCELLANEOUS EXAMPLES

Example 1 : Find the value of expression $1.(2-\omega)(2-\omega^2)+2.(3-\omega)(3-\omega^2)+....+(n-1)(n-\omega)(n-\omega^2)$, where ω is an imaginary cube root of unity.

Solution : We have, $(z-1)(z-\omega)(z-\omega^2) \equiv z^3-1$

$\therefore \quad 1(2-\omega)(2-\omega^2)+2(3-\omega^2)+....+(n-1)(n-\omega)(n-\omega^2)$

$$=\sum_{r=2}^{n}(r-1)(r-\omega)(r-\omega^2)=\sum_{r=2}^{n}(r^3-1)=\sum_{r=2}^{n}r^3-\sum_{r=2}^{n}1$$

$$=\left(\sum_{r=1}^{n}r^3\right)-1-\left(\sum_{r=2}^{n}1\right)=\left\{\frac{n(n+1)}{2}\right\}^2-1-(n-1)=\left\{\frac{n(n+1)}{2}\right\}^2-n$$

Example 2 : Find the square root of $8-15i$.

Solution : Here $y=-15<0$

$$\Rightarrow \quad \sqrt{8-15i}=\pm\left(\sqrt{\frac{\text{Re}(z)+|z|}{2}}-i\sqrt{\frac{|z|-\text{Re}(z)}{2}}\right)=\pm\frac{1}{\sqrt{2}}(5-3i)$$

Example 3 : The values of x and y satisfying the equation $\frac{(1+i)x-2i}{3+1}+\frac{(2-3i)y+i}{3-i}=i$ are

(a) $x=-1, y=3$ (b) $x=3, y=-1$ (c) $x=0, y=1$ (d) $x=1, y=0$

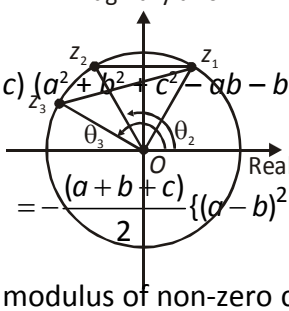
Solution : **(b)** $\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y+i}{3-i}=i \Rightarrow (4+2i)x+(9-7i)y-3i-3=10i$

Equating real and imaginary parts, we get $2x-7y=13$ and $4x+9y=3$.

Example 4 : z_1, z_2 and z_3 are three non-zero numbers such that $z_1 \neq z_2 \neq z_3$, and $a=|z_1|, b=|z_2|, c=|z_3|$.

If $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=0$, then show that $\arg\left(\frac{z_3}{z_2}\right)=\arg\left(\frac{z_3-z_1}{z_2-z_1}\right)^2$.

Solution : We have $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=-(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$



$$=-\frac{(a+b+c)}{2}\{(a-b)^2+(b-c)^2+(c-a)^2\}=0$$

Now a, b and c are positive, as modulus of non-zero complex numbers

$\Rightarrow a=b=c$

So z_1, z_2 and z_3 lie on a circle with centre at origin.

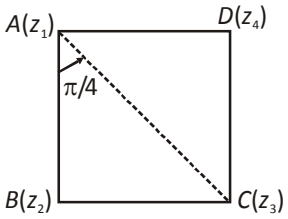
Now $\arg\left(\frac{z_3}{z_2}\right) = \arg(z_3) - \arg(z_2) = \theta_3 - \theta_2$

$$\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)^2 = 2\angle BAC = 2\theta$$

Now $2\theta = (\theta_3 - \theta_2)$ (As we know that angle subtended by a chord on centre of a circle is double the angle subtended by the same chord at any point ont eh circumference of the circle). Hence Proved.

Example 5 : Consider a square *ABCD* such that z_1, z_2, z_3 and z_4 represent its vertices *A*, *B*, *C* and *D* respectively. Express 'z_3' and 'z_4' in terms of z_1 and z_2.

Solution : Consider the rotation of *AB* about *A* through an angle $\pi/4$.



We get $\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} e^{i\pi/4} = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

$\Rightarrow z_3 = z_1 + (z_2 - z_1)(1 + i)$

Similarly, $\frac{z_4 - z_1}{z_2 - z_1} = \frac{|z_4 - z_1|}{|z_2 - z_1|} e^{i\pi/2} = i$

$\Rightarrow z_4 + i(z_2 - z_1)$

Example 6 : In a geometrical progression, first term and common ratio are both $\frac{1}{2}\left(\sqrt{3} + i\right)$. Then find the absolute value of the nth term of the progression.

Solution : Here $a = \frac{1}{2}\left(\sqrt{3} + i\right). r = \frac{1}{2}\left(\sqrt{3} + i\right)$

$\therefore \quad T_n = ar^{n-1} = \left[\frac{1}{2}\left(\sqrt{3} + i\right)\right]^n$

Put $\frac{\sqrt{3}}{2} = r\cos\theta, \frac{1}{2} = r\sin\theta$.

These give $r = 1$ and $\theta = \frac{\pi}{6}$,

$\therefore \quad T_n = [r(\cos\theta + i\sin\theta)]^n = \left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)^n = \cos\frac{n\pi}{6} + i\sin\frac{n\pi}{6}$

$$|T_n| = \sqrt{\cos^2\frac{n\pi}{6} + i\sin^2\frac{n\pi}{6}} = 1$$

Example 7 : Consider a quadratic equation $az^2 + bz + c = 0$ where *a*, *b*, *c* are complex numbers. Find the condition that the equation has

(i) One purely imaginary root
(ii) One purely real root
(iii) Two purely imaginary roots
(iv) Two purely real roots

Solution : (i) Let z_1 (purely imaginary) be a root of the given equation $\Rightarrow z_1 = -\bar{z}_1$

and $az_1^2 + bz_1 + c = 0$ ..... (i)

$\Rightarrow \overline{az_1^2 + bz_1 + c = 0} \Rightarrow \overline{az_1}^2 + \bar{b}\bar{z}_1 + \bar{c} = 0 \Rightarrow \bar{a}z_1^2 - \bar{b}z_1 + \bar{c} = 0$ (as $\bar{z}_1 = -z_1$) ..... (ii)

Now (i) and (ii) must have one common root. Then required condition becomes

$$\Rightarrow \frac{(b\bar{c} + c\bar{b})}{-(a\bar{b} + \bar{a}b)} = \frac{(b\bar{c} + a\bar{c})^2}{(a\bar{b} + \bar{a}b)^2} \Rightarrow (b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b) + (c\bar{a} - a\bar{c})^2 = 0$$

(ii) Let z_1 (purely real) be a root of the given eqution $\Rightarrow z_1 = \bar{z}_1$ and $az_1^2 + bz_1 + c = 0$

.... (1)

$\Rightarrow \overline{az_1^2 + bz_1 + c = 0} \Rightarrow \overline{az_1}^2 + \bar{b}\bar{z}_1 + \bar{c} = 0 \Rightarrow \bar{a}z_1^2 + \bar{b}z_1 + \bar{c} = 0$ ..... (2)

Now (1) and (2) must have one common root.

$$\frac{z_1^2}{(b\bar{c} - \bar{b}c)} = \frac{z_1}{-(a\bar{c} - \bar{a}c)} = -\frac{1}{(a\bar{b} - \bar{a}b)}$$

Required codition is

$$(b\bar{c} - c\bar{b})(a\bar{b} - \bar{a}b) + (c\bar{a} - a\bar{c})^2 = 0$$

(iii) Let z_1 and z_2 be two purely imaginary roots then

$\bar{z}_1 = -z_1, \bar{z}_2 = -z_2$

Now $az^2 + bz + c = 0$ ..... (i)

$\Rightarrow \overline{az^2 + bz + c = 0} \Rightarrow \overline{az}^2 + \bar{b}\bar{z} + \bar{c} = 0 \Rightarrow \bar{a}z^2 - \bar{b}z + \bar{c} = 0$ ..... (ii)

Equation (i) and (ii) must be identical as their root are same

$$\Rightarrow \frac{a}{a} = -\frac{b}{b} = \frac{c}{c}$$

(iv) Let z_1 and z_2 be two purely real roots then

$\bar{z}_1 = z_1, \bar{z}_2 = z_2$

In this case $az^2 + bz + c = 0$

$\Rightarrow \overline{az}^2 + \bar{b}\bar{z} + \bar{c} = 0 \Rightarrow \bar{a}z^2 + \bar{b}z + \bar{c} = 0$

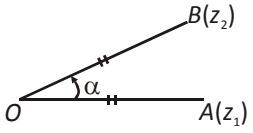
Equation (i) and (ii) must be identical, as their root are same

$$\Rightarrow \frac{a}{a} = -\frac{b}{b} = \frac{c}{c}$$

Example 8 : Let z_1 and z_2 be roots of the equation $z^2 + pz + q = 0$, where the coefficients p and q may be complex numbers. Let A and B represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and $OA = OB$, where O is the origin, then establish a relation between p, q and α.

Solution : From diagram,

$z_2 = z_1 e^{i\alpha}$



z_1 and z_2 are roots of $z^2 + pz + q = 0$

$\Rightarrow z_1 + z_2 = -p$ and $z_1 z_2 = q$

$z_1 + z_1 e^{i\alpha} = -p$

$$z_1 = \frac{-p}{1 + e^{i\alpha}} \qquad \therefore z_2 = -\frac{pe^{i\alpha}}{1 + e^{i\alpha}}$$

$$\Rightarrow z_1 z_2 = \frac{p^2 e^{i\alpha}}{(1 + e^{i\alpha})^2} = q$$

$$p^2 e^{i\alpha} = q[1 + \cos\alpha + i\sin\alpha]^2 = q\left[2\cos^2\frac{\alpha}{2} + i2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}\right]^2$$

$$= q\left[2\cos\frac{\alpha}{2}\left(\cos\frac{\alpha}{2} + i\sin\frac{\alpha}{2}\right)\right]^2$$

$$\Rightarrow p^2 e^{i\alpha} = q.4\cos^2\frac{\alpha}{2}e^{i\alpha} \Rightarrow p^2 = 4q\cos^2\frac{\alpha}{2}$$

Example 9 : Find the minimum value of $|a + b\omega + c\omega^2|$, where a, b, and c are all not equal integers and ($\omega \neq 1$) is a cube root of unity.

Solution : Let x = $|a + b\omega + c\omega^2|$

$\Rightarrow x^2 = |a + b\omega + c\omega^2|^2 = (a + b\omega + c\omega^2)\ (a + b\bar{\omega} + c\bar{\omega}^2)$ [a, b, c are integer , $\bar{a} = a$]

= $(a + b\omega + c\omega^2)\ (a + b\omega^2 + c\omega)$; $\bar{\omega} = \omega^2$ and $\bar{\omega}^2 = \omega$

$$\Rightarrow x^2 = a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$\Rightarrow x^2 = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

Since a, b, c are integers but not all simultaneously equal hence we may assume two of them equal,

say $b = c$ but $a \neq b$ and $a \neq c$.

$\Rightarrow (b-c)^2 = 0$, $(a-b)^2 \geq 1$ and $(a-c)^2 \geq 1$, as the difference between two consecutive integers is ± 1.

$$\Rightarrow x^2 \geq \frac{1}{2}[1 + 0 + 1] = 1$$

$\Rightarrow |x| \geq 1$ or $|x| \leq -1$ But $|x| \neq$ –ve $\Rightarrow$ min $|x| = 1$.

UNSOLVED EXERCISE

SECTION-A
(Single Correct Option)

(I) COMPLEX NUMBER IN IOTA FORM

1. If $\left(\frac{3}{2}+i\frac{\sqrt{3}}{2}\right)^{48}=3^{24}(x+iy)$, then the ordered pair (x, y) is
(a) (0, 2) (b) (0, 1)
(c) (1, 0) (d) (1, 1)

2. $1 + i^2 + i^4 + ...i^{2n}$ is
(a) Positive
(b) Negative
(c) 0
(d) Cannot be determined

3. If $z = (1 + 3i)/(1 – 2i)$, then the value of z^{24} is
(a) $2^{24}(1 + i)$ (b) 2^{24}
(c) $2^{24} i$ (d) None of these

4. One of the values of i^i is
(a) $e^{-\pi/2}$ (b) $e^{\pi/2}$
(c) e^{π} (d) $e^{-\pi}$

5. Convert $(i + 1)/\left(\cos\frac{\pi}{4}-i\sin\frac{\pi}{4}\right)$ in polar form.
(a) $\cos(\pi/4) + i\sin(\pi/4)$
(b) $\cos(\pi/2) – i\sin(\pi/2)$
(c) $\sqrt{2}[\cos(\pi/4)+i\sin(\pi/4)]$
(d) $\sqrt{2}[\cos(\pi/2)+i\sin(\pi/2)]$

6. If $z = e^{2\pi/3}$, then $1 + z + 3z^2 + 2z^3 + 2z^4 + 3z^5$ is equal to
(a) $-3e^{\pi i/3}$ (b) $3e^{\pi i/3}$
(c) $3e^{2\pi i/3}$ (d) $-3e^{2\pi i/3}$

7. If value of $(1 + i)^3 + (1 – i)^3$ is equal to
(a) 1 (b) –2
(c) 0 (d) –4

8. If $(\sqrt{3}i+1)^{100}=2^{99}(a+ib)$, then $a^2 + b^2$ is equal to
(a) 4 (b) 3
(c) 2 (d) 0

9. The value of $(1+\sqrt{3}i)^4+(1-\sqrt{3}i)^4$ is
(a) –16 (b) 16
(c) 14 (d) –14

10. If $2x = 3 + 5i$, then the value of $2x^3 + 2x^2 – 7x + 72$ is
(a) 4 (b) –4
(c) 8 (d) –8

11. The value of $\left|\frac{1+i\sqrt{3}}{\left(1+\frac{1}{i+1}\right)^2}\right|$ is
(a) 20 (b) 9
(c) 5/4 (d) 4/5

12. If $x=\frac{-3+i\sqrt{3}}{2}$ is a complex number, then the value of $(x^2 + 3x)^2 (x^2 + 3x + 1)$ is
(a) –9/8 (b) 6
(c) –18 (d) 36

13. If $(x + iy)^{1/3} = 2 + 3i$, then $3x + 2y$ is equal to
(a) –20 (b) –60
(c) –120 (d) –18

14. The value of $\frac{\cos 30^\circ+i\sin 30^\circ}{\cos 60^\circ-i\sin 60^\circ}$ is equal to
(a) i (b) – i
(c) $\frac{1+\sqrt{3}i}{2}$ (d) $\frac{1-\sqrt{3}i}{2}$

(II) CONJUGATE & MODULUS OF A COMPLEX NO.

15. The number of solutions of the equation $z^2 + |z|^2 = 0$, where $z \in C$ is
(a) One (b) Two
(c) Three (d) Infinitely many

16. If $iz^3 + z^2 – z + i = 0$ then
(a) $|z| = 0$ (b) $|z| = 1$
(c) $|z| = \sqrt{2}$ (d) None of these

17. If α and β are different complex numbers with $|\beta| = 1$ then $\left|\frac{\beta-\alpha}{1-\bar{\alpha}\beta}\right|$
(a) 0 (b) 1/2
(c) 1 (d) 2

18. Among the complex numbers z satisfying the condition $|z + 1 – i| \le 1$, the number having the least positive argument is
(a) $1 – i$
(b) $–1 + i$
(c) $–i$
(d) None of these.

19. If $z = x + iy$ and $w = (1 – iz)/(z – i)$, then $|w| = 1$ implies that, in the complex plane
(a) z lies on the imaginary axis
(b) z lies on the real axis
(c) z lies on the unit circle
(d) None of these

20. The minimum value of $|z| + |z - i|$ is
(a) 0 (b) 1
(c) 2 (d) None of these

21. z_1 and z_2 are two complex numbers with $|z_1| = |z_2|$. If the real part of z_1 is positive and the imaginary part of z_2 is negative, then $\frac{z_1 + z_2}{z_1 - z_2}$ is
(a) 0 (b) Real and positive
(c) Real and negative (d) Purely imaginary

22. If $\frac{5z_2}{7z_1}$ is purely an imaginary number, then $\left|\frac{2z_1 + 3z_2}{2z_1 - 3z_2}\right|$ is equal to
(a) 5/7 (b) 7/9
(c) 25/49 (d) None of these

23. The distances of the roots of the equation $|\text{Cos}\,\theta_1|z^3 + |\text{Cos}\,\theta_2|z^2 + |\text{Cos}\,\theta_3|z + |\text{Cos}\,\theta_4| = 3$, from $z = 0$, are
(a) Greater than 2/3
(b) Less than 2/3
(c) Greater than $|\text{Cos}\,\theta_1| + |\text{Cos}\,\theta_2| + |\text{Cos}\,\theta_3| + |\text{Cos}\,\theta_4|$
(d) Less than $|\text{Cos}\,\theta_1| + |\text{Cos}\,\theta_2| + |\text{Cos}\,\theta_3| + |\text{Cos}\,\theta_4|$

24. For any two complex numbers z_1, z_2; $\left|1 - \bar{z}_1 z_2\right|^2 - \left|z_1 - z_2\right|^2$ is equal to
(a) $\left(1 + |z_1|^2\right)\left(1 + |z_2|^2\right)$
(b) $\left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right)$
(c) $\left(1 + |z_1|^2\right)\left(1 - |z_2|^2\right)$
(d) $\left(1 - |z_1|^2\right)\left(1 + |z_2|^2\right)$

25. Complex numbers whose real and imaginary parts, x and y are inegers and satisfy the equaion $3x^2 - |xy| - 2y^2 + 7 = 0$
(a) Do not exist
(b) Exist and form a conjugate pair
(c) Form two conjugate pairs
(d) Do not form conjugate pairs

26. The greatest and the least absolute value of $z + 1$, where $|z + 4| \leq 3$ are respectively
(a) 6 and 0 (b) 10 and 6
(c) 4 and 3 (d) None of these

27. If $|z| \geq 3$, the least value of $|z + 1/z|$ is
(a) 8/3 (b) 3/8
(c) 10/3 (d) None of these

28. Let $z_1 = 6 + i$ and $z_2 = 4 - 3i$. Let z be a complex number such that $\left(\frac{z - z_1}{z_2 - z}\right) = \frac{\pi}{2}$
(a) $|z - (5 - i)| = 5$ (b) $|z - (5 - i)| = \sqrt{5}$
(c) $|z - (5 + i)| = 5$ (d) $|z - (5 + i)| = \sqrt{5}$

29. If $\left|\log_{(1/\sqrt{3})} \frac{|z|^2 - |z| + 1}{2 + |z|}\right| < 2$, then he possible values of $|z|$ are given by
(a) $|z| < 1/3$ (b) $|z| = 1$
(c) $|z| = 5$ (d) $1 < |z| < 5$

30. If α and β are two different complex numbers with $|\beta| = 1$, then $\left|\frac{\beta - \alpha}{1 - \bar{\alpha}\beta}\right|$ is equal to
(a) 1/2 (b) 0
(c) –1 (d) 1

31. If $z = \frac{(\sqrt{3} + i)^3 (3i + 4)^2}{(8 + 6i)^2}$, then $|z|$ is equal to
(a) 8 (b) 2
(c) 5 (d) 4

32. Let $w \neq 1$ be complex number. If $|w| = 1$ and $z = \frac{w - 1}{w + 1}$, then Re(z) is equal to
(a) 1 (b) $\frac{1}{|w + 1|}$
(c) Re(w) (d) 0

33. The value of $|z|^2 + |z - 3|^2 + |z - i|^2$ is minimum, when z equals
(a) $2 - \frac{2}{3}i$ (b) 45 + 3i
(c) $1 + \frac{i}{3}$ (d) $1 - \frac{i}{3}$

34. Let $z_1 \neq z_2$ and $|z_1| = |z_2|$. If z_1 has positive real part and z_2 has negative imaginary part. Then, $\frac{z_2 + z_1}{z_1 - z_2}$ may be
(a) 0
(b) real and positive
(c) real and negtive
(d) None of these

35. The complex number z = x + iy, which satisfies the equation $\left|\frac{z - 3i}{z + 3i}\right| = 1$, lie on
(a) the X-axis
(b) the straight line y = 3
(c) a circle passing through the origin
(d) None of the above

36. If $z_1 = 2\sqrt{2}(1+i)$ and $z_2 = 1 + i\sqrt{3}$, then $z_1^2 z_2^3$ is equal to

(a) 128 i (b) 64i

(c) –64 i (d) –128 i

37. If z_1, z_2 and z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$, $|z_1 + z_2 + z_3|$ is

(a) 3 (b) 1

(c) greater than 3 (d) less than 1

38. If z_1 and z_2 are two complex numbers such that $|z_1| = |z_2| + |z_1 - z_2|$, then

(a) $\text{Im}\left(\frac{z_1}{z_2}\right) = 0$ (b) $\text{Re}\left(\frac{z_1}{z_2}\right) = 0$

(c) $\text{Re}\left(\frac{z_1}{z_2}\right) = \text{Im}\left(\frac{z_1}{z_2}\right)$ (d) None of these

39. If $\left(\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)^{50} = 3^{25}(x + iy)$, where x and y are real, then the ordered pair (x, y) is

(a) (–3, 0) (b) (0, 3)

(c) (0, –3) (d) $\left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)$

40. If $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$, then $\frac{z_1}{z_2}$ is

(a) purely real

(b) purely imaginary

(c) zero of purely imaginary

(d) neither real nor imaginary

41. If $z_1, z_2, \ldots\ldots, z_n$ are complex numbers such that

$|z_1| = |z_2| = \ldots\ldots\ldots = |z_n| = 1$, then

$|z_2 + z_2 + \ldots\ldots\ldots\ldots + z_n|$

is equal to

(a) $|z_1 z_2 z_{3\ldots\ldots} z_n|$

(b) $|z_1| + |z_2| + \ldots\ldots\ldots + z_n|$

(c) $\left|\frac{1}{z_1} + \frac{1}{z_2} + \ldots\ldots\ldots\ldots + \frac{1}{z_n}\right|$

(d) n (e) $\sqrt{n}$

42. If $z_1 = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ and

$z_2 = \sqrt{3}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, then $|z_1 z_2|$ is

(a) 6 (b) $\sqrt{2}$

(c) $\sqrt{6}$ (d) $\sqrt{3}$

(e) $\sqrt{2} + \sqrt{3}$

43. If $z = r(\cos\theta + i\sin\theta)$, then the value of $\frac{z}{\bar{z}} + \frac{\bar{z}}{z}$ is

(a) cos 2θ (b) 2 cos 2θ

(c) 2 cosθ (d) 2 sinθ

(III) ARGUMENT OF A COMPLEX NUMBER

44. The locus of *z* moving in the Argand plane such that $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{2}$ is

(a) A straight line

(b) A circle, centre origin, radius 2

(c) A circle, centre origin, radius $\sqrt{2}$

(d) None of these

45. In Argand's diagram, the locus given by arg

$$\left[(1+2i)z - 1 + 3i\right] = \frac{\pi}{4}$$

(a) Is a straight line through the origin

(b) Is the straight line passing through the points $2 - 2i$ and $5 - 3i$

(c) Is a circle drawn with the line joining $1 + 2i$ and $1 - 3i$ as a diameter

(d) Is none of the above

46. Given that $|z - a| = a$, where *z* is a point in the Argand's plane, then $\frac{z-2a}{z}$ equals

(a) *i* tan (arg *z*) (b) *i* cot (arg *z*)

(c) tan (arg *z*) (d) None of these

47. The point of intersection of the curves $\arg(z - 3i) = 3\pi/4$ and $\arg(2z + 1 - 2i) = \pi/4$ is

(a) $\frac{3}{4} + i\frac{9}{4}$ (b) $1 + 3i$

(c) $1 + i$ (d) None of these

48. If $z \neq 0$, then $\int_{x=0}^{100} \left[\arg|z|\right] dx$ is (Where $|z|$ denotes modulus and [.] denotes greatest integer function)

(a) 0 (b) Not defined

(c) 100 (d) None of these

49. The value of $\text{Arg}\left[i\,ln\left(\frac{a-ib}{a+ib}\right)\right]$, where *a* and *b* are real numbers, is

(a) 0 or π (b) π/2

(c) Not defined (d) None of these

50. If $\arg\frac{z-i}{z+i} = \alpha$ and $\frac{z-2i}{z+2i} = \beta$, then equations, are;

(a) 0, if $\alpha > 0, \beta < 0$ (b) 0, if $\alpha < 0, \beta > 0$

(c) 0, if $\alpha > 0, \beta > 0$ (d) 0, if $\alpha < 0, \beta < 0$

51. If $z(2-i2\sqrt{3})^2 = i(\sqrt{3}+i)^4$, then amplitude of z is

(a) $\frac{-\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{6}$ (d) None of these

52. Among the complex number z satisfying condition $|z+1-i| \le 1$, the number having least positive argument is

(a) $1-i$ (b) $1+i$

(c) $-i$ (d) None of the above

53. If the conjugate of $(x+iy)(1-2i)$ is $1+i$, then

(a) $x-iy = \frac{1-i}{1-2i}$ (b) $x+iy = \frac{1-i}{1-2i}$

(c) $x = \frac{1}{5}$ (d) $x = -\frac{1}{5}$

54. If z_1 and z_2 are two non-zero complex numbers such that $|z_1+z_2| = |z_1|+|z_2|$, then arg $\left(\frac{z_1}{z_2}\right)$ is

(a) 0 (b) $-\pi$

(c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{2}$

55. If $z = \frac{4}{1-i}$, then $\bar{z}$ is (where, $\bar{z}$ is complex conjugate of z)

(a) $2(1+i)$ (b) $(1+i)$

(c) $\frac{2}{1-i}$ (d) $\frac{4}{1-i}$

56. If $-\pi < \arg(z) < -\frac{\pi}{2}$, then arg $(\bar{z})$ – arg $(-\bar{z})$ is

(a) π (b) $-\pi$

(c) $\pi/2$ (d) $-\pi/2$

(IV) ROTATION OF A COMPLEX NUMBER

57. A triangle with vertics represented by complex numbers z_0, z_1, z_2 has opposite side lengths in ratio $2:\sqrt{6}:\sqrt{3}-1$ respectively. Then

(a) $(z_2-z_0)^4 = -9(7+4\sqrt{3})(z_1-z_0)^4$

(b) $(z_2-z_0)^4 = 9(7+4\sqrt{3})(z_1-z_0)^4$

(c) $(z_2-z_0)^4 = (7+4\sqrt{3})(z_1-z_0)^4$

(d) None of these

58. The area of the triangle on the Argand diagram formed by the complex numbers z, iz and $z+iz$ is

(a) $|z|^2$ (b) $1/2\,|z|^2$

(c) $|z|^2/3$ (d) None of these

59. If the vertices of a square are z_1, z_2, z_3 and z_4 taken in the anticlockwise order, then

(a) $z_3 = -iz_1 + (1+i)z_2$ (b) $z_4 = (1-i)z_1 + iz_2$

(c) $z_3 = iz_1 + (1-i)z_2$ (d) $z_3 = (1+i)z_1 + z_2$

60. Let z_1 be a fixed point on the circle of radius 1 centred at the origin in the argand plane and $z_1 \neq 1$. Consider an equilateral triangle inscribed in the circle with z_1, z_2 and z_3 as the vertices taken in the counter clockwise direction. Then, $z_1 z_2 z_3$ is equal to

(a) z_1^2 (b) z_1^3

(c) z_1^4 (d) z_1

61. Suppose that z_1, z_2 and z_3 are three vertices of an equilatral triangle in the argand plane. Let $\alpha = \frac{1}{2}(\sqrt{3}-i)$ and β be a non-zero complex numbers. The points $\alpha z_1+\beta$, $\alpha z_2+\beta$, $\alpha z_3+\beta$ will be

(a) the vertices of an equilateral triangle

(b) the vertices of an isosceles triangle

(c) collinear

(d) the vertices of a scalene triangle

(V) Demoiver's Theorem, Cube roots & n^{th} roots of unity

62. If ω ($\neq 1$) is a cube root of unity, then

$$\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -i+\omega-1 & -1 \end{vmatrix}$$ equals

(a) 0 (b) 1

(c) i (d) ω

63. The value of the expression

$$\left(\frac{i+\sqrt{3}}{2}\right)^8 + \left(\frac{i-\sqrt{3}}{2}\right)^8$$ is

(a) –1 (b) 0

(c) 1 (d) None of these

64. The value of $\sum_{k=1}^{10}\left(-\cos\frac{2k\pi}{11} + i^3 \sin\frac{2k\pi}{11}\right)$ is

(a) 1 (b) –1

(c) $-i$ (d) i

65. If α and β are the roots of $x^2 - x + 1 = 0$, then the equation whose roots are α^{100} and β^{100} are

(a) $x^2 - x + 1 = 0$ (b) $x^2 + x - 1 = 0$

(c) $x^2 - x - 1 = 0$ (d) $x^2 + x + 1 = 0$

66. The value of $\frac{\left(-1+i\sqrt{3}\right)^{15}}{\left(1-i\right)^{20}} + \frac{\left(-1-i\sqrt{3}\right)^{15}}{\left(1+i\right)^{20}}$ is

(a) -2^6 (b) 2^6

(c) 1 (d) $1-\sqrt{3}$

67. If $i = \sqrt{-1}$, then $4+5\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{365}$ is equal to

(a) $1 - i\sqrt{3}$ (b) $-1 + i\sqrt{3}$
(c) $i\sqrt{3}$ (d) $-i\sqrt{3}$

68. If ω, ω^2 are the cube roots of unity, then roots of equation $(x-1)^3 + 5 = 0$ are
(a) $-5, -5\omega, 5\omega^2$ (b) $-4, 1-5\omega, 1-5\omega^2$
(c) $6, 1-5\omega, 1+5\omega^2$ (d) None of the above

69. If the fourth roots of unity are z_1, z_2, z_3 and z_4, then $z_1^2 + z_2^2 + z_3^3 + z_4^2$ is equal to
(a) 0 (b) 2
(c) 3 (d) –14

70. $\left[\dfrac{1+\cos\left(\frac{\theta}{2}\right) - i\sin\left(\frac{\theta}{2}\right)}{1+\cos\left(\frac{\theta}{2}\right) + i\sin\left(\frac{\theta}{2}\right)}\right]^{4n}$ is equal to
(a) $\cos n\theta - i \sin\theta$ (b) $\cos n\theta + i \sin\theta$
(c) $\cos 2n\theta - i \sin 2n\theta$ (d) $\cos 2n\theta + i \sin\theta\, 2n\theta$

(VI) GEOMETRY IN COMPLEX NUMBER'S

71. The centre of a regular hezagon is *i*. One vertex is $(2 + i)$. *z* is an adjacent vertex. Then *z* is equal to
(a) $1 + i(1 \pm \sqrt{3})$ (b) $i + 1 \pm \sqrt{3}$
(c) $2 + i(1 \pm \sqrt{3})$ (d) None of these

72. The equation $\bar{b}z + b\bar{z} = c$, where *b* is a non-zero complex constant and *c* real, represents
(a) A circle (b) A straight line
(c) Hyperbola (d) None of these

73. Let $a, b \in R$ such that $0 < a < 1$, $0 < b < 1$. The values of *a* and *b*, such that the complex numbers $z_1 = a + i$, $z_2 = -1 + bi$ and $z_3 = 0$ form an equilateral triangle; are
(a) $a = b = 2 - \sqrt{3}$
(b) $a = 2 - \sqrt{3}$, $b = 2 + \sqrt{3}$
(c) $a = \sqrt{3}$, $b = -\sqrt{3}$
(d) None of these

74. The locus of the centre of the circle which touches the circles $|z - z_1| = a$ and $|z - z_2| = b$ externally (z, z_1 & z_2 are complex numbers) will be
(a) An ellipse (b) A circle
(c) A hyperbola (d) None of these

75. If Arg $(z_2 - z_1)$ + Arg $(z_3 - z_1)$ = 2 Arg $(z - z_1)$, then locus of *z* is a
(a) Straight line (b) Circle
(c) Parabola (d) Hyperbola

76. Let z_1 and z_2 be non-zero complex numbers satisfying the equation $z_1^2 - 2z_1z_2 + 2z_2^2 = 0$. The geometrical nature of the triangle whose vertices are the origin and the points representing z_1 and z_2 is
(a) An isosceles right angled triangle
(b) A right angled riangle
(c) An equilateral triangle
(d) None of these

77. Points z_1 and z_2 are adjacent vertices of a regular octagon. The vertex z_3 adjacent to z_2 ($z_3 \neq z_1$) is represented by
(a) $z_2 + \dfrac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$
(b) $z_1 + \dfrac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$
(c) $z_2 + \dfrac{1}{\sqrt{2}}(1 \pm i)(z_2 - z_1)$
(d) None of these

78. The point *z* in Argand's plane moves such that $\text{Re}\dfrac{iz+1}{iz-1} = 2$. The locus of *z* is
(a) A straight line (b) A circle
(c) An ellipse (d) None of these

79. If the complex numbers z_1, z_2 and z_3 denote the vertices of an isosceles triangle, right angled at z_1, then $(z_1 - z_2)^2 + (z_1 - z_3)^2$ is equal to
(a) 0 (b) $(z_2 + z_3)^2$
(c) 2 (d) 3

80. Let z_1 and z_2 be two fixed complex numbers in the argand plane and z be an arbitrary point satisfying $|z - z_1| + |z - z_2| = 2|z_1 - z_2|$. Then , the locus of z will be
(a) an ellipse
(b) a straight line joining z_1 and z_2
(c) a parabola
(d) a bisector of the line segment joining z_1 and z_2

81. In the argand plane, the distinct roots of $1 + z + z^3 + z^4 = 0$ (z is a complex number) represent vertices of
(a) a square
(b) an equilateral triangle
(c) a rhombus
(d) a rectangle

82. If $|z - \bar{z}| + |z + \bar{z}| = 2$, then z lies on
(a) circle (b) a square
(c) an ellipse (d) a line

83. If $\omega \neq 1$ is a cube root of unity, then the sum of the series $S = 1 + 2\omega + 3\omega^2 + ...+3n\omega^{3n-1}$ is
(a) $\dfrac{3n}{\omega - 1}$ (b) $3n(\omega - 1)$
(c) $\dfrac{\omega - 1}{3n}$ (d) 0

84. The modulus of the complex number z such that $|z + 3 - i| = 1$ and arg $(z) = \pi$, is equal to

(a) 1 (b) 2
(c) 9 (d) 3

SECTION-B
(Assertion-Reason)

1. **Statement – 1 :** If $\alpha = \cos\left(\frac{2\pi}{7}\right) + i \sin\left(\frac{2\pi}{7}\right)$, $p = \alpha + \alpha^2 + \alpha^4$, $q = \alpha^3 + \alpha^5 + \alpha^6$, then the equation whose roots are p and q is $x^2 + x + 2 = 0$

Statement – 2 : If α is root of $Z^7 = 1$, then $1 + \alpha + \alpha^2 + ... + \alpha^6 = 0$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

2. **Statement-1:** $3 + ix^2y$ and $x^2 + y + 4i$ are conjugate numbers, then $x^2 + y^2 = 3$.

Statement-2: If sum and product of wo complex numbers is real then they are conjugate complex numbers.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

3. **Statements-1:** If $x^2 + x + 1 = 0$ then the value of $\left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)^2 + ... + \left(x^{27} + \frac{1}{x^{27}}\right)^2$ is 54.

Statements-2: ω, ω^2 are the roots of given equation and

$$x + \frac{1}{x} = -1, x^2 + \frac{1}{x^2} = -1, \; x^3 + \frac{1}{x^3} = 2 .$$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

4. **Statement–1 :** If $|z| < \sqrt{2} - 1$, then $|z^2 + 2z\cos\alpha|$ is less than 1.

Statement–2 : $|z_1 + z_2| < |z_1| + |z_2|$ also $|\cos\alpha| \le 1$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

5. **Statement–1 :** The number of complex numbers z satisfying $|z|^2 + a|z| + b = 0$ ($a, b \in R$) is at the most 2.

Statement–2 : A quadratic equation in which all the coefficients are non-zero have at the most two roots.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Years Questions)

Q.1 A complex number z is said to be unimodular, if $|z| \neq 1$. If z_1 and z_2 are complex numbers such that $\frac{z_1 - 2z_2}{2 - z_1 z_2}$ is unimodular and z_2 is not unimodular. Then, the point z_1 lies on a

(a) straight line parallelel to X-axis
(b) straight line parallel to Y-axis
(c) circle of radius 2
(d) circle of radius $\sqrt{2}$

Q.2 If z is a complex number such that $|z| \ge 2$, then the minimum value of $\left|z + \frac{1}{2}\right|$

(a) is equal to $\frac{5}{2}$
(b) lies in the interval (1, 2)
(c) is strictly greater than $\frac{5}{2}$
(d) is strictly greater than $\frac{3}{2}$ but less than $\frac{5}{2}$

Q.3 Let z be a omplex number such that the

imaginary part of z is non-zero and $a = z^2 + z + 1$ is real. Then, *a* cannot take the value

(a) -1 (b) $\frac{1}{3}$

(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

Q.4 Let $z = x + iy$ be a complex number where, x and y are integers. Then, the area of the rectangle whose vertices are the root of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$, is

(a) 48 (b) 32

(c) 40 (d) 80

Q.5 If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on

(a) a line not passing through the origin

(b) $|z| = \sqrt{2}$

(c) the X-axis

(d) the Y-axis

Q.6 If $w = \alpha + i\beta$, where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\left(\frac{w - \bar{w}z}{1-z}\right)$ is purely real, then the set of values of z is

(a) $|z| = 1, z \neq 2$

(b) $|z| = 1$ and $z \neq 1$

(c) $z = \bar{z}$

(d) None of the above

Q.7 If $|z| = 1$ and $w = \frac{z-1}{z+1}$ (where, $z \neq -1$), then Re (w) is

(a) 0 (b) $\frac{1}{|z-1|^2}$

(c) $\left|\frac{1}{z+1}\right| \cdot \frac{1}{|z+1|^2}$ (d) $\frac{\sqrt{2}}{|z+1|^2}$

Q.8 For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is

(a) 0 (b) 2

(c) 7 (d) 17

Q.9 If $z_1 z_2$ and z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$, then $|z_1 + z_2 + z_3|$ is

(a) equal to 1 (b) less than 1

(c) greater than 3 (d) equal to 3

Q.10 For positive integers n_1, n_2 the value of expression $(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2}$, here $i = \sqrt{-1}$ is a real number, if and only if

(a) $n_1 = n_2 + 1$ (b) $n_1 = x_2 - 1$

(c) $n_1 = n_2$ (d) $n_1 > 0, n_2 > 0$

Q.11 The complex numbers sin x +i cos2x and cosx -i sin2x are conjugate to each other, for

(a) $x = n\pi$ (b) x = 0

(c) $x = (n + 1/2)\pi$ (d) no value of x

Q.12 The points z_1, z_2, z_3 and z_4 in the complex plane are the vertices of a parallelogram taken in order, if and only if

(a) $z_1 + z_4 = z_2 + z_3$ (b) $z_1 + z_3 = z_2 + z_4$

(c) $z_1 + z_2 = z_3 + z_4$ (d) None of the above

Q.13 If $z = x + iy$ and $w = (1-iz)/(z-i)$, then $|w| = 1$ implies that, in the complex plane

(a) z lies on the imaginary axis

(b) z lies on the real axis

(c) z lies on the unit circle

(d) None of the above

Q.14 The inequality $|z-4| < |z-2|$ repreasents the region given by

(a) Re (z) ≥ 0 (b) Re (z) < 0

(c) Re (z) > 0 (d) None of these

Q.15 If $z = \left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^5 + \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^2$, then

(a) Re (z) =0 (b) In (z) = 0

(c) Re (z) > 0, Im (z) > 0 (d) Re(z)>0, Im (z) < 0

Q.16 The complex numbers $z = x + iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$, lie on

(a) the X -axis

(b) the straight line y = 5

(c) a circle passing through the origin

(d) None of these

Q.17 Let z_1 and z_2 be complex numbers such that $z_1 \neq z_2$ and $|z_1| = |z_2|$. If z_1 has postiove real part and $|z_1| = |z_2|$. and z_2 has negative imaginary

part, then $\frac{z_1+z_2}{z_1-z_2}$ may be

(a) zero
(b) real and positive
(c) real and negative
(d) purely imaginary

Q.18 If $z_1 = a + ib$ and $z_2 = c + id$ are complex numbers such that $|z_1| = |z_2| = 1$ and $Re(z_1\bar{z}_2) = 0$, then the pair of complex numbers $w_1 = a + ic$ and $w_2 = b + id$ satisfies

(a) $|w_1| = 1$ (b) $|w_2| = 1$
(c) $Re(w_1\bar{w}_2) = 0$ (d) None of these

Q.19 If z is a complex number of unit modulus and argument θ, then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ is equal to

(a) $-\theta$ (b) $\frac{\pi}{2}-\theta$
(c) θ (d) $\pi-\theta$

Q.20 If arg (z) < 0, then arg (– z) – arg (z) equals

(a) π (b) $-\pi$
(c) $-\pi/2$ (d) $\pi/2$

Q.21 Let z and w be two non-zero complex numbers such that |z| = |w| and arg (z) + arg (w) = π, then z equals

(a) w (b) – w
(c) $\bar{w}$ (d) $-\bar{w}$

Q.22 If z_1 and z_2 are two non-zero complex numbers suh that $|z_1 + z_2| = |z_1| + |z_2|$, then arg (z_1) – arg (z_2) is equal to:

(a) $-\pi$ (b) $-\frac{\pi}{2}$
(c) 0 (d) $\frac{\pi}{2}$

Q.23 Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t)z_1 + iz_2$ for some real number *t* with 0 < t < 1. If arg (w) denotes the principle argument of a non-zero complex number w, then

(a) $|z - z_1| + |z - z_2| = |z_1 - z_2|$
(b) $\arg(z - z_1) = \arg(z - z_2)$
(c) $\begin{vmatrix} z-z_1 & \bar{z}-\bar{z}_1 \\ z_2-z_1 & \bar{z}_2-\bar{z}_1 \end{vmatrix} = 0$
(d) $\arg(z - z_1) = \arg(z_2 - z_1)$

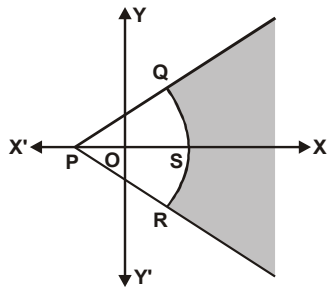
Q.24 A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1. From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i}+\hat{j}$ and then it moves through an angle $\pi/2$ in anti-clockwise direction on a circle with centre at origin, to reach a point z_2. The point z_2 is given by

(a) 6 + 7i (b) – 7 + 6i
(c) 7 + 6i (d) – 6 + 7i

Q.25 A man walks a distance of 3 units from the origin towards the North-East (N 45° E) direction. From there, he walks a distance of 4 units towards the North-West (N 45° W) direction to reach a point P. Then, the position of P in the Argand plane is

(a) $3e^{i\pi/4} + 4i$ (b) $(3 - 4i)e^{i\pi/4}$
(c) $(4 + 3i)e^{i\pi/4}$ (d) $(3 + 4i)e^{i\pi/4}$

Q.26 The shaded region, where P = (– 1, 0), Q = $(-1+\sqrt{2}, \sqrt{2})$, R = $(-1+\sqrt{2}, -\sqrt{2})$, S = (1, 0) is represent by:



(a) $|z + 1| > 2$, $|\arg(z + 1)| < \frac{\pi}{4}$

(b) $|z + 1| < 2$, $|\arg(z + 1)| < \frac{\pi}{2}$

(c) $|z + 1| > 2$, $|\arg(z + 1)| > \frac{\pi}{4}$

(d) $|z + 1| < 2$, $|\arg(z + 1)| > \frac{\pi}{2}$

Q.27 The complex numbrs z_1, z_2 and z_3 satisfying $\frac{z_1-z_3}{z_2-z_3} = \frac{1-i\sqrt{3}}{2}$ are the vertices of a triangle which is

(a) of area zero
(b) right angled isosceles
(c) equilateral

(d) obtuse angled isosceles

Q.28 Let $z = \cos\theta + i\sin\theta$. Then, the value of $\sum_{m=1}^{15} \text{Im}(z^{2m-1})$ at $\theta = 2°$ is

(a) $\frac{1}{\sin 2°}$ (b) $\frac{1}{3\sin 2°}$

(c) $\frac{1}{2\sin 2°}$ (d) $\frac{1}{4\sin 2°}$

Q.29 The minimum value of $|ia + b\omega + c\omega^2|$, where a, b and c are all not equal integers and $\omega(\neq 1)$ is a cube root of unity, is

(a) $\sqrt{3}$ (b) $\frac{1}{2}$

(c) 1 (d) 0

Q.30 If $\omega(\neq 1)$ be a cube root of unity and $(1 + \omega^2)^n = (1 + \omega^4)^n$, then the least positive value of *n* is

(a) 2 (b) 3

(c) 5 (d) 5

Q.31 If $i = \sqrt{-1}$, then $4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}$ is equal to:

(a) $1 - i\sqrt{3}$ (b) $-1 + i\sqrt{3}$

(c) $i\sqrt{3}$ (d) $-i\sqrt{3}$

Q.32 If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^7$ is equal to

(a) 128ω (b) -128ω

(c) $128\omega^2$ (d) $-1288\omega^2$

Q.33 The value of $\sum_{k=1}^{n}\left(\sin\frac{2\pi k}{7} - i\cos\frac{2\pi k}{7}\right)$ is

(a) – 1 (b) 0

(c) – i (d) i

Section-D
(School /Board Pattern)

Q.1 A Δ with vertices represented by complex numbers z_1, z_2, z_3 has angles 90°, 30°, 60° respectively. Express

(i) z_1 in terms of z_2 & z_3; (ii) z_2 in terms of z_3 & z_1; (iii) z_3 in terms of z_1 & z_2.

Q.2 Let z_1 and z_2 be roots of the equation $z^2 + pz + q = 0$, where the coefficients *p* and *q* may be complex numbers. Let *A* and *B* represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and $OA = OB$, where *O* is the origin, prove that $p^2 = 4q\cos^2\alpha/2$.

Q.3 If *A*, *B*, *C* represent the complex numbers z_1, z_2, z_3 respectively on the complex plane and the angles at *B* and *C* are each equal to $1/2(\pi - \alpha)$, then prove that $(z_2 - z_3)^2 = 4(z_3 - z_1)(z_1 - z_2)\sin^2\alpha/2$.

Q.4 Let $\alpha = \text{cis}2\pi/11$, $\beta = \alpha^2$, $\gamma = \alpha^3$, $\delta = \alpha^4$, $\varepsilon = \alpha^5$, $\gamma = \alpha^6$, $\mu = \alpha^7$, $v = \alpha^8$. Calculate the values of

(i) $(1 - v)(1 - v^2)(1 - v^3)(1 - v^4)(1 - v^5)(1 - v^6)(1 - v^7)(1 - v^8)(1 - v^9)(1 - v^{10})$

(ii) $(i - \beta)(i - \beta^2).....(i - \beta^9)(i - \beta^{10})$, where $i = \sqrt{-1}$

(iii) $(\omega - \varepsilon)(\omega - \varepsilon^2).....(\omega - \varepsilon^9)(\omega - \varepsilon^{10})$, where $\omega = \frac{-1 + \sqrt{-3}}{2}$

(iv) $1 + \gamma + \gamma^2 + \gamma^3 + ... + \gamma^{10}$

(v) Re $(\delta + \delta^2 + \delta^3 + \delta^4 + \delta^5)$

(vi) Re $(1 + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5)$

(vii) $(\mu - \beta)(\mu - \beta^2)(\mu - \beta^3).....(\mu - \beta^9)(\mu - \beta^{10})$

Q.5 If 1, α_1, α_2, α_3, α_4 be the roots of $x^5 - 1 = 0$ then prove that

$$\frac{\omega - \alpha_1}{\omega^2 - \alpha_1}\cdot\frac{\omega - \alpha_2}{\omega^2 - \alpha_2}\cdot\frac{\omega - \alpha_3}{\omega^2 - \alpha_3}\cdot\frac{\omega - \alpha_4}{\omega^2 - \alpha_4} = \omega$$

where ω is a rotational non real complex root of unity.

Q.6 If z_1 and z_2 are two complex numbers such that $z_2 \neq 0$ and $\left|\frac{z_1 - z_2}{z_1 + z_2}\right| = 1$, prove that $\frac{iz_1}{z_2} = k$, where k is a real number. Find the angle between the lines from the origin to the points $z_1 + z_2$ and $z_1 - z_2$ in erms of *k*.

Q.7 Prove that $|z_1| + |z_2| = |1/2(z_1 + z_2) + \sqrt{(z_1 z_2)}| + |1/2(z_1 + z_2) - \sqrt{(z_1 z_2)}|$.

Q.8 If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then the maximum value of $|2z - 6 + 5i|$ is.....

Q.9 Suppose z_1, z_2, z_3 are the vertices of an equilateral triangle inscribed in the circle $|z| = 2$. If $z_1 = 1 + i\sqrt{3}$, then z_2 = ..., z_3.....

Q.10 If a and b are real numbers between 0 and 1 such that the points $z_1 = a + i$, $z_2 = 1 + bi$ and $z_3 = 0$ form an equilateral triangle, then a = ... and b = ...

CHAPTER 3

PERMUTATION AND COMBINATIONS

1. THE FACTORIAL

Factorial notation : Let n be a +ve integer, then the continued product of first n natural numbers is called factorial n, to be denoted by $n!$ or $\lfloor n$. Also, we define 0! = 1. when n is negative or fractional then $n!$ is meaningless.

i.e., $n!$ is meaningful only fo $n \in W$.

Thus,

$n! = n(n-1)(n-2)(n-3)$ 4.3.2.1

Illustration 1: Find n, if $(n+1)! = 12 \times (n-1)!$

Solution : $(n+1)! = 12 \times (n-1)!$

$\Rightarrow (n+1) \times n \times (n-1)! = 12 \times (n-1)!$

$\Rightarrow n(n+1) = 12 \Rightarrow n = -4, 3$

but $(n+1)!$ is meaningful only for $n = 3$

Illustration 2: Prove that $(n!)^2 < n^n \cdot n! < (2n)!$ for all +ve integers n.

Solution : We have,

$(n!)^2 = n!\, n! = (1 \times 2 \times 3 \times 4 \times \ldots\ldots \times n)\, n!$

Now, $1 \le n$

$2 \le n$

$3 \le n$

.....

.....

$n \le n$

$\Rightarrow 1 \times 2 \times 3 \times \ldots\ldots \times n \le n \times n \times n \times \ldots\ldots \times n = n^n$

$\Rightarrow n! \le n^n$

$\Rightarrow (n!)(n!) \le (n!)n^n \Rightarrow (n!)^2 \le n^n (n!)$ .... (i)

Also,

$(2n)! = 1 \times 2 \times 3 \times \ldots\ldots \times n \times (n+1) \times \ldots\ldots \times (2n-1) \times (2n)$

Now, $n + 1 > n$

$n + 2 > n$

$n + 3 > n$

.....

.....

$n + n > n$

$\Rightarrow (n+1)(n+2) \ldots\ldots (n+n) > n^n$

$\Rightarrow n!\,(n+1)(n+2) \ldots\ldots (2n) > n!\, n^n$

$\Rightarrow (2n)! > n! \cdot n^n \Rightarrow n!\, n^n < (2n)!$ .... (ii)

From (i) and (ii), we get $(n!)^2 \le n^n \cdot n! < (2n)!$

Student Task

Q.1 Find the sum of the series $\sum_{r=1}^{n} r \times r!$

Q.2 Prove that $\frac{n!}{r!} = n(n-1)(n-2) \ldots\ldots (r+1)$; $r \le n$

Q.3 Find the last digit of $(1! + 2! + 3! + ... + 2010!)^{500}$

Q.4 Prove that : $P = 21(21^2 - 1^2)(21^2 - 2^2)(21^2 - 3^2) ..... (21^2 - 10^2)$ is divisible by 21!

Q.5 Prove that : $\frac{(2n)!}{n!} = \{1 . 3 . 5 . 7 . ..... (2n - 1)\} . 2^n$

Q.6 Prove that $(n! + 1)$ is not divisible by any natural number between 2 and n.

Q.7 Prove that the product of r consecutive positive integers is divisible by $r!$

Q.8 Find the number of integral values of x such that the factorial $(3 - x^2)!$ is meanigul

Answers

1. $(n + 1)! - 1$ **3.** 1 **8.** 3

2. EXPONENT OF A PRIME *p IN n!*

Let p be a prime number and n be a +ve integer. Then the last integer amongst 1, 2, 3, ... , $(n - 1)$, n which is divisible by p is $\left[\frac{n}{p}\right]p$, where $\left[\frac{n}{p}\right]$ denotes the greatest integer less than or equal to $\frac{n}{p}$.

Let $E_p(n)$ denote the exponent of prime p in the +ve integer n, then

$$E_p(n!) = E_p(1 . 2 . 3 . ..... (n - 1) . n)$$

$$= E_p\left(p . 2p . 3p . .....\left[\frac{n}{p}\right]p\right) = \left[\frac{n}{p}\right] + E_p\left(1 . 2 . 3 .....\left[\frac{n}{p}\right]\right)$$

Now, the last integer amongst 1, 2, 3, ... $\left[\frac{n}{p}\right]$ which is divisible by p is $\left[\frac{\frac{n}{p}}{p}\right]p = \left[\frac{n}{p^2}\right]p$

$$\therefore \quad E_p(n!) = \left[\frac{n}{p}\right] + E_p\left(p . 2p . 3p . .....\left[\frac{n}{p^2}\right]p\right)$$

$$= \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + E_p\left(1 . 2 . 3 .....\left[\frac{n}{p^2}\right]\right)$$

Continuing in this manner we get

$$E_p(n!) = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + ... + \left[\frac{n}{p^s}\right]$$ where s is the largest +ve integer such that $p^s \le n < p^{s+1}$

Note :- *(a)* $E_p\left(\frac{n!}{r!}\right) = E_p(n!) - E_p(r!)$

(b) $E_p({}^nP_r) = E_p(n!) - E_p((n - r)!)$

(c) $E_p({}^nC_r) = E_p(n!) - E_p(r!) - E_p((n - r)!)$

Illustration 3: Find the largest integer n such that 33! is divisible by 2^n

Solution : $E_2(33!) = \left[\frac{33}{2}\right] + \left[\frac{33}{2^2}\right] + \left[\frac{33}{2^3}\right] + \left[\frac{33}{2^4}\right] + \left[\frac{33}{2^5}\right]$

$= 16 + 8 + 4 + 2 + 1 = 31$

Illustration 4: Find the exponent of 30 in 100!

Solution : We have $30 = 2 \times 3 \times 5$

Now, $E_2(100!) = \left[\frac{100}{2}\right] + \left[\frac{100}{2^2}\right] + \left[\frac{100}{2^3}\right] + \left[\frac{100}{2^4}\right] + \left[\frac{100}{2^5}\right] + \left[\frac{100}{2^6}\right]$

$= 50 + 25 + 12 + 6 + 3 + 1 = 97$

Similarly,

$E_3(100!) = 48$ and $E_5(100!) = 24$

$\therefore$ Exponent of 30 in 100! = min (97, 48, 24) = 24

Illustration 5: Find the number of zeroes at the end of 100!

Solution : $E_2(100!) = 97$

$E_5(100!) = 24$

Therefore,

$100! = 2^{97} \times 3^b \times 5^{24} \times 7^d \times .....$

$= 2^{73} \times (2 \times 5)^{24} \times 7^d \times .....$

$= 10^{24} \times 2^{73} \times 3^b \times 7^d \times .....$

Thus, the number of zeroes at the end of 100! is 24

Student Task

Q.1 Find :

(a) $E_2\,(^{100}C_{40})$ (b) $E_2\,(^{100}P_{50})$

Q.2 Find :

(a) $E_4\,(100!)$ (b) $E_8\,(100!)$

Q.3 Find :

(a) $E_{20}\,(100!)$ (b) $E_{100}(100!)$

Answers :

1. (a) 3, (b) 50 **2.** (a) 48, (b) 32 **3.** (a) 24, (b) 12

3. NUMBER OF DIVISORS AND SUM OF DIVISORS

Let $N = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot p_3^{\alpha_3} \cdot .....\cdot p_k^{\alpha_k}$, where $p_1, p_2, p_3, ..., p_k$ are natural numbers then

(a) The total number of divisors of N including 1 and N is = $(\alpha_1 + 1)(\alpha_2 + 1)(\alpha_3 + 1) ..... (\alpha_k + 1)$

(b) The total number of divisors of N excluding 1 and N (proper divisors) is

$= (\alpha_1 + 1)(\alpha_2 + 1) ..... (\alpha_k + 1) - 2$

(c) The total number of divisors of N excluding 1 or N is = $(\alpha_1 + 1)(\alpha_2 + 1) ..... (\alpha_k + 1) - 1$

(d) The sum of all divisors of N is

$$= (p_1^0 + p_1^1 + p_1^2 + ... + p_1^{\alpha_1})(p_2^0 + p_2^1 + p_2^2 + ... + p_2^{\alpha_2}).....(p_k^0 + p_k^1 + p_k^2 + ... + p_k^{\alpha_k})$$

(e) The number of ways in which N can be resolved as a product of two factors is

$$\begin{cases} \frac{1}{2}(\alpha_1 + 1)(\alpha_2 + 1)....(\alpha_k + 1), \text{If } N \text{ is not a perfect square} \\ \frac{1}{2}\left[(\alpha_1 + 1)(\alpha_2 + 1)....(\alpha_k + 1) + 1\right], \text{If } N \text{ is a perfect square} \end{cases}$$

(f) The number of ways in which a composite number N can be resolved into two factors which are relatively prime (or co-prime) to each other is equal to 2^{n-1} where n is the number of different factors in N.

Illustration 6: Find the number of divisors of 9600 including 1 and 9600.

Solution : Since $9600 = 2^7 \times 3^1 \times 5^2$

Hence number of divisors = (7 + 1) (1 + 1) (2 + 1) = 48

Illustration 7: Find the number of proper divisors of 38808.

Solution : Since $38808 = 2^3 \times 3^2 \times 7^2 \times 11$

So, the number of proper divisors = (3 + 1) (2 + 1) (2 + 1) (2 + 1) – 2 = 70

Student Task

Q.1 If a, b, c, d, e are prime numbers, then find the number of divisors of ab^2c^2de excluding 1 as a factor.

Q.2 Find the number of proper divisors of 1800 which are also divisible by 10.

Q.3 Find the number of odd proper divisors of $3^p \cdot 6^m \cdot 21^n$.

Q.4 Find the number of even proper divisors of 1008.

Q.5 Find the number of divisors of the form $4n + 2$ ($n \geq 0$) of the number 240.

Answers :

1. 71 **2.** 18 **3.** $(p + q + 1)(q + r + 1)(r + 1) - 2$ **4.** 23

5. 4

4. FUNDAMENTAL PRINCIPLES OF COUNTING

(a) Multiplication Principle

If an operation can be performed in 'm' different ways, following which a second operation can be performed in 'n' different ways, then the two operations in succession can be performed in $m \times n$ ways. This can be extended to any finite number of operations.

Illustration 8: Anish wants to go from Dehradun to Delhi via Roorkee. There are 3 routes from Dehradun to Roorkee and 2 routes from Roorkee to Delhi. In how many ways can he travel from Dehradun to Delhi ?

Solution : He can go from Dehradun to Roorkee in 3 ways and Roorkee to Delhi in 2 ways so number of ways of travel from Dehradun to Delhi via Roorkee is $3 \times 2 = 6$ ways.

Illustration 9: A mint prepares metallic callendars specifying months, dates and days in the form of monthly sheets (one-plate for each month). How many types February callendars should it prepare to serve for all the possibilities in the future years ?

Solution : The mint has to perform three operations

(i) Selecting the number of days in the February months (there can be 28 days or 29 days or 30 days), and

(ii) Selecting the first day of the February months.

The first operation can be completed in 3 ways while the second can be performed in 7 ways.

Thus, the required number of plates = $3 \times 7 = 21$

Illustration 10: Find the total number of ways of answering 25 straight objective type questions, each question having 4 choices.

Solution : Since each questions can be answered in 4 ways. So, the total number of ways of answering 25 questions = $4 \times 4 \times 4 \times ..... \times 4 = 4^{25}$

Student Task

Q.1 (i) Find the number of four letter word that can be formed from the letters of the word HISTORY. (each letter to be used at most once)

(ii) How many of them contain only consonants ?

(iii) How many of them begin and end in a consonants ?

(iv) How many of them begin with a vowel ?

(v) How many contains the letter *Y* ?

(vi) How many begin with *T* and end in a vowel ?

(vii) How many begin with *T* and also contain S ?

(viii) How many contain both vowels ?

Q.2 If repetitions are not allowed, then
(i) How many three digit numbers can be formed from the six digits 2, 3, 5, 6, 7 and 9 ?
(ii) How many of these are less than 400 ?
(iii) How many are even ?
(iv) How many are odd ?
(v) How many are multiples of 5 ?

Q.3 How many car number plates can be made if each plate contains 2 different letters of English alphabet, followed by 3 different digits ?

Q.4 Find the number of natural numbers between 100 and 1000 such that at least one of their digits is 9 ?

ANSWER KEY :

1.	(i)	840	(ii)	120	(iii)	400	(iv)	240	(v)	480	(vi)	40
	(vii)	60	(viii)	240								
2.	(i)	120	(ii)	40	(iii)	40	(iv)	80	(v)	20		
3.	468000			**4.**	252							

(b) Addition Principle

If there are two operations such that they can be performed in dependently in m and n ways respectively, then either of the two operations can be performed in $(m + n)$ ways.

Illustration 11: In a class there are 60 boys and 40 girls. The teacher wants to select either a boy or a girl to represent the class in a function. In how many ways the teacher can make this selection ?

Solution : Here the teacher is to perform either of the following two operations :
(i) Selecting a boy among 60 boys or
(ii) Selecting a girl among 40 girls
So the teacher can select either a boy or a girls in 60 + 40 = 100 ways.

Illustration 12: A college offers 10 courses in the morning and 6 in the evening. Find the number of ways a student can select exactly one course, either in the morning or in the evening.

Solution : The student has ten choices from the morning courses out of which he can select one course in 10 ways. For the evening course, he has 6 choices out of which he can select one course in 6 ways. Hence he has total number of 10 + 6 = 16 choices.

Student Task

Q.1 There are 10 candidates for an Engineering, 8 for a Medical and 4 for a MBA scholarship. In how many ways one of these scholarships be awarded ?

Q.2 There are 11 doors in a hall; 6 on one side and 5 on the other. In how many ways a man can go out of the room.

Q.3 There are 2012 locks and 2012 matching keys. If all the locks and keys are to be perfectly matched, find the maximum number of trials required to open a lock.

ANSWER KEY :

1. 22 **2.** 11 **3.** 2025078

5. DEFINITION OF PERMUTATION

Each of the different arrangement that can be made by taking some or all of a number of given things or objects at a time is called a permutation. In permutation, order of appearance of things is taken into account.

For example : 4 different things *a*, *b*, *c* and *d* are given, then different arrangements which can be made by taking 2 things from 4 given things are *ab*, *ac*, *ad*, *bc*, *bd*, *cd*, *ba*, *ca*, *da*, *cb*, *db*, *dc*. Therefore the number of permutations will be 12.

6. NUMBER OF PERMUTATIONS WITHOUT REPETITION

Arranging *n* objects, taken *r* at a time equivalent to filling *r* places from *n* things

r-places :

1	2	3	4	.....	*r*

Number of choices : $n(n-1)(n-2)(n-3)\ldots\ldots\ldots(n-(r-1))$

The number of ways of arranging is = The number of ways of filling *r* places.

$= n(n-1)(n-2)\ldots\ldots\ldots(n-r+1)$

$$= \frac{n(n-1)(n-2)\ldots\ldots\ldots(n-r+1)\cdot(n-r)!}{(n-r)!} = \frac{n!}{(n-r)!} = {}^{n}p_{r}$$

The number of arrangements of *n* different objects taken all at a time = ${}^{n}p_{n} = n!$

Note :- *(i)* ${}^{n}p_{0} = \frac{n!}{n!} = 1$ *(ii)* ${}^{n}p_{r} = n \cdot {}^{n-1}p_{r-1}$

Illustration 13: How many numbers of 6 digits formed by using the digits 0, 1, 2, 3, 4, 5 when repetition of digit is not allowed.

Solution : Numbers can be formed = Total – Those beginning with zero = 6! – 5! = 600

Illustration 14: If ${}^{10}p_{r} = 5040$, find the value of *r*.

Solution : ${}^{10}p_{r} = 5040 = 10 \times 9 \times 8 \times 7 = {}^{10}p_{4} \Rightarrow r = 4$

Illustration 15: How many different signals can be given using any number of flags from 4 flags of different colours ?

Solution : The total number of signals when *r* flags are used at a time from 4 flags is equal to the number of arrangements of 4, taking *r* at a time, i.e., ${}^{4}p_{r}$. Here *r* may be 1, 2, 3 or 4. The total number of signals

$= {}^{4}p_{1} + {}^{4}p_{2} + {}^{4}p_{3} + {}^{4}p_{4}$ {by the fundamental principle of addition} = 64

Student Task

Q.1 If ${}^{9}p_{5} + 5.\ {}^{9}p_{4} = {}^{10}p_{r}$ then find the value of *r*.

Q.2 Prove that ${}^{n}p_{s}$ is divisible by ${}^{n}p_{r}$ where $r \le s \le n$.

Q.3 How many 4-letter words, with or without meaning, can be formed out of the letters in the word 'LOGARITHMS', if repetition of letters is not allowed.

Q.4 How many six-digit odd numbers, greater than 6,00,000 can be formed by using the digits 5, 6, 7, 8, 9 and 0 if repetition of digits is not allowed ?

Q.5 Prove that ${}^{n}p_{r} = {}^{n-1}p_{r} + r \cdot {}^{n-1}p_{r-1}$

Q.6 Find the total number of 5-digit numbers of different digits in which the digit in the middle is the largest.

Q.7 Find the number of ways in which 6 boys and 6 girls can be seated in a row so that

(i) All the girls sit together and all the boys sit together,

(ii) All the girls are never sitted together.

Q.8 If the best and the worst papers never appear together, find in how many ways 8 examination papers can be arranged.

ANSWER KEY :

1. 5 **3.** 5040 **4.** 240 **6.** $3 \times {}^{3}P_{3} + 4 \times {}^{4}P_{3} + 5 \times {}^{5}P_{3} + 6 \times {}^{6}P_{3} + 7 \times {}^{7}P_{3} + 8 \times {}^{8}P_{3}$

7. (i) $2 \times (6!)^2$ (ii) 12! – 7! 6! **8.** 8! – 7! ×2

Note :- *Divisibility of Numbers*

Divisible by	Condition
2	Whose last digit is even
3	Sum of whose digits is divisible by 3
4	Whose last two digits number is divisible by 4
5	Whose last digit is either 0 or 5
6	Which is divisible by both 2 and 3
8	Whose last three digits number is divisible by 8
9	Sum of whose digits is divisible by 9
11	The difference between the sum of digits at even places and the sum of digits at the odd places is divisible by 11.
25	Whose last two digits are divisible by 25

7. NUMBER OF PERMUTATIONS WITH REPETITION

(a) The number of permutations of n different objects, taken r at a time, when each object may occur once, twice thrice, ... upto r times in any arrangement = The number of ways of filling r places where each place can be filled by any one of n objects

r-places	1	2	3	4	.....	r

Number of choices $n\ n\ n\ n$ .....n

The number of permutations = The number of ways of filling r places = n^r

(b) The number of arrangements that can be formed by using n objects out of which p are identical (and of one kind) q are identical (and of another kind), r are identical (and of another kind) and the rest are distinct is $\frac{n!}{p!\,q!\,r!}$

Illustration 16: How many words can be formed (with or without meaning) from the letters of the word 'COMMITTEE'.

Solution : Number of words = $\frac{9!}{2!\,2!\,2!} = \frac{9!}{(2!)^2}$

{Since the total number of letters is 9 and 2M's, 2T's and 2E's}

Illustration 17: Find the total number of permutations of n different things taken not more than r at any number of times.

Solution : Here, we have to arrange p things out of n, $1 \le p \le r$, and repetition is allowed when $p = 1$, the number of permutations is n.

when $p = 2$, the number of permutations is $n \times n = n^2$

when $p = r$, the number of permutations = n^r

Hence, total number of permutations is

$= n + n^2 + n^3 + ... + n^r$

$= \frac{n(n^r - 1)}{n - 1}$

Student Task

Q.1 In how many ways the letters of the word "COMBINATORICS" can be arranged if
(a) All the vowels are always grouped together to form a contiguous block.
(b) All vowels and all consonants are alphabetically ordered.

Q.2 How many different arrangements are possible with the factor of the form $a^2b^4c^5$ written in full length ?

Q.3 Find the number of four digit numbers starting with 1 and having exactly two identical digits.

Q.4 Find the number of different words that can be formed using all the letters of the word "DEEPMALA" if two vowels are together and the other two are also together but separated from the first two.

Q.5 10 IIT and 2 PET students sit in a row. Find the number of ways in which exactly 3 IIT students sit between 2 PET students.

Q.6 There are six periods in each working day of a school. Find the number of ways in which 5 subjects can be arranged if each subject is allotted at least one period and no period remains vacant.

Q.7 Find the number of all possible selections of one or more questions from 10 given questions, each question having an alternative.

ANSWER KEY

1. (a) $\frac{(9!)(5!)}{(2!)^3}$ (b) $\frac{(13!)}{(8!)(5!)}$ **2.** 6930 **3.** 432 **4.** 1440 **5.** 16 (10!)

6. 1800 **7.** $3^{10} - 1$

8. CONDITIONAL PERMUTATION

(a) Number of permutations of n different objects, taken r at a time, when a particular object is to be always included in each arrangement, is $r \cdot {}^{n-1}p_{r-1}$

(b) Number of permutations of n different things, taken r at a time, when a particular thing is never taken in each arrangement is ${}^{n-1}P_r$.

(c) Number of permutations of n dissimilar things taken r at a time when P particular things always occur = ${}^{n-P}C_{r-P} \cdot r!$.

(d) Number of permutations of n dissimilar things taken r at time when p particular things never occur = ${}^{n-P}C_r \cdot r!$.

(e) Number of permutations of n different things, taken all at a time, when m specified things always come together is $m! \times (n - m + 1)!$.

(f) Number of permutations of n different things, taken all at a time, when m specified things never come together is $n! - m! \times (n - m + 1)!$.

(g) The total number of permutations of n different things taken not more than r at a time, when each thing may be repeated any number of times, is $\frac{n(n^r - 1)}{n-1}$.

Illustration 18: All the letters of the word 'EAMCET' are arranged in all possible ways. Find the number of such arrangements in which two vowels are not adjacent to each other.

Solution : First we arrange 3 consonants in 3! ways and then at four places 3 vowels can be placed in ${}^4P_3 \times \frac{1}{2!}$ ways. Hence the required ways = $3! \times {}^4P_3 \times \frac{1}{2!} = 72$

Illustration 19: p men and q women are to be seated in a row, so that no two women sit together. If $p > q$, then find the number of ways in which they can be seated.

Solution : First arrange p men, in a row in $p!$ ways. Since $q < p$ and no two women can sit together, in any one of the $p!$ arrangement, there are $(p + 1)$ places in which q women can be arranged in ${}^{p+1}P_q$ ways.

$\therefore$ By the fundamental theorem, the required number of arrangement = $p! \, {}^{p+1}P_q = \frac{p!(p+1)!}{(p-q+1)!}$

Illustration 20: We are to form different words with the letters of the word 'INTEGER'. Let n_1 be the number of words in which I and N are never together, and n_2 be the number of words which begin with I and end with R. Then find $\frac{n_1}{n_2}$.

Solution : We have 5 letters other than 'I' and 'N' of which two are identical (E's). We can arrange these letters in a line in $\frac{5!}{2!}$ ways. In any such arrangement 'I' and 'N' can be placed in 6 available gaps in ${}^6p_2$ ways, so required number $= \frac{5!}{2!} \, {}^6p_2 = n_1$. Now if word start with I and end with R then the remaining letters are 5. So, the total number of ways $= \frac{5!}{2!} = n_2$

$\therefore \quad \frac{n_1}{n_2} = 30$

Student Task

Q.1 Find the number of 4-digit numbers that can be formed from the digits, 0, 1, 2, 3, 4, 5, 6, 7 so that each number contain digit 1.

Q.2 How many numbers greater than 40000 can be formed from the digit 2, 4, 5, 5, 7 ?

Q.3 Find the total number of 9 digit numbers which have all the digit different.

Q.4 Let a denotes the number of permutations of $x + 2$ things taken all at a time, b the number of permutations of x things taken 11 at a time and c the number of permutations of x–11 things taken all at a time such that $a = 182$ bc, then find the value of x.

Q.5 Let A {x : x is a prime number and $x < 30$}. Find the number of different rational numbers whose numerator and denominator belong to A.

Q.6 How many ways are there to arrange the letters of the word 'GARDEN' with the vowels in alphabetical order ?

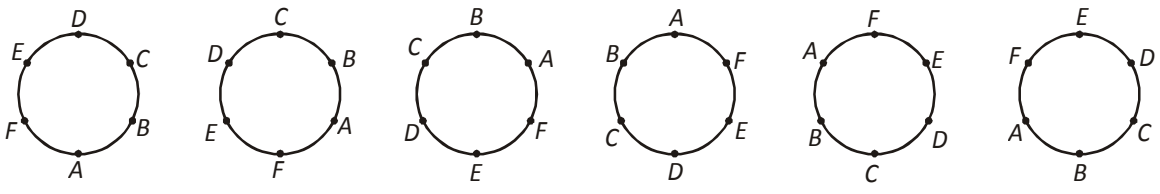
Q.7 Find the number of ways in which ten candidates $A_1, A_2, \ldots\ldots, A_{10}$ can be ranked such that A_1 is always above A_{10}.

ANSWER KEY

1. 480 **2.** 48 **3.** 9 × 9! **4.** 12 **5.** 91 **6.** 360 **7.** $\frac{1}{2}(10!)$

9. CIRCULAR PERMUTATIONS

Consider six persons A, B, C, D, E, F to be seated on the circumference of a circular table in order (which has no head). Now, shifting A, B, C, D, E and F one position in anticlockwise direction we will get arrangement as follows :



(i) (ii) (iii) (iv) (v) (vi)

We see that the arrangements in all figures are same.

∴ The number of circular permutations of n things taken all at a time is $\frac{{}^nP_n}{n} = (n-1)!$, if clockwise and anticlockwise orders are taken as different.

Note :-

(a) The number of circular permutations of n different object is $(n-1)!$

(b) The number of ways in which n persons can be seated to form a necklace, is $\frac{1}{2}(n-1)!$

(c) The number of ways in which n different beads can be arranged to form a necklace, is $\frac{1}{2}(n-1)!$

(d) When the positions are numbered, circular arrangement is treated as a linear arrangement.

(e) In a linear arrangements, it does not make difference whether the positions are numbered or not.

Illustration 21: In how many ways can 8 boys and 8 girls sit in a circle so that no boys sit together ?

Solution : Since total number of ways in which boys can occupy any place is $(8-1)! = 7!$ and the 8 girls can be sit accordingly in 8! ways. Hence the required number of ways are $7! \times 8!$

Illustration 22: Find the number of ways in which 7 beads of different colours form a necklace.

Solution : The number of ways in which 7 beads of different colours can be arranged in a circle to form a necklace are $(7-1) = 6!$. But the clockwise and anticlockwise arrangement are not different. Hence the total number of ways of arranging the beads = $\frac{1}{2}(6!) = 360$

Note :-

(i) Number of circular permutations of n different things taken r at a time is $\frac{{}^nP_r}{r}$, if clockwise and anticlockwise orders are taken as different.

(ii) Number of circular permutations of n different things taken r at a time is $\frac{{}^nP_r}{2r}$ if clockwise and anticlockwise orders are taken as same.

Student Task

Q.1 If eleven members of a committee sit at a round table so that the president and secretary always sit together, then find the number of such arrangement.

Q.2 2012 persons were invited to a party. In how many ways can they and the host be seated at a circular table ? In how many of these ways will two particular persons be seated on either side of the host ?

Q.3 Consider 2013 different pearls on a necklace. How many ways can the pearls be placed in on this necklace such that 222 specific pearls always remain together ?

Q.4 How many necklace of 7 beads each can be made from 11 beads of various colours ?

Q.5 In how many ways can 10 persons sit, when 6 persons sit on round table and 4 sit on the other round table ?

Q.6 In how many ways can 15 members of a council sit along a circular table, when the secretary is to sit on one side of the chairman and the Deputy Secretary on the other side ?

Q.7 There are 20 persons among whom two are brothers. Find the number of ways in which we can arrange them round a circle so that there is exactly one person between the two brothers.

Answer Key :

1.	$9! \times 2$	**2.**	2012!, $2 \times 2011!$	**3.**	1/2 (1791!) (222!)	**4.**	${}^{11}p_7/14$
5.	${}^{10}C_6 \times 5! \times 3!$			**6.**	$2 \times 12!$	**7.**	2 (18!)

EXERCISE - 3.1

Q.1 Find the sum of the digits in the unit place of all numbers formed with the help of 3, 4, 5, 6 taken all at a time.

Q.2 All the letters of the word AGAIN are permuted in all possible ways and the word so formed (with or without meaning) are written as in dictionary then find the 50th word.

Q.3 A dictionary is printed consisting of 7 lettered words only that can be made with the letter of the word 'CRICKET'. If the words are printed at the alphabetical order, as in an dictionary. Find the number of word before the word 'CRICKET'.

Q.4 Find the number of ways of arranging the letter AAAAABBBCCCDEEF in a row when no two $C's$ are together.

Q.5 Find the number of 4 digit numbers that can be made with the digit 1, 2, 3, 4 and 5 in which at least two digit are identical.

Q.6 Find the number of ways in which 6 rings can be worn on the four fingers of one hand.

Q.7 In how many ways can 4 prizes be distributed among 3 students, if each student can get all the 4 prizes ?

Q.8 In how many ways 3 letters can be posted in 4 letter-boxes, if all the letters are not posted in the same letter-box.

Q.9 For some natural N, find the number of +ve integral 'x' satisfying the equation, $1! + 2! + 3! + ... + x! = N^2$.

Q.10 Find the number of six digit numbers that can be formed from the digit 1, 2, 3, 4, 5, 6 and 7 so that digits do not repeat and the terminal digits are even.

Q.11 Find the number of ways in which letters of the word 'VALEDICTORY' be arranged so that the vowels may never be separated.

Q.12 Find the product of all odd +ve integers less than 10000.

Q.13 Find the number of three digit numbers having only two consecutive digits identical.

Q.14 Find the number of 3-digit numbers in which the digit at hundreath's place is greater than the other two digit.

Q.15 Find the number of permutations of 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time are such that the digit 1 appearing somewhere to the left of 2, 3 appearing to the left of 4 and 5 somewhere to the left of 6. e.g. (815723946 would be one such permutation).

ANSWER KEY

1. 108 **2.** NAAIG **3.** 530 4. $\frac{12!}{5!\,3!\,2!}\times\frac{^{13}P_3}{3!}$ **5.** 505 **6.** 4^6

7. 3^4 **8.** 60 **9.** 2 **10.** 720 **11.** 967680 **12.** $\frac{(10000)!}{2^{5000}.(5000)!}$

13. 162 **14.** 285 **15.** 9.7!

10. DEFINITION OF COMBINATION

Each of the different groups or selections which can be formed by taking some or all of a number of objects, irrespective of their arrangements, is called a combination. Suppose we want to select three out of four persons A, B, C, D we may choose ABC or ABD or ACD or BCD. Clearly ABC, ACB, BCA, BAC, CAB, CBA represents the same selection or group but they give rise to different arrangements. Clearly, in a group or selection, the order in which the objects are arranged is immaterial.

Notation : The number of all combinations of n things, taken r at a time is denoted by

$$C(n, r) \text{ or } {}^nC_r \text{ or } \left(\frac{n}{r}\right)$$

Difference between permutations and combination

(a) In a combination only selection is made whereas in a permutation not only a selection is made but also an arrangement in a definite order is considered.

(b) In a combination, the ordering of the selected objects is immaterial where as in a permutation, the ordering is essential.

For example AB and *BA* are same as combination but different in permutations.

(c) Practically to find the permuation of *n* different items, taken *r* at a time, we first select *r* items from *n* items and then arrange them. So usually the number of permutations exceeds the number of combinations.

(d) Each combination corresponds to many permutations.

For example the permutations *ABC* , *BCB*, *BCA*, *BAC*, *CAB*, *CBA* corresponds to the same combination *ABC*.

11. NUMBER OF COMBINATIONS WITHOUT REPETITION

The number of combinations (selections or groups) that can be formed from *n* different objects taken *r* ($0 \le r \le n$) at a time is ${}^nC_r = \frac{n!}{r!(n-r)!}$.

Explanation : Let the total number of selections (or groups) = *x*. Each group contains *r* objects, which can be arranged in *r*! ways. Hence the number of arrangements of *r* objects = $x \times (r!)$. But the number of arrangements = ${}^nP_r \Rightarrow x \times r! = {}^nP_r \Rightarrow x = \frac{{}^nP_r}{r!} = {}^nC_r$

Note :-

(a) ${}^nC_r$ is a natural number.

(b) ${}^nC_r = {}^nC_{n-r}$

(c) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(d) ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$

(e) $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$

(f) ${}^nC_r = \frac{n}{r}.{}^{n-1}C_{r-1}$

(g) If *n* is even then the greatest value of ${}^nC_r$ is ${}^nC_{n/2}$ and ${}^nC_{\frac{n+1}{2}}$ or ${}^nC_{\frac{n-1}{2}}$ when *n* is odd.

(h) ${}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_h = 2^n$

(i) ${}^nC_0 + {}^nC_2 + {}^nC_4 + ... = {}^nC_1 + {}^nC_3 + {}^nC_5 + ... = 2^{n-1}$

(j) ${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + ... + {}^{2n+1}C_n = 2^{2n}$

(k) ${}^nC_n + {}^{n+1}C_n + {}^{n+2}C_n + ... + {}^{2n-1}C_n = {}^{2n}C_{n+1}$

(l) ${}^nC_r = \frac{{}^nP_r}{r!} = \frac{n(n-1)(n-2).....(n-r+1)}{r(r-1)(r-2).....2.1}$; $n \in N, r \in W$

Illustration 23: In a conference of 2012 persons, if each person shake hand with the other one only, then find the total number of shake hands.

Solution : Total number of shake hands when each person shake hands with the other one only

$= {}^{2012}C_2 = \frac{2012 \times 2011}{2} = 1006 \times 2011 = 2023066$

Illustration 24: If ${}^nC_{r-1} = 36$, ${}^nC_r = 84$ and ${}^nC_{r+1} = 126$ then find the value of *r*.

Solution : Here $\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{36}{84}$ and $\frac{{}^nC_r}{{}^nC_{r+1}} = \frac{84}{126}$

$\Rightarrow$ $3n - 10r = -3$ and $4n - 10r = 6$

On solving we get $n = 9$ and $r = 3$

Illustration 25: From 4 gentleman and 6 ladies, a committee of five is to be selected. Find the number of ways in which the committee can be formed so that gentlemen are in majority.

Solution : The committee will consist of 4 gentleman and 1 lady or 3 gentlemen and 2 ladies

$\therefore$ the number of committees = ${}^4C_4 \times {}^6C_1 + {}^4C_3 \times {}^6C_2 = 66$.

Student Task

Q.1 There are 3 books of mathematics, 4 of science, and 5 of literature. How many different collections can be made such that each collection consists of
(i) One of each subject, (ii) At least one book of each subject, (iii) At least one book of literature

Q.2 In how many ways we can select 4 letters from the letters of the word MISSISSIPPI.

Q.3 If ${}^{15}C_{3r} = {}^{15}C_{r+3}$, then find the value of r.

Q.4 If ${}^{n+1}C_3 = 2.\ {}^nC_2$, then find the value of n.

Q.5 Evaluate : $${}^{47}C_4 + \sum_{j=0}^{3} {}^{50-j}C_3 + \sum_{k=0}^{5} {}^{56-k}C_{53-k}$$

Q.6 If ${}^{n+1}C_{r+1} : {}^nC_r : {}^{n-1}C_{r-1} = 11 : 6 : 3$, then find the value of n and r.

Q.7 Prove that ${}^nC_r + 4\ .\ {}^nC_{r-1} + 6\ .\ {}^nC_{r-2} + 4.\ {}^nC_{r-3} + {}^nC_{r-4} = {}^{n+4}C_r$

Q.8 Prove that $({}^7C_0 + {}^7C_1) + ({}^7C_1 + {}^7C_2) + ... + ({}^7C_6 + {}^7C_7) = 2^8 - 2$.

Answer Key

1. (i) 60, (ii) 3255, (iii) 3968 **2.** 21 **3.** 3
4. 5 **5.** ${}^{57}C_4$ **6.** $n = 10, r = 5$

12. NO. OF COMBINATIONS WITH REPETITION AND ALL POSSIBLE SELECTION

(a) The number or combinations of n different things taken r at a time, when k particular objects occur is ${}^{n-k}C_{r-k}$. If k particular objects never occur is ${}^{n-k}C_r$.

(b) The number of ways of n different things selecting at least one of them is ${}^nC_1 + {}^nC_2 + {}^nC_3 + ... + {}^nC_n = 2^n - 1$.

(c) The number of combinations of n identical things taking r ($r \le n$) at a time is 1.

(d) The number of ways of selecting r things out of n alike things is $n + 1$ (where $r = 0, 1, 2, ..., n$).

(e) If out of $(p + q + r + s + t)$ things, p are alike of one kind, q are alike of second kind, r are alike of third kind, s are alike of fourth kind and t are different, then the total number of combinations is $(p + 1)(q + 1)(s + 1)\ 2^t - 1$.

(f) Number of ways in which at least one object may be selected from 'n' objects where 'p' are alike of one type 'q' are alike of second type and 'r' are alike of third type and rest $n-(p + q + r)$ are different is $(p + 1)(q + 1)(r + 1)\ 2^{n-(p+q+r)} - 1$.

(g) The number of combinations of n objects, of which p are identical, taken r at a time is
$= {}^{n-p}C_r + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + ... + {}^{n-p}C_0$ if $r \le p$ and
$= {}^{n-p}C_r + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + ... + {}^{n-p}C_{r-p}$ is $r \ge p$.

Illustration 26: Shivam has 11 friends. In how many ways he can invite one or more of them to a party?

Solution : Required number of ways = $2^{11} - 1$ (since the case that no friend is invited i.e., ${}^{10}C_0$ is excluded).

Illustration 27: In how many ways a team of 10 players out of 22 players can be made if 6 particular players are always excluded.

Solution : 6 particular players are always to be included and 4 are always excluded. Therefore, the total number of selection, now 4 players out of 12.
Hence the number of ways = ${}^{12}C_4$.

Student Task

Q.1 In the 13 cricket players 4 are bowlers, then how many ways can form a cricket team of 11 players in which at least 2 bowlers included ?

Q.2 There are 16 lamps in a hall. Each one of them can be switched on independently. Find the number of ways in which the hall can be illuminated.

Q.3 There are *n* different books and *p* copies of each in a library. Find the number of ways in which one or more than one book can be selected.

Q.4 How many different selections of 6 books can be made from 11 different books, if

(a) Two particular books are always selected;

(b) Two particular books are never selected ?

Q.5 A student is allowed to select at most *n* books from a collection of (2*n* + 1) books. If the total number of ways in which he can select a book is 63, find the value of *n*.

Q.6 Find the number of ways in which one or more letters can be selected from the letters.
A B C DDDDD EEEE FFF GG.

Q.7 Find the number of combinations that can be formed with 3 orranges, 4 mangoes and 5 bananas when it is essential to take

(a) at least one fruit (b) one fruit of each kind.

Q.8 Prove ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + ... + {}^nC_2) = \frac{n(n+1)(2n+1)}{6}$.

Q.9 Ten persons, amongst whom are *A*, *B* and *C* to speak at a function. Find the number of ways in which it can be done if *A* wants to speak before *B* and *B* wants to speak before *C*.

Q.10 Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{r}$ be a variable vector such that $\vec{r}.\hat{i}, \vec{r}.\hat{j}$ and $\vec{r}.\hat{k}$ are positive integers. If $\vec{r}.\vec{a} \le 12$ then find the number of values of $\vec{r}$.

Answer Key :

1. 78 **2.** $2^{16} - 1$ **3.** $(1 + p)^n - 1$ **4.** (a) 126 (b) 84 **5.** 3
6. 2879 **7.** (a) 119 (b) 60 **9.** 10!/6 **10.** ${}^{12}C_3$

13. DIVISION INTO GROUPS

(a) The number of ways in which (*m* + *n*) different things can be divided into two groups which contain *m* and *n* things respectively is ${}^{m+n}C_m \cdot {}^nC_n = \frac{(m+n)!}{m!\,n!}$, $m \neq n$. If $m = n$, then the groups are of equal size. Division of these groups can be given by two types

(i) If order of group is important : The number of ways in which 2*n* different things can be divided equally into two distinct groups is $\frac{(2n)!}{2!(n!)^2}$

(ii) If order of group is not important : The number of ways into which 2*n* different things can be divided equally into two distinct groups is $\frac{(2n)!}{2!(n!)^2} \times 2! = \frac{(2n)!}{(n!)^2}$

(b) The number of ways in which (*m* + *n* + *p* + *q*) different things can be divided into four groups which contain *m*, *n*, *p* and *q* things respectively is

$${}^{m+n+p+q}C_m \cdot {}^{n+p+q}C_n \cdot {}^{p+q}C_p \cdot {}^qC_q = \frac{(m+n+p+q)}{m!\,n!\,p!\,q!},\ m \neq n \neq p \neq q$$

If $m = n = p = q$, then the groups are of equal size. Division of these groups can be given by two types

(i) **If order of group is not important :** The number of ways in which $4p$ different things can be divided equally into 4 groups is $\frac{(4p)!}{4!(p!)^4}$

(ii) **If order of group is important :** The number of ways in which $4p$ different things can be divided equally into 4 groups is $\frac{(4p)!}{4!(p!)^4} \times 4! = \frac{(4p)!}{(p!)^4}$

Note :-

(i) If order of group is not important, then the number of ways in which mn different things can be divided equally into m groups is $\frac{(mn)!}{(n!)^m m!}$

(ii) If order of group is important, then the number of ways in which mn different things can be divided equally into m distinct groups is $\frac{(mn)!}{(n!)^m m!} \times m! = \frac{(mn)!}{(n!)^m}$

Illustration 28: In how many ways can a pack of 52 cards be divided in 4 sets, three of them having 17 cards each and fourth just one card.

Solution : First we divide 52 cards into two sets which contains 1 and 51 cards respectively is $\frac{52!}{1!\,51!}$. Now 51 cards can be divided equally in three sets each contains 17 cards (here order of sets is not in portant) in $\frac{51!}{3!(17!)^3}$ ways

Hence the required number of ways $= \frac{52!}{1!\,51!} \times \frac{51!}{3!(17!)^3} = \frac{52!}{(17!)^3\,3!}$

Illustration 29: In how many ways can 12 balls can be divided into grops of 5, 4 and 3 balls respectively?

Solution : Here order is not important, the number of ways in which 12 different balls can be divided into three groups of 5, 4 and 3 balls respectively, is = $\frac{12!}{5!\,4!\,3!} = 27720$

Student Task

Q.1 The number of ways of dividing 52 cards amongst four players so that three players have 17 cards each and the fourth player just one card, is

(a) $\frac{52!}{(17!)^3}$ (b) 52 ! (c) $\frac{52!}{17!}$ (d) None of these

Q.2 The number of ways in which five identical balls can be distributed among ten identical boxes such that no box contains more than one ball, is

(a) 10 ! (b) $\frac{10!}{5!}$ (c) $\frac{10!}{(5!)^2}$ (d) None of these

Q.3 The number of ways in which mn students can be distributed equally among n sections is

(a) (mn^n) (b) $\frac{(mn)!}{(m!)^n}$ (c) $\frac{mn}{m!}$ (d) $\frac{mn}{m!n!}$

Answers : **1.** (a) **2.** (c) **3.** (b)

Principle of inclusion and exclusion

If $A_1, A_2, \ldots, A_m$ are m sets and $n(S)$ denotes the number of elements in the set S, then

$$n\left(\bigcup_{k=1}^{m} A_k\right) = \sum_{k=1}^{m} n(A_k) - \sum_{1\le i<j\le m} n(A_i \cap A_j) + \ldots + (-1)^{s-1} \sum_{i<i_1<i_2\ldots<i_s\le m} n\left(\bigcap_{k=1}^{s} A_{i_k}\right) + \ldots + (-1)^{m-1} n\left(\bigcap_{k=1}^{m} A_k\right)$$

Note that if $x \in \bigcup_{k=1}^{m} A_k$, then x belongs to at least one of A_k, $1 \le k \le m$.

As another application of the principle of inclusion and exclusion, number of derangement of n objects (number of ways in which n numbered balls (from 1 to n) can be placed in n numbered boxes (from 1 to n), one in each box, so that no ball goes to its corresponding numbered box) is given by $n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + \frac{(-1)^n}{n!}\right)$.

Illustration 30: There are 4 balls of different colour and 4 boxes of colours same as those of the balls. Find the number of ways to place two balls in boxes with respect to their colour.

Solution: The required number of ways $\frac{4!}{2!}\left[1 - \frac{1}{1!} + \frac{1}{2!}\right] = 4.3\left[1 - 1 + \frac{1}{2}\right] = 6$

Illustration 31: Find the number of ways in which two Americans, two British, one Chinese, one Dutch and one Egyptian can sit on a round table so that persons of the same nationality are separated.

Solution : Total = 6!

$n(A)$ = when $A_1 A_2$ together = 5! 2! = 240

$n(B)$ = when $B_1 B_2$ together = 5! 2! = 240

$\Rightarrow$ $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 240 + 240 - 96 = 384$

Hence $n(\overline{A} \cap \overline{B})$ = Total $- n(A \cup B) = 6! - 384 = 720 - 384 = 336$

Illustration 32: There are 3 letters and 3 envelopes, find the number of ways in which all letters are put in the wrong envelopes.

Solution : The required number of ways = $3!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right] = 3 - 1 = 2$

SUM OF NUMBERS

(i) For given n different digits a_1, a_2, a_3 ..... a_n the sum of the digits in the unit place of all numbers formed (if numbers are not repeated) is $(a_1 + a_2 + a_3 + \ldots + a_n)(-1)!$

(ii) Sum of the total numbers which can be formed with given n different digits $a_1, a_2, \ldots\ldots a_n$ is $(a_1 + a_2 + a_3 + \ldots\ldots + a_n)(n-1)!$. (111 .....n times)

Illustration 33: Find the sum of all 4 digit numbers formed with the digits 1,2,4 and 6.

Solution : Here total 4-digit numbers will be 4 = 24. So every digit will occur 6 times at every one of the four place. Now since the sum of the given digits = 1 + 2 + 4 + 6= 13. So the sum of all the digits at every place of all 24 numbers = 13 x 6 = 78

So the sum of the values of all digits

at first place =78 ; at ten place = 780 ; at hundred place = 7800 ; at thousand place = 78000

$\therefore$ the required sum = 78 + 780 + 7800 + 78000 = 86658

Important Results about points

If there are n points in a plane of which m (< n) are collinear, then

Total number of different straight lines obtain by joining these n points is ${}^{n}C_2 - {}^{m}C_2 + 1$

Total number of different triangles formed by joining these n points is ${}^nC_3 - {}^mC_3$

Number of diagonals in polygon of *n* sides is ${}^nC_2 - n$ i.e $\frac{n(n-3)}{2}$

if m parallel lines in a plane are intersected by a family of other n parallel lines. Then total number of parallelograms so formed is

${}^nC_2 \times {}^nC_2$ i.e $\frac{mn(m-1)n-1)}{4}$

14. MULTINOMIAL THEOREM

Consider the equation $x_1 + x_2 + \ldots + x_r = n$, where $a_i \le x_i \le b_i; a_i, b_i, x_i \in I; i = 1,2,\ldots,r.$ In order to find the number of solutions of the given equation satisfying the given conditions we observe that the number of solutions is the same as the coefficient of x^n in the product

$$\left(x^{a_1} + x^{a_1+1} + x^{a_1+2} + \ldots x^{b_1}\right)\left(x^{a_1} + x^{a_1+1} + x^{a_1+2} + \ldots x^{b_1}\right)$$

$$\left(x^{a_3} + x^{a_3+1} + x^{a_3+2} + \ldots x^{b_3}\right) \ldots\ldots \left(x^{a_r} + x^{a_r+1} + x^{a_r+2} + \ldots x^{b_r}\right).$$

For example, if we have to find the number of nonnegative integral solutions of $x_1 + x_2 + \ldots + x_r =$ n, then as above the required number is the coefficient of x^n in

$$\left(x^0 + x^1 + \ldots + x^n\right)\left(x^0 + x^1 + \ldots + x^n\right)\ldots\left(x^0 + x^1 + \ldots + x^n\right)(r - brackets)$$

= Coefficient of x^n in $(1 + x + x^2 + \ldots + x^n)^r$

= Coefficient of x^n in $(1 + x + x^2 + \ldots)^r$

= Coefficient of x^n in $(1 - x)^{-r}$

= Coefficient of x^n in $\left[1 + (-r)(-x) + \frac{(-r)(-r-1)}{2!}(-x)^2 + \frac{(-r)(-r-1)(-r-2)}{3!}(-x)^3 \ldots\right]$

= Coefficient of x^n in $\left[1 + {}^rC_1 x + {}^{r+1}C_2 x^2 + {}^{r+2}C_3 x^3 + \ldots\right] \Rightarrow {}^{n+r-1}C_n = {}^{n+r-1}C_{r-1}$.

If there are ℓ objects of one kind, m objects of second kind, *n* objects of third kind and so on; then the number of ways of choosing r objects out of these objects is the coefficient of x^r in the expansion of

$(1 + x + x^2 + x^3 + \ldots + x^{\ell})\ (1 + x + x^2 + \ldots + x^m)\ (1 + x + x^2 + \ldots + x^n)$.

Further if one object of each kind is to be included, then the number of ways of choosing r objects out of these objects is the coefficient of x^r in the expansion of

$(x + x^2 + x^3 + \ldots + x^{\ell})\ (x + x^2 + x^3 + \ldots + x^m)\ (x + x^2 + x^3 + \ldots + x^n) \ldots$

Illustration 34: Find the number of ways of filling three boxes (named *A*, *B* and *C*) by 12 or less number of identical balls, if no box is empty, box *B* has at least 3 balls and box *C* has at most 5 balls.

Solution : Suppose box *A* has x_1 balls, box *B* has x_2 balls and box *C* has x_3 balls. Then,

$x_1 + x_2 + x_3 \le 12, x_1 \ge 1, x_2 \ge 3, 1 \le x_3 \le 5$

Let $x_4 = 12 - (x_1 + x_2 + x_3)$. Then

$x_1 + x_2 + x_3 + x_4 = 12$ $(1 \le x_1 \le 8, 3 \le x_2 \le 10, 1 \le x_3 \le 5$ and $0 \le x_4 \le 7)$

The required number = coefficient of x^{12} in

$$\left(x^1 + x^2 + \ldots + x^8\right)\left(x^3 + x^4 + \ldots + x^{10}\right)\left(x^1 + x^2 + \ldots + x^5\right)\left(x^0 + x^1 + \ldots + x^7\right)$$

= Coefficient of x^{12} in

$$\left(x + x^2 + x^3 + \ldots\right)\left(x^3 + x^4 + x^5 + \ldots\right)\left(x + x^2 + \ldots + x^5\right)\left(1 + x + x^2 + \ldots\right)$$

= Coefficient of x^7 in

$$\left(1+x+x^2+\ldots\right)\left(1+x+x^2+\ldots\right)\left(1+x+x^2+x^3+x^4\right)\left(1+x+x^2+\ldots\right)$$

= Coefficient of x^7 in $(1-x)^{-4}(1-x^5)$

= Coefficient of x^7 in $(1-x^5)(1+{}^4C_1x+{}^5C_2x^2+{}^6C_3x^3+\ldots)$

= ${}^{10}C_7 - {}^5C_2 = 110$

Illustration 35: Find the number of non negative integral solutions of $x_1 + x_2 + x_3 + 4x_4 = 20$.

Solution : Number of non negative integral solutions of the given equation

= coefficient of x^{20} in $(1-x)^{-1}(1-x)^{-1}(1-x)^{-1}(1-x^4)^{-1}$

= coefficient of x^{20} in $(1-x)^{-3}(1-x^4)^{-1}$

= coefficient of x^{20} in $(1+{}^3C_1x+{}^4C_2x^2+{}^5C_3x^3+{}^6C_4x^4+\ldots)(1+x^4+x^8+\ldots)$

= $1+{}^6C_4+{}^{10}C_8+{}^{14}C_{12}+{}^{18}C_{16}+{}^{22}C_{20} = 536$.

Illustration 36: There are 10 points in a plane and 4 of them are collinear. Find the number of straight lines joining any two of them.

Solution : A straight line can be drawn joining two points, so there will be ${}^{10}C_2$ straight lines joining 10 points. But 4 of them are collinear, so we shall get only one line joining any two of these 4 points. Hence the total number of lines = ${}^{10}C_2 - {}^4C_2 + 1 = 40$

Illustration 37: Total number of ways in which six '+' and four '–' sings can be arranged in a line such that no two '–' sings occur together, is

(A) 35 (B) 18 (C) 15 (D) 42

Solution (A) : First we write six '+' sings at alternate places i.e by leaving one place vacant between two successive '+' sings. Now there are 5 places vacant between these sings and these are two places vacant at the ends. If we write 4 '–' sings these 7 places then no two '–' will come together. Hence total number of ways ${}^7C_4 = 35$

Illustration 38: In an examination of 9 papers a candidate has to pass in more papers than the number of papers in which he fails in order to be successful. Find the number of ways in which he can be unsuccessful.

Solution: The candidate is unsuccessful if he fails in 9 or 8 or 7 or 6 or 5 papers.

$\therefore$ The number of was to be unsuccessful

$= {}^9C_9 + {}^9C_8 + {}^9C_7 + {}^9C_6 + {}^9C_5 = {}^9C_0 + {}^9C_1 + {}^9C_2 + {}^9C_3 + {}^9C_4$

$= \frac{1}{2}({}^9C_0 + {}^9C_1 + \ldots\ldots\ldots + {}^9C_9) = \frac{1}{2}.2^9 = 2^8 = 256$

Illustration 39: Find the number of 6 digit numbers that can be made with the digits 1, 2, 3 and 4 and having two pairs of digits.

Solution : The number will have 2 pairs and 2 different digits

The number of selections = ${}^4C_2 \times {}^2C_2$, and for each selection, number of arrangements

$= \frac{6!}{2!2!}$.

Therefore, the required number of number = ${}^4C_2 \times {}^2C_2 \times \frac{6!}{2!2!} = 1080$

Student Task

Q.1 Find the number of ways of distributing 50 identical things among 8 persons in such a way that three of them get 8 things each, two of them get 7 things each and remaining 3 get 4 things each.

Q.2 Five balls are to be placed in three boxes. Each can hold all the five balls. In how many different ways can we place the balls so that no box remains empty, if

(i) balls and boxes are all different

(ii) balls are identical but boxes are different

(iii) balls are different but boxes are identical

(iv) balls as well as boxes are identical

Q.3 Show that n letters in n corresponding envelopes can be put such that none of the letters goes to the correct envelop is $n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+.....+\frac{(-1)^n}{n!}\right)$ ways.

Q.4 Find the number of onto functions from a set containing 6 elements to a set containing 3 elements.

Q.5 Find the total number of positive integral solutions of $15 < x_1 + x_2 + x_3 \le 20$.

Q.6 In an examination, the maximum marks for each of the three papers are 50 each. Maximum marks for the fourth paper is 100. Find the number of ways in which a candidate can score 60% marks on the whole.

Q.7 Find the number of non negative integer solutions of $x_1x_2x_3x_4 = 300$.

Q.8 Let $n = p_1^{\alpha 1}.p_2^{\alpha 2}.p_3^{\alpha 3} \ldots p_k^{\alpha k}$ where ($p_1, p_2 \ldots p_k$ are primes and $\alpha_1, \alpha_2, \alpha_3 \ldots \alpha_k \in N$) then find the number of ways in which n can be expressed as the product of two factors which are prime to one another.

Q.9 Find the exponent of 2 in $x = 20 \times 19 \times 18 \times \ldots \times 11$.

Q.10 The number of prime numbers among the numbers 105! + 2, 105! + 3, 105! + 4 . . . 105! + 104 and 105! + 105, is

(a) 31 (b) 32 (c) 33 (d) None of these

ANSWERS

(1) $\frac{(8!)}{(3!)^2(2!)}$ (2) (i) 150 (ii) 6 (iii) 50 (iv) 2 (4) 540 (5) 685

(6) 110551 (7) 400 (8) 2^{k-1} (9) 10 (10) (d)

Note :-

1. In an examination the maximum marks for each of the three papers are n and for the fourth paper is 2n, then the number of ways in which a candidate can get 3n marks is 1/6(n + 1) ($5n^2 + 10n + 6$).

2. The number of words that can be formed out of the letter a, b, c, d, e and f taken three together, each word containing at least one vowel atleast is 96.

3. In a certain test, a_i students gave wrong answers to at least i question, where i = 1, 2, . . . k. No student gave more than k wrong answers then the total number of wrong answers given is $a_1 + a_2 + a_3 \ldots + a_k$

4. The side AB, BC and CA of a triangle ABC have 3, 4 and 5 interior points respectively on them, then the number of triangles that can be constructed using these interior points as vertices is 205

5. Total number of ways in which six '+' and four '–' signs occur together is 35

6. There are four balls of different colours and four boxes same as these of the balls, then the number of ways in which the balls, one in each box could be placed such that a ball does not go in a box of its own colour is 9.

7 Let n and k be positive integers such that $n \ge \frac{k(k+1)}{2}$, then the number of solutions ($x_1, x_2, ...x_k$), $x_1 \ge 1, x_2 \ge 2, ...x_k \ge k$, all integers, satisfying $x_1 + x_2 + ...+ x_k = n$, is ${}^mC_{k-1}$ where m = (1/2) ($2n - k^2 + k - 2$)

8 The product of any r consecutive natural numbers is always divisible by r!.

9. Number of circular permutations of n things when p alike and the rest different taken all at a time distinguish clockwise and anticlockwise arrangement is $\frac{(n-1)!}{p!}$.

10. Number of ways in which distinct things can be distributed to p persons if there is no restriction to the number of things received by men = p^n.

EXERCISE - 3.2

Q.1 Number of divisors of the form $4n + 2$ ($n \geq 0$) of the integer 240 is
(A) 4 (B) 8 (C) 10 (D) 3

Q.2 In a college of 300 students, every student reads 5 news papers & every paper is read by 60 students. The number of news papers is
(A) Atleast 30 (B) Almost 20 (C) Exactly 25 (D) None of these

Q.3 Find the number of divisors of the form $4n + 2$ ($n \geq 0$) of the integer 240?

Q.4 How many different nine digit numbers can be formed from the number 223355888 by rearranging its digits so that odd digits occupy even positions ?
(A) 16 (B) 36 (C) 60 (D) 180

Q.5 How many different 9 digit numbers can be formed from the number 223355888 by rearranging its digits, so that the odd digits occupy even position ?

Q.6 Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon of 'n' sides. If $T_{n+1} - T_n = 21$, then 'n' equals
(A) 5 (B) 7 (C) 6 (D) 4

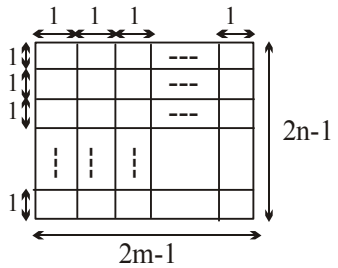
Q.7 Let $E = \{1, 2, 3, 4\}$ & $F = \{1, 2\}$. Then the number of onto functions from E to F is
(A) 14 (B) 16 (C) 12 (D) 8

Q.8 The number of arrangements of the letters of the word BANANA in which the two N's do not appear adjacently is
(A) 40 (B) 60 (C) 80 (D) 100

Q.9 Using permutation or otherwise, prove that $\frac{(n^2)!}{(n!)^n}$ is an integer, where n is a positive integer.

Q.10 Prove that $\frac{(n^2)!}{(n!)^n}$ is an integer, using permutations or otherwise.

Q.11 A rectangle with sides $2m - 1$ and $2n - 1$ is divided into square of unit length by drawing parallel lines as shown in diagram, then the number of rectangles possible with odd side length is



(A) $(m + n - 1)^2$ (B) 4^{m+n-1} (C) m^2n^2 (D) $m(m + 1)\, n(n + 1)$

Q.12 If total number of runs scored in n matches is $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$, where $n > 1$, and the runs scored in the kth match are given by $k \cdot 2^{n+1-k}$, where $1 \leq k \leq n$. Find n.

Q.13 If r, s, t are prime numbers and p, q are the positive integers such that LCM of p, q is $r^2t^4s^2$, then the number of ordered pair (p, q) is
(A) 252 (B) 254 (C) 225 (D) 224

Q.14 The letters of the COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN is

(A) 360 (B) 192 (C) 96 (D) 48

Q.15 Consider all possible permutations of the letters of the word ENDEANOEL.

Match the Statements / Expressions in Column I with the Statements / Expressions in Column II and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in the ORS.

	Column I		**Column II**
(A)	The number of permutations containing the word ENDEA is	(p)	5!
(B)	The number of permutations in which the letter E occurs in the first and the last positions is	(q)	2 × 5!
(C)	The number of permutations in which none of the letters D, L, N occurs in the last five positions is	(r)	7 × 5!
(D)	The number of permutations in which the letters A, E, O occur only in odd positions is	(s)	21 × 5!

ANSWER KEY

1. (a) 2. (c) 3. 4 4. (c) 5. 60 6. (b) 7. (a)
8. (a) 11. (c) 12. 7 13. (c) 14. (c)
15. A → (p), B → (s), C → (q), D → (q)

MISCELLANEOUS EXAMPLES

Example 1 : Find the number of 4-letter words, that can be formed from the letters of the word "ALLAHABAD"

Solution : We have four A's, two L's, and one each of H, B and D. Four letters from the letters of the word ALLAHABAD would be one of the following types; (i) all same (ii) three same (iii) two same, two same (iv) two same, two different and (v) all different

Now number of words of type (i) is 1

number of words of type (ii) is $\frac{4!}{3!}=16$

number of words of type (iii) is $\frac{4!}{2!2!}=6$

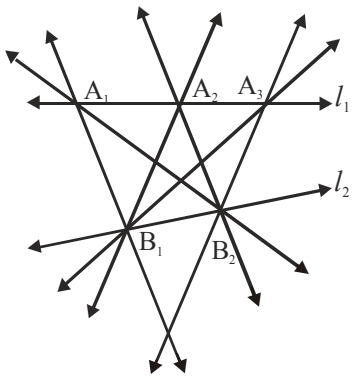
number of words of type (iv) is ${}^{2}C_{1}\times {}^{4}C_{2}\times \frac{4!}{2!}=144$

number of words of type (v) is ${}^{5}C_{4}\times 4! = 120$

Thus the required number = 1 + 16 + 6 + 144 + 120 = 287.

Example 2 : Straight lines are drawn by joining m points on a straight line to n points on another line. Then excluding the given points, prove that the lines drawn will intersect at 1/2 mn(m – 1) (n– 1) points. (No two lines drawn are parallel and no three lines are concurrent.)

Solution : Let $A_1, A_2, \ldots, A_m$ be the points on the first line (say ℓ_1) and let $B_1, B_2, \ldots, B_n$ be the points on the second line (say ℓ_2). Now any point on ℓ_1 can be chosen in m ways and any point on ℓ_2 can be chosen in n ways. Hence number of ways of choosing a point ℓ_1 and a point on ℓ_2 is mn.



Hence number of lines obtained on joining a point on ℓ_1 and a point on ℓ_2 is mn. Now any point of intersection of these lines, which can be done in ${}^{mn}C_2$ ways. Hence number of point is ${}^{mn}C_2$. But points counted many times. For example the point A_1 has been counted ${}^{n}C_2$ times. Hence required number of points is

$${}^{mn}C_2 - m.\,{}^{n}C_2 - n.\,{}^{m}C_2 = \frac{1}{2}mn(m-1)(n-1)$$

Example 3 : Find the number of nonnegative integral solutions of $2x + y + z = 21$.

Solution : Clearly $x = 0, 1, 2, 3, \ldots, 10$. Let $x = k$; then $0 \le k \le 10$

When $x = k$, $y + z = 21 - 2k$

The number of nonnegative integral solutions

= the number of ways to distribute $(21 - 2k)$ identical things (each thing is the number 1) among 2 persons

$= {}^{21-2k+2-1}C_{2-1} = {}^{22-2k}C_1 = 22 - 2K$

$\therefore$ The required number of solutions $= \sum_{k=0}^{10}(22-2k) = 22+20+18+\ldots+2$

$= 2\,(1 + 2 + 3 + \ldots + 11) = 2 \times \frac{11\times12}{2} = 132$

Example 4 : Find the number of numbers between 300 and 3000 that can be formed with the digits 0, 1, 2, 3, 4 and 5, no digit being repeated in any number.

Solution : Any number between 300 and 3000 must be of 3 or 4 digits

Case - I When number is of 3 digits

Hundreds place can be filled up by any one of the three digits 3, 4 and 5 in 3 ways.

Remaining two places can be filled up by remaining five digits in ${}^5P_2$ ways.

$\therefore$ Number of numbers formed in this case $= 3 \times {}^5P_2 = 3 \times \frac{5!}{3!} = 60$

Case - II When number is of 4 digits number of numbers formed $= 2 \times {}^5P_3 = 2 \times \frac{5!}{2!} = 120$

$\therefore$ Required number = 60 + 120 = 180

Example 5 : In how many ways can 13 persons out of 24 persons be seated around a table.

Solution : In case of circular table the clockwise and anticlockwise orders are different, thus the required number of circular permutations

$$= \frac{{}^{24}P_{13}}{13} = \frac{24!}{13\times11!}$$

Example 6 : Between two junction stations A and B, there are 12 intermediate stations. The number of ways in which a train can be made to stop at 4 of these stations so that no two of these halting stations are consecutive, is

(a) ${}^8C_4$ (b) ${}^9C_4$ (c) ${}^{12}C_4 - 4$ (d) None of these

Solution (b) : Let x_1 be the number of stations before the first halting station, x_2 between first and second, x_3 between second and third, x_4 between third and fourth and x_5 on the right of 4th stations. Then

$x_1 \geq 0, x_5 \geq 0, x_2, x_3, x_4 \geq 1$ satisfying $x_1 + x_2 + x_3 + x_4 + x_5 = 8$ ..... (i)

The total number of ways is the number of solution of the above equation

Let $y_2 = x_2 - 1$, $y_3 = x_3 - 1$, $y_4 = x_4 - 1$.

Then (i) reduces to $x_1 + y_2 + y_3 + y_4 + x_5 = 5$, where $y_2, y_3, y_4 \geq 0$.

The number of solution of this equation is ${}^{5+5-1}C_{5-1} = {}^{9}C_4$.

Example 7 : Find the number of positive integers from 1 to 1000, which are divisible by at least one of 2, 3 or 5.

Solution : Let A_k be the set of positive integers from 1 to 1000, which are divisible by k.

Obviously we have to find $n(A_2 \cup A_3 \cup A_5)$. If [.] denotes the greatest integer function, then $n(A_2)$,

$$\left[\frac{1000}{2}\right] = 500, n(A_3) = \left[\frac{1000}{3}\right] = 333, n(A_5) = \left[\frac{1000}{5}\right] = 200,$$

$$n(A_2 \cap A_3) = 166, n(A_3 \cap A_5) = 66, n(A_2 \cap A_5) = 100, n(A_2 \cap A_3 \cap A_5) = 33.$$

Hence $n(A_2 \cup A_3 \cup A_5) = 500 + 333 + 200 - 166 - 66 - 100 + 33 = 734$

Note that number of positive integers from 1 to 1000, which are not divisible by any of 2, 3 or 5 is $1000 - n(A_2 \cup A_3 \cup A_5) = 266$

Example 8 : Find the sum of all the four digits numbers that can be formed with the digits 0, 1, 2 and 3.

Solution : The number of four digits numbers that can be formed using 0, 1, 2 and 3 is $3 \times 4 \times 4 \times 4 = 192$

When we add all these numbers, then number of units = $\frac{192}{4}$ (0 + 1 + 2 + 3) = 288.

Number of tens = 288, number of hundreds = 288.

Finally number of thousands = $\frac{192}{3}$ (1 + 2 + 3) = 384.

Hence required sum = $384 \times 1000 + 288 \times 100 + 288 \times 10 + 288 = 415968$.

Example 9 : A person writes letters to six friends and address the corresponding envelopes. In how many ways can the letters be placed in the envelopes so that (i) at least two of them are in the wrong envelopes (ii) all the letters are in the wrong envelopes

Solution :

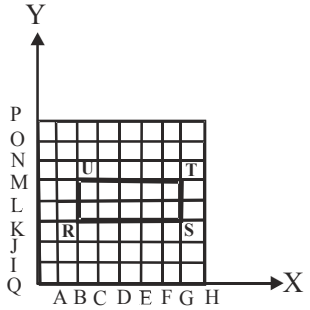
(i) The number of all the possible ways of putting 6 letters into 6 envelopes is 6!. There is only one way of putting all the letters correctly into the corresponding envelopes. Hence if there is a mistake, at least 2 letters will be in the wrong envelope.
Hence the required answer is 6! – 1 = 719.

(ii) Using the result of derangements, the required number of ways

$$= 6!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!}\right) = 720\left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} + \frac{1}{720}\right)$$

= 360 – 120 + 30 – 6 + 1 = 265.

Example 10 : Show that the number of rectangles of any size on a chess board is $\sum_{k=1}^{8} k^3$.



Solution : A rectangle can be fixed on the chess board if and only if we fix two points on x-axis and two points on y-axis. For example in order to fix the rectangle RSTU, we fix B and G on x-axis and K and M on y-axis and vice versa.
Hence total number of rectangles on the chess board is the number of ways of choosing two points on x-axis (which can be done in ${}^9C_2$ ways) and two points on y-axis (which can also be done is ${}^9C_2$ ways). Hence require number is $({}^9C_2)^2 = \sum_{k=1}^{8} k^3$.

Example 11 : The sum of all the four digit even numbers which can be formed by using the digits 0, 1, 2, 3, 4 and 5 if repetition of digits is allowed is
(a) 1765980 (b) 1756980 (c) 1769580 (d) 1759680

Solution (c) : $\square\square\square\square$
Last place can be filled by 0, 2, 4
So total sum = 5 × 6 × 6 (0 + 2 + 4) + 5 × 6 × 3 × 10 (0 + 1 + 2 + 3 + 4 + 5) + 5 × 6 × 3 × 100 (0 + 1 + 2 + 3 + 4 + 5) + 6 × 6 × 3 × 1000 (0 + 1 + 2 + 3 + 4 + 5)
= 180 × 6 + 900 × 15 + 9000 × 15 + 10800 × 15
= 1080 + 13500 + 135000 + 1620000 = 1769580

Example 12 : In a unique hockey series between India & Pakistan, they decide to play till a team wins 5 matches . The number of ways in which the series can be won by India, if no match ends in a draw is
(a) 126 (B) 252 (C) 225 (D) None of those

Solution (a) : India wins exactly in 5 matches $\Rightarrow$ looses in none $\Rightarrow$ ${}^5C_0$ ways
India wins exactly in 6 matches $\Rightarrow$ wins the 6th and looses anyone in the 1st five $\Rightarrow$ ${}^5C_1$ ways and so on.
Total number of ways = ${}^5C_0 + {}^5C_1 + {}^6C_2 + {}^7C_3 + {}^8C_4 = 126$

Example 13 : Five balls of different colours are to be placed in three boxes of different sizes. Each box can hold all five balls. Find the number of ways in which we can place the balls in the boxes (order is not considered in the box) so that no box remains empty.

Solution: One possible arrangement is $\boxed{2}\boxed{2}\boxed{1}$
Three such arrangements are possible. Therefore, the number of ways is $({}^5C_2)({}^3C_2)({}^1C_1)(3) = 90$
The other possible arrangements $\boxed{1}\boxed{1}\boxed{3}$
Three such arrangements are possible. In this case, the number of ways is $({}^5C_1)({}^4C_1)({}^3C_3)(3) = 60$
Hence, the total number of ways is 90 + 60 = 150

Example 14 : The number of ways of choosing a committee of 2 women and 3 men from 5 women and 6 men, if Mr. A refuses to serve on the committee if Mr. B is a member and Mr. B can only serve, if Miss C is the member of the committee, is
(a) 60 (b) 84 (c) 124 (d) None of these

Solution (c) : **(i) Miss C is taken**
(a) *B* included $\Rightarrow$ *A* excluded $\Rightarrow$ ${}^4C_1 \cdot {}^4C_2 = 24$
(b) *B* excluded $\Rightarrow$ ${}^4C_1 \cdot {}^5C_3 = 40$
(ii) Miss C is not taken
$\Rightarrow$ *B* does not comes ; ${}^4C_2 \cdot {}^5C_3 = 60$ $\Rightarrow$ Total = 124

Example 15 : In how many ways can a pack of 52 cards be
(i) distributed equally among four players in order?
(ii) divided into 4 groups of 13 cards each?
(iii) divided into four sets of 20, 15, 10, 7 cards?
(iv) divided into four sets, three of them having 15 cards each and the fourth having 7 cards?

Solution : **(i)** From 52 cards of the pack, 13 cards can be given to the first player in ${}^{52}C_{13}$ ways.
From the remaining 39 cards, 13 cards can be given to the second player in ${}^{39}C_{13}$ ways.
From the remaining 26 cards, 13 cards can be given to the second player in ${}^{26}C_{13}$ ways.
The remaining 13 cards can be given to the fourth player in ${}^{13}C_{13} = 1$ way.
By fundamental theorem, the number of ways of dividing 52 cards equally among four players

$$= {}^{52}C_{13} \times {}^{39}C_{13} \times {}^{26}C_{13} \times {}^{13}C_{13} = \frac{52!}{13!39!} \times \frac{39!}{13!26!} \times \frac{26!}{13!13!} \times 1 = \frac{52!}{(13!)^4}$$

(ii) By standard result, the number of ways of forming 4 groups, each of 13 cards = $\frac{52!}{4!(13!)^4}$

(iii) Here the sets have unequal number of cards, hence the required number of ways

$$= {}^{52}C_{20} \times {}^{32}C_{15} \times {}^{17}C_{10} \times {}^{7}C_{7} = \frac{52!}{20!32!} \times \frac{32!}{15!17!} \times \frac{17!}{10!7!} \times 1 = \frac{52!}{20!\,15!\,10!\,7!}$$

(iv) By standard result, the required number of ways $= \frac{52!}{15!15!15!7!3!} = \frac{52!}{(15!)^3.3!7!}$

Example 16 : A bag contains 3 one rupee coins, 4 fifty paisa coins and 5 ten paisa coins. How many selection of money can be formed by taking atleast one coin from the bag.

Solution : Here are 3 things of first kind, 4 things of second kind and 4 things of third kind so the total number of selections

$= (3 + 1)(4 + 1)(5 + 1) - 1 = 119$

Example 17 : Messages are conveyed by arranging 4 white , 1 blue and 3 red flags on a pole . Flags of the same colour are alike . If a message is transmitted by the order in which the colours are arranged then the total number of messages that can be transmitted if exactly 6 flags are used is –

(a) 45 (b) 65 (c) 125 (d) 185

Solution (d) : We will consider the following cases,

I case : 4 alike and 2 others alike
II case : 4 alike and 2 different
III case : 3 alike and 3 others alike
IV case : 3 alike and 2 other alike and 1 different

Total ways $= 1 \times \frac{6!}{4!\ 2!} + 1 \times \frac{6!}{4!} + 1 \times \frac{6!}{3!\ 3!} + {}^{2}C_{1} \times \frac{6!}{3!\ 2!} = 15 + 30 + 20 + 120 = 185.$

Example 18 : Find the total number of ways of selecting five letters from letters of the word INDEPENDENT.

Solution : There are 11 letters in the given word which are as follows (NNN) (EEE) (DD)IPT
Five letters can be selected in the following manners :

(i) All letters different : ${}^{6}C_{5} = 6$
(ii) Two similar and three different : ${}^{3}C_{1}, {}^{5}C_{3} = 30$
(iii) Three similar and two different : ${}^{2}C_{1}. {}^{5}C_{2} = 20$
(iv) Three similar and two similar : ${}^{2}C_{1}. {}^{2}C_{1} = 4$
(v) Two similar, two similar and one different : ${}^{3}C_{2}. {}^{4}C_{1} = 12$
$\therefore$ Total selections = 6 + 30 + 20 + 4 + 12 = 72

Example 19 : Find The sum of the divisors of $2^5. 3^4. 5^2$

Solution : Any divisor of $2^5. 3^4. 5^2$ is of the form $2^a, 3^b. 5^c$ where $0 \le a \le 5, 0 \le b \le 4$ and $0 \le c \le 2$.

Hence the sum of the divisors $= \sum_{a=0}^{5}\sum_{b=0}^{4}\sum_{c=0}^{2} 2^a\, 3^b\, 5^c$

$= (1 + 2 + \ldots . 2^5)\,(1 + 3 + \ldots . + 3^4)\,(1 + 5 + 5^2) = \left(\frac{2^6-1}{2-1}\right)\left(\frac{3^5-1}{3-1}\right)\left(\frac{5^3-1}{5-1}\right)$

$= 63 \times 121 \times 31 = 3^2.7^1.11^2.\,31$.

Example 20 : There are 20 questions in a question paper. If no two students solve the same combination of questions but solve equal number of questions then the maximum number of students who appeared in the examination.

Solution : If r questions are solved by each student then the number of possible selections of questions is ${}^{20}C_r$

$\therefore$ the number of students = ${}^{20}C_r$

($\therefore$ each student has solved different combinations of questions)

$\therefore$ the maximum number of students = maximum value of ${}^{20}C_r = {}^{20}C_{10}$

because ${}^{20}C_{10}$ is the largest among ${}^{20}C_0$, ${}^{20}C_{20}$ being the made one

Example 21 : Find the number of integers which lie between 1 and 10^6 and which have the sum of the digits equal to 12 ?

Solution : Consider the product $(x^0 + x^1 + x^2 \ldots + x^9)\,(x^0 + x^1 + x^2 \ldots + x^9)$ . . . 6 factors. The number of ways in which the sum of the digits will be equal to 12 is equal to the coefficient of x^{12} in the above product.

So, required number of ways = coefficient of x^{12} in $(x^0 + x^1 + x^2 + \ldots + x^9)^6$

= coefficient of x^{12} in $(1 - x^{10})^6\,(1 - x)^{-6}$ = coefficient of x^{12} in $(1 - x)^{-6}\,(1 - {}^6C_1\, x^{10} + \ldots)$

= coefficient of x^{12} in $(1 - x)^{-6} - {}^6C_1$. coefficient of x^2 in $(1 - x)^{-6}$

$= {}^{12+6-1}C_{6-1} - {}^6C_1 \times {}^{2+6-1}C_{6-1} = {}^{17}C_5 - 6 \times {}^7C_5 = 6062$.

Example 22 : There are three coplanar parallel lines. If any p points are taken on each the lines, find the maximum number of triangles with vertices at these points.

Solution : Maximum no. of triangles $= {}^{3p}C_3 - 3\,({}^pC_3) = \frac{3p\,(3p-1)\,(3p-2) - 3p\,(p-1)\,(p-2)}{6}$

$= \frac{p}{2}[9p^2 - 9p + 2 - p^2 + 3p - 2] = p\,[4p^2 - 3p] = p^2\,[4p - 3]$

Example 23 : Show that the number of combinations of n letters out of 3n letters of which n are a' s, n are b' s and the rest are unequal is $(n + 2).\ 2^{n-1}$.

Solution: From na's we have 0, 1, 2, 3 . . ., n a's From nb' we may have 0, 1, 2, 3 . . . , n b's, while for each of the rest n letters we may have 2 combinations 0 or 1. Thus the required number of combinations is thus = Coefficient of x^n in

$(1 + x + x^2 + \ldots + x^n)\,(1 + x + x^2 + \ldots + x^n)\,(1 + x)\,(1 + x) + \ldots (1 + x)$

= Coefficient of x^n in $\frac{\left(1 - x^{n+1}\right)^2}{\left(1 - x\right)^2}.\left(1 + x\right)^n$

= Coefficient of x^n in $(1 - x^{n+1})^2\,(1 + x)^n\,(1 - x)^{-2}$

Since $(1 - x^{n+1})^2$ will not contain x^n, we have required number of combinations

= Coefficient of x^n in $(1 + x)^n.\,(1 - x)^{-2}$

= Coefficient of x^n in $[2 - (1 - x)]^n\,(1 - x)^{-2}$

= Coefficient of x^n in $2^n\,(1 - x)^{-2} - {}^nC_1\,2^{n-1} + {}^nC_1.\,2^{n-2}.\,(1 - x)^0 - {}^nC_3.\,2^{n-3}\,(1 - x) + \ldots + (-1)^n.\,{}^nC_n\,(1 - x)^{n-2}$

= Coefficient of x^n in $2^n\,(1 - x)^{-2} - n.\,2^{n-1}.\,(1 - x)^{-1}$

$= 2^n.\ \frac{(n+1)!}{n!} - n.2^{n-1} = 2^n.(n+1) - n.2^{n-1} = 2^{n-1}.(n+2)$

Example 24 : Find the sum of all the numbers of four different digits that can be made by using the digits 0, 1, 2 and 3.

Solution : The number of numbers with 0 in the units place = 3! = 6

The number of numbers with 1 or 2 or 3 in the units place = 3! – 2! = 4

$\therefore$ The sum of the digits in the units = $6 \times 0 + 4 \times 1 + 4 \times 3 = 24$

Similarly for the tens and the hundreds places.

The number of numbers with 1 or 2 or 3 in the thousands place = 3!

$\therefore$ the sum of the digits in the thousands place = $6 \times 1 + 6 \times 2 + 6 \times 3 = 36$

$\therefore$ the required sum $36 \times 1000 + 24 \times 1000 + 24 \times 10 + 24 = 38664$

Example 25 : Find the number of times the digit 3 will be written when listing the integers from 1 to 1000.

Solution : A three digit block from 000 to 999 mean 1000 numbers, each number constituting 3 digits. Hence, total digit which we have to write is 3000 . Since the total number of digits is 10 (0 to 9) no digit is filled preferentially.

$\Rightarrow$ number of times we write 3 $= \frac{3000}{10} = 300$.

Alternative :

Any one block can be selected in ${}^3C_1$ ways and the digit 3 can be filled in it. Now the remaining two blocks can be filled in $9 \times 9 = 81$ ways (excluding the digit 3)

Total number of times digit 3 is used = ${}^3C_1 \cdot 9 \cdot 9 = 243$

Similarly any two blocks can be selected in ${}^3C_2$ ways and the digit 3 can be put in both of them. Remaining one block can be filled in 9 ways .

Total number of times digit 3 is used = ${}^3C_2 \cdot 2 \cdot 9$

When all the blocks are taken we have used digit 3 in ${}^3C_3$ .3 times

Thus the total number of times digit 3 will be written = $({}^3C_1 \cdot 9^2)\, 1 + ({}^3C_2 \cdot 9)\, 2 + {}^3C_3 \cdot 3 = 300$

Example 26 : If a denotes the number of permutations of $x + 2$ things taken all at a time, b the number of permutations of x thins taken 11 at a time and c the number of permutations of $x = 11$ things taken all at a time such that $a = 182\, bc$, then find the value of x.

Solution : ${}^{x+2}P_{x+2} = a \Rightarrow a = (x+2)!$

${}^xP_{11} = b \Rightarrow b = \frac{x!}{(x-11)!}$ and ${}^{x+11}P_{x+11} = c = \Rightarrow c\,(x-11)!$ $\quad \therefore a = 182$ bc

$(x+2)! = 182\, \frac{x!}{(x-11)!}(x-11)! \Rightarrow (x+2)(x+1) = 182 = 14 \times 13 \;\therefore x + 1 = 13 \quad \therefore x = 12$

Example 27 : Natural numbers less than 10^4 and divisible by 4 and consisting of only the digits 0, 1, 2, 3, 4 & 5 (no repetition) are formed . Find the number of ways of formation of such number is :

Solution : 4 digit number $\square\square$XY last two digit can be

X	0	2	4	1	2	3	5
Y	4	0	0	2	4	2	2

Total 4 digit number = $4 \times 3 \times 3 + 3 \times 3 \times 4 = 72$

3 digit number $\square$XY

Total 3 digit number = $4 \times 3 + 3 \times 4 = 24$

Total 2 digit number = 6

Total 1 digit number = 1

Total numbers = 72 + 24 + 6 + 1 = 103

Example 28 : How many numbers can be formed with the digits 1,2,3,4,3,2,1 so that the odd digits always occupy the odd place.

Solution : There are 4 odd digits (1,1,3,3) and 4 odd place (first, third, fifth and seventh).
At these places the odd digits can be arranged in 6 ways
then at the remaining 3 places, the remaining three digits (2,2,4) can be arranged in 3 ways
$\therefore$ The required of number of numbers = 6 × 3 =18

Example 29 : There are three coplanar parallel lines. If any p points are taken on each of the lines, find the maximum number of triangles with vertices at these points.

Solution : The number of triangles with vertices on different lines = ${}^{p}C_1 \times {}^{p}C_1 \times {}^{p}C_1 = p^3$
The number of triangles with 2 vertices on one line and the third vertex on any one of the other two lines

$$= {}^{3}C_1 \{{}^{p}C_2 \times {}^{2p}C_1\} = 6p.\ \frac{p(p-1)}{2}$$

$\therefore$ the required number of triangles = $p^3 + 3p^2(p-1) = p^2(4p-3)$
The word "maximum" ensures that no selection of points from each of the three lines are collinear.

Example 30 : The number of different ways in which five 'alike dashes' and eight 'alike dots' can be arranged, using only seven of these 'dashes' & 'dots' is –
(a) 1287 (b) 119 (c) 120 (d) 1235520

Solution (C) :

Dashes	dots	arrangements
5	2	${}^{7}C_2$
4	3	${}^{7}C_3$
3	4	${}^{7}C_4$
2	5	${}^{7}C_5$
1	6	${}^{7}C_6$
0	7	${}^{7}C_7$

$= {}^{7}C_2 + {}^{7}C_3 + {}^{7}C_4 + {}^{7}C_5 + {}^{7}C_6 + {}^{7}C_7 = 2^7 - 8 = 120.$

Example 31 : Find the nuber of ways of selecting 5 letters of the word KHARAHARAPRIYA.

Solution : The word consists of the 14 letters AAAAA, HH, RRR, I, K, P, Y. The five letters selected can consists of

(i) 5 letters alike : the number of selections = 1
(ii) 4 letters alike + 1 distinct letter : the number of selections = 1.6
(iii) 3 letters alike + 2 2 letters alike of second kind : the number of selections = 2.2
(iv) 3 letters alike + 2 letters distinct : the number of selections = 2. ${}^{6}C_2$
(v) a pair of two letters alike + 1 letter distinct : the number of selections = ${}^{3}C_2 . {}^{5}C_1$
(vi) 2 letters allike + 3 letters distinct : the number of selections = ${}^{3}C_1 . {}^{6}C_3$
(vii) all the five letters distinct : the number of selections = ${}^{7}C_5$

The total number of selection = 1 + 6 + 4 + 30 + 15 + 60 + 21 = 137.

Example 32 : In how many ways can n letters be arranged in a row so that two particular letters do not occur either at the beginning or at the end of the row ?

Solution : First, remove the two particular things. The remaining $(n-2)$ things can be arranged in $n-2!$ ways.
In any one such arrangement of $(n-2)$ things, either the place to the left of all or the place to the right of all should not be occupied by the two particular things. In the first case, there are $(n-3)$ in between gaps and one outside on the right where the two particular things can be placed. Similar is the analysis for leaving the extreme right.
Hence the total number of ways = $2 \times {}^{(n-2)}C_2 \times (n-2)! = (n-2)(n-1)(n-2)!$

Example 33 : The different letters of the alphabet are given, Out of which five letter words are formed. Then the numbers of words in which at least one letter is repated is

(a) 50400 (b) 840 (c) 30240 (d) 69760

Solution (d) : Number of words in which no letter is repeated = 10 x 9 x 8 x 7 x 6 = 30240

Hence, the number of words in which at least one letter is repeated = $10^5 - 30240 = 69760$

Example 34 : In how many ways 15 books be distributed among 5 libraries so that every library receives at least two books ?

Solution : The required number = the number of ways in which 15 books can be distributed among the libraries without imposing any condition i.e. a library may receive no book or it may receive all the 15 books

= the coefficient of x^{15} in the expansion of $(1 + x + x^2 + \ldots\ldots\ldots + x^{15})^5$

= the coefficient of x^{15} in $\left(\frac{1-x^{16}}{1-x}\right)^5$ = the coefficient of x^{15} in $(1-x)^{-5}$

$= {}^{5+15-1}C_{15} = {}^{19}C_4 = 3876$

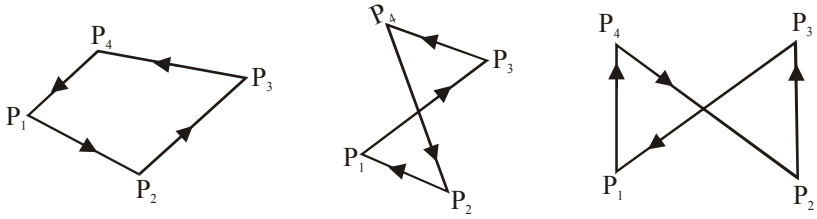
Example 35 : Find the number of n-sided polygons that can be drawn by joining n points in a plane no three of which are collinear. Verify the result when $n = 4$.

Solution : If we call the points $P_1, P_2, \ldots\ldots, P_n, P_1$ can be joined to any one of the $(n - 1)$ points. Assuming that p_1 is joined to P_2, P_2 in ist turn can be joined to any one of the remaining $(n - 2)$ points and so on. The total number of polygons produced is $(n - 1)!$. In this process

(i) every point is joined to every other point so that if we start with P_{21} say, no new polygons are added, P_2

(ii) a polygon got by joining $P_1 P_2 \ldots P_{n-1} P_1$ in this order is produced again by joining the points in the reverse order, i.e., $P_1 P_n P_{n-1} \ldots P_2 P_1$ so that every polyogn is produced twice. Thus the number of distinct polygons = $\frac{1}{2}(n-1)!$

With n = 4, the only three possiblities are



Example 36 : There are 16 things of which three are alike of one kind, four alike of another kind, five alike of a third kind and the rest are different. Show that the number of different combinations, taking one or more at a time is 1919.

Solution : 3A's, 4B's, 5 C's and D,E,F, G are the 16 things

In any combination , we may have A, 0 time, 1 time, 2 times, 3 times i.e. A can be included in 4 ways: similarly, B in 5 ways and C in 6 ways and D, E, F, G each in 2 ways.

Hence the total number of ways of choosing these 16 things = 4 × 5 × 6 × 2 × 2 × 2 × 2 = 1920.

But this counting includes the way in which each one of them is selected 0 time i.e. not selecting any of them.

Hence the required number of combinations = 1920 – 1 = 1919

Example 37 : How many different seven digit numbers are there, the sum of whose digits is even ?

Solution : Let there be 7 blank spaces × × × × × × ×.

The position on the left can be filled in 9 ways (0 cannot be used) and each of the other in 10 ways.

Thus, the total number of numbers that can be formed = 9×10^6.
1000000 is a number, the sum of whose digits is odd.
The next number 1000001 is a number, the sum of whose digits is even.
Thus, the alternate numbers belong to two different categories.
Hence there are $1/2 \times 9 \times 10^6 = 45 \times 10^5$ numbers, the sum of whose digits is even.

Example 38 : All the digits 1, 2, ...9 are permuted. For any permutation the nine digits occupy positions 1 to 9 some order.

(i) Ih how many permutations is the product of the digits in all sets of six consecutive positions divisible by 7 ?

(ii) in how many of the permutations is the product of the digits in all sets of six consecutive positions is divisible by 35 ?

Solution :

1	2	3	4	5	6	7	8	9

(i) The two extreme sets of six consecutive positions are 1 to 6 and 4 to 9 so that the product of the digits occupying all sets of six consecutive position is divisible by seven if the digit 7 occupies any of the three positions - 4th, 5th and 6th. The required number of permutations = 3.8!

(ii) The product of six digits will be divisible by 35 provide both 7 and 5 appear among the six. The given condition implies that both 5 and 7 should occupy two of the three positions mentioned in (i) above. Thus the required number of permutations = ${}^3P_2$. 7!.

Example 39 : There are $p + q + r$ books in which there are p copies of the same title, q copies of another title and one copy each of r different titles. The number of ways in which one or more books can be selected is-

(a) $2^{p+q+r} - 1$ (b) $(p + 1)(q + 1) 2^r - 1$

(c) $2^{p+q-1}(2^r - 1)$ (d) $2^{p+q}(2^r + 1) - 1$

Solution (b) : Out of the p books that are alike, m($0 \le m \le p$) books can be selected in only one way, so that the total number of selection of no book or one or more books out of p copies of the same title is $p + 1$.

Similarly the total number of selection of no books or one or more books out of q copies of the same title is $q + 1$. The remaining r books are of r different titles and the total number of selection of no book or one or more books is 2^r as each book can be dealt with in two ways; rejections or acceptance.

The total number of selections is $(p + 1)(q + 1) 2^r - 1$ as total rejection is excluded.

Example 40 : The number of 5 digit numbers of the form xyzyz in which x < y is

(a) 350 (b) 360 (c) 380 (d) 390

Solution (b) : The first digit x can be any one of 1 to 8 whereas z can be any one of 0 to 9.

When x is 1, y can assume the values 2 to 9 ;

when x is 2, y can assume the values 3 to 9 and so on.

Thus the total number = $(8 + 7 + \ldots\ldots\ldots\ldots + 1) \times 10 = \frac{8.9}{2} . 10 = 360$.

Example 41: Three are n points in a plane, no three being collinear except m of them which are collinear. The number of triangles that can be drawn with their vertices at three of the given points is-

(a) ${}^{n-m}C_3$ (b) ${}^nC_3 - {}^mC_3$ (c) ${}^nC_3 - m$ (d) None of these

Solution (b) : Given n points the number of triangles that can be drawn by joining any three non-collinear points ${}^nC_3$ out of this ${}^mC_3$ is to be subtracted as m points are collinear and no triangle is possible within the m points.

Example 42 : With 17 consonants and 5 vowels the number of words of four letters that can be formed having two different vowels in the middle and one consonant, repeated or different at each end is-

(a) 5780 (b) 2890 (c) 5440 (d) 2720

Solution (a) : The two letters, the first and the last of the four lettered word can be chosen in $(17)^2$ ways, as repetition is allowed for consonants. The two vowels in the middle are distinct so that the number of ways of filling up the two places is ${}^5P_2 = 20$. The number of different words $= (17)^2 \cdot 20 = 5780$.

Example 43 : Words are formed utilising all the letters of the word TUTORIALS and arranged as in a dictionary. Find the rank of the word in this arrangement.

Solution : The nine letters of the word in the alphabetical order are *A*, *I*, *L*, *O*, *R*, *S*, *T*, *T*, *U*.

The number of words with one of *A*, *I*, *L*, *O*, *R*, *S* in the first place $= 6.\frac{8!}{2!} = 120960$.

With *T* in the first place and anyone of *A*,*I*,*L*,*O*,*R*,*S*,*T* in the second place, the number of words = 7.7! = 35280. With *T*, *U* in the first two places and anyone of *A*,*I*,*L*,*O*,*R*,*S* in the third, the number of words = 6.6! = 4320.

With *T*,*U*,*T* in the first three places and any one of *A*,*I*,*L* in the fourth place, the number of words = 3.5! = 360.

With *T*,*U*,*T*,*O* in the first four places and any one of *A*,*I*,*L* in the fifth place, the number of words = 3.4! = 72.

With *T*,*U*,*T*,*O*,*R* in the first five places and *A* in the sixth place, the number of words = 3! = 6.

The very next word is *TUTORIALS*.

Hence the rank is = 120960 + 35280 + 4320 + 360 + 72 + 6 + 1 = 160999.

Example 44 : In how many ways can 5 letters be selected out of the letters of the word CHUNCHANAKATTE?

Solution : The word consists of the 14 letters *AAA*, *CC*, *HH*, *NN*, *TT*, *E*, *K*, *U*

The 5 letters selected may consist of

(i) 3 letters alike + 2 letters alike : the number of selections = 1.4 = 4

(ii) 3 letters alike + 2 letters distinct : the number of selections = $1.{}^7C_2 = 21$

(iii) a pair of 2 letters alike + 1 letter distinct : the number of selections = ${}^5C_2 \cdot {}^6C_1 = 60$

(iv) 2 letters alike + 3 letters distinct : the number of selections = ${}^5C_1 \cdot {}^7C_3 = 175$

(v) all the five letters distinct: the number of selections = ${}^8C_5 = 56$

Total number of selections = 316

Example 45 : How many different four digit numbers can be formed with the digits 0, 1, 2, 3,....., 9 so that each number contains the digit 1 exactly once ?

Solution : 

Case I : Let 1 occupy the left position. Then each of the remaining four positions can be filled in 9 ways. Thus, there are 9^3 four-digit numbers with 1 as starting digit.

Case II : Let 1 occupy the 2nd or 3rd or 4th position. For each of these, the left position can be filled in 8 ways. (0 excluded) and the other two in 9 ways.

Thus, the number of ways in this case $= 3 \times 8 \times 9^2$.

$\therefore$ total number of ways $= 9^3 + 3 \times 8 \times 9^2 = 9^2 \times 33 = 2673$.

Example 46 : The sum of 5 digit numbers in which only odd digits occur without any repetition is

(a) 277775 (b) 555550 (c) 1111100 (d) None of these

Solution (D) : The digits that make the numbers are 1, 3, 5, 7 and 9.

The number of numbers with one of these in the first place = 4!

Every one of the digits appear in all the 5 places – digital, 10th, 100th, etc.

in one or the other 5-digit number. The required sum of all the numbers is, therefore,

$$= 25\,(10^4 + 10^3 + 10^2 + 10 + 1) \times 4! = 600 \times \frac{10^5 - 1}{10 - 1} = 600 \times 11111 = 6666600.$$

Example 47 : Find the number of 5 digit numbers that can be made of one 1, two 2's, three 3's, four 4's and five 5's.

Solution : The selection can be made up as follows :

(a) All different digits (1, 2, 3, 4, 5). Number of such numbers = 5! = 120

(b) 2 like and 3 different digits.
The number of selection is ${}^4C_1 \times {}^4C_3$ and therefore the number of such numbers
$= {}^4C_1 \times {}^4C_3 \times \frac{5!}{2!} = 960$

(c) 3 like and two different digits. Number of such numbers = ${}^3C_1 \times {}^4C_2 \times \frac{5!}{3!} = 360$

(d) 4 like and one different digit. Number of such numbers = ${}^2C_1 \times {}^4C_1 \times \frac{5!}{4!} = 40$

(e) All like digits $\rightarrow$ 1

(f) 2 like digits, 2 like digits of another kind and 1 different.
The number of selections = ${}^4C_2 \times {}^3C_1$ and
the number of such numbers = ${}^4C_2 \times {}^3C_1 \times \frac{5!}{2!2!} = 540$

(g) 3 like digits and 2 like digits of another kind.
Number of selections = ${}^3C_1 \times {}^3C_1$ and the number of such numbers
$= {}^3C_1 \times {}^3C_1 \times \frac{5!}{3!2!} = 90$

All possible cases are exhausted and the total number = 2111.

Example 48 : How many words of three letters can be formed from the word "PROPOSAL" a vowel being always in the middle ?

Solution : **Case I** : Let '*o*' occupy the middle place .....x o x.....
The letters that are left are *p*, *r*, *s*, *l*, *a*, *o* and hence the remaining two places can be filled in ${}^6P_2 = 30$ ways.
Since there are two *p*'*s*, we get an additional word pop. Thus, in this case there are 31 words.
Case II : Let 'a' occupy the middle place *x a x*.
The letters that are left are *p*, *r*, *s*, *l*, o and hence we obtain ${}^5P_2 = 20$ words. Since there are two *p*'*s* and two *o*'*s*, we get two more words *pap* and oao thus totalling 22 words.
Hence the overall number of words = 31 + 22 = 53.

Example 49 : Six apples and six oranges are distributed among ten boys so that each boy receives at least one fruit. What is the number of ways of distributions ?

Solution : 12 fruits are to be distributed, so that some boy receives three and each of the other nine receives one fruit (or) some two boys receive two each and each of the other eight receives one.
Case I : One boy can receive 3 fruits in the following manner.

Apples	**Oranges**	**Distribution to others**	
1	2	${}^9C_5$	(5 left over apples distributed)
2	1	${}^9C_4$	
3	0	${}^9C_3$	
0	3	${}^9C_6$	

Since one boy can be selected in ${}^{10}C_1$ ways, the total number of ways of distribution in this case is ${}^{10}C_1 \times ({}^9C_5 + {}^9C_4 + {}^9C_3 + {}^9C_6) = 4200$.
Case II : Two boys A and B receive two fruits each and 8 others receive one each.

A	**B**	**Distribution to others**	
2(*A*)	2(*A*)	${}^8C_2$	(2 left over apples distributed)

2(*O*)	2(*O*)	$^8C_6$
2(*A*)	1(*A*) 1(*O*)	$^8C_3$
2(*O*)	1(*A*) 1(*O*)	$^8C_5$
1(*A*) 1(*O*)	2(*A*)	$^8C_3$
1(*A*) 1(*O*)	2(*O*)	$^8C_5$
2(*A*)	2(*O*)	$^8C_4$
2(*O*)	2(*A*)	$^8C_4$
1(*A*) 1(*O*)	1(*A*) 1(*O*)	$^8C_4$

Since the two boys can be selected in $^{10}C_2$ ways, the total number of ways in this case = $^{10}C_2 \times \{2 \times {}^8C_2 + 4 \times {}^8C_3 + 3 \times {}^8C_4\} = 22050$.

Hence, the total number of ways of distribution = 4200 + 22050 = 26250.

Example 50 : Find the sum of all the numbers of five digits which can be formed with 0 , 1, 2, 3, 4, no digit being repeated in any numbers.

Solution : Total number of five digit numbers that can be formed with 0, 1, 2 ,3, 4 is 4 x 4 x 3 x 2 x 1 = 96.

Corresponding to say, 4 in the 10^4 the place, the other four places can be arranged in 4 or 24 ways

Consider the unit placein these 24 numbers 1, 2, 3, 0 each will occur 6 times similar pattern occurs for the numbers 1, 2, 3, 0 in the 10^4 th place.

Hence there will be 18 fours, 18 threes, 18 twos, 18 ones and 24 zeros in the unit place of these 96 numbers.

$\therefore$ Sum of the digits in the unit place = 18 (1 + 2 + 3 + 4) = 180

Similarly the sum of the digits in the tenth, hundredth and thousandth place is 180. But 0 does not occur in the fen thousandth place and 1, 2 ,3 , 4 will each occurs 24 times. Therefore the sum of the digits in the ten thousandth place = 24 (1 + 2 +3 +4) = 240.

Hence the sum of all the 96 numbers = 240 x 10^4 + 180 ($10^3 + 10^2 + 10^1 + 1$) = 2599980

Example 51 : On how many nights may a watch of 4 men be arranged by a security agency from a crew of 16 so that no two watches are identical? On how many of these nights would a particular person be off-duty ?

Solution : The required number of nights = number of ways of selecting 4 men from 16 = $^{16}C_4$ = 1820.

In this, the number of nights a particular person is on-duty = $^{15}C_3$ = 455.

The number of nights a particular person is off-duty = 1820 – 455 = 1365.

Example 52 : In a shooting competition a man can score 5, 4, 3, 2, or 0 points. Find the number of ways he can score 30 in 7 shots.

Solution : The number of ways of making 30 in 7 shots is the coefficient of x^{30} in the expansion of $(x^0 + x^2 + x^3 + x^4 + x^5)^7$, because this coefficient arises out of the different ways in which 7 of the indices 0, 2, 3, 4, 5 combine to make 30.

Now $(x^0 + x^2 + x^3 + x^4 + x^5)^7 = \{x^4(x + 1) + (x^3 + x^2 + 1)\}^7$

$= x^{28} (x + 1)^7 + {}^7C_1\, x^{24} (x + 1)^6 (x^3 + x^2 + 1) + {}^7C_2\, x^{20} (x + 1)^5 (x^3 + x^2 + 1)^2$

$+ {}^7C_3\, x^{16} (x + 1)^4 (x^3 + x^2 + 1)^3 + \ldots\ldots$

Now coefficient of x^{30} in $(1 + x^2 + x^3 + x^4 + x^5)^7 = {}^7C_5 + {}^7C_1 \{{}^6C_3 + {}^6C_2 + 1\} + {}^7C_2 \{{}^5C_1 + 2\} = 420$



UNSOLVED EXERCISE

SECTION-A
(Single Correct Option)

GENERAL ARRANGEMENT

1. There are m points on a straight line AB and n points on another line *AC*, none of them being the point *A*. Triangles are formed from these points as vertices when (i) *A* is excluded (ii) *A* is included.

Then the ratio of the number of triangles in the two cases is

(a) $\frac{m+n-2}{m+n}$ (b) $\frac{m+n-2}{2}$

(c) $\frac{m+n-2}{m+n+2}$ (d) None of these

2. Set *A* contains *n* elements. *A* subset *P* of *A* is choosen. The set *A* is reconstructed by replacing the elements of *P*. *A* subset *Q* of *A* is again choosen. The number of ways of choosing *P* and *Q* so that $P \cap Q = \phi$ is

(a) 2^n (b) 3^n

(c) $2^n - 1$ (d) None of these

3. Number of ways of selection of 8 letters from 24 letters of which 8 are *a*, 8 are *b* and the rest unlike, is given by

(a) 2^7 (b) 8.2^8

(c) 10.2^7 (d) None of these

4. The number of words formed by any 4 letters of word MEERUT is equal to

(a) 180 (b) 120

(c) 72 (d) 192

5. If *x* lies in the smallest interval [*a*, *b*) for which $2[x/8]^2 + 3[x/8] = 20$, max. $({}^{(b-a)}C_r)$ =

(a) 56 (b) 70

(c) 20 (d) None of these

6. Let *m*, *n* denotes two integers chosen out of the set {1, 2,....., 10}. The number of pairs (*m*, *n*) such that $2^m + 2^n$ is divisible by 5 is

(a) 12 (b) 24

(c) 8 (d) 16

7. *n* is selected from the set {1, 2, 3,....,100} and the number $2^n + 3^n + 5^n$ is formed. Total numer of ways of selecting n so that the formed number is divisible by 4, is equal to

(a) 50 (b) 48

(c) 49 (d) None of these

8. Let $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$. Total number of functions *f* that are onto and $f(i) \neq i$, is equal to

(a) 20 (b) 36

(c) 16 (d) 44

9. The number of different matrices that can be formed with elements 0, 1, 2 or 3, each matrix having 4 elements, is

(a) 3×2^4 (b) 2×4^4

(c) 3×4^4 (d) None of these

10. Total number of integers '*n*' such that $2 \leq n \leq 2000$ and H.C.F. of n and 36 is one, is equal to

(a) 664 (b) 667

(c) 665 (d) 666

11. The number of ways in which three distinct number in A.P. can be selected from the set {1, 2, 3,...., 24} is equal to

(a) 66 (b) 132

(c) 264 (d) None of these

12. How many 5-digit telephone numbers can be constructed using the digits 0 to 9, if each number starts with 67 and no digit appears more than once?

(a) 335 (b) 336

(c) 338 (d) 337

13. If ${}^nC_r$ denotes the number of combinations of n things taken r at a time, then the expression ${}^nC_{r+1} + {}^nC_{r-1} + 2 \times {}^nC_r$ equals

(a) ${}^{n+2}C_r$ (b) ${}^{n+2}C_{r+1}$

(c) ${}^{n+1}C_r$ (d) ${}^{n+1}C_{r+1}$

14. The number of ways in which 6 men and 5 women can dine at a round table. if no two women an to sit together, is given by

(a) 6! x 5! (b) 30

(c) 5! x 4! (d) 7! x 5!

15. The number of ways of distributing 8 identical balls in 3 distinct boxes. so that none of the boxes is empty, is

(a) 5 (b) 21

(c) 3^8 **(d) ${}^8C_3$**

PROPERTIES OF COMBINATION & GENERAL SELECTIONS

16. Total number of 3 letter words that can be formed from the letters of the word 'SAHARANPUR' is equal to

(a) 210 (b) 237

(c) 247 (d) 227

17. The number of bijective functions $f : A \to A$, where $A = \{1, 2, 3, 4\}$ such that $f(1) \neq 3$, $f(2) \neq 4$, $f(3) \neq 2$, $f(4) \neq 1$, is

(a) 9 (b) 12

(c) 23 (d) None of these

18. The number of eight digits natural numbers '*n*', whose last three digits are 124 such that $x^4 - y^4 = n$, $x, y \in N$, has at least one solution, is

(a) $10^2 \times 2$ (b) $10^2 \times 3$

(c) $10^4 \times 3$ (d) None of these

19. Number of ordered triplets of natural numbers (x,y,z) satisfying $x.y.z. \leq 10$ is
(a) 53 (b) 16
(c) 76 (d) None of these

20. If ${}^{n-1}C_3 + {}^{n-1}C_4 > {}^{n}C_3$, then n is just greater than integer
(a) 5 (b) 6
(c) 4 (d) 7

21. Find ${}^{n}C_{21}$, if ${}^{n}C_{10} = {}^{n}C_{11}$.
(a) 1 (b) 0
(c) 11 (d) 10

22. Determine n, if ${}^{2n}C_2 : {}^{n}C_2 = 9:2$.
(a) 5 (b) 4
(c) 3 (d) 2

23. The number of positive integers which can be formed by using any number of digits from 0, I, 2. 3, 4.5 but using each digit not more than once in each number is
(a) 1200 (b) 1500
(c) 1600 (d) 1630

24. The total number of ways in which five '+' and three '–' signs can be arranged in a line such that no two '-' sign occur together, is
(a) 10 (b) 20
(c) 15 (d) None of the above

25. A student is to answer 10 out of 13 questions in an examination such that he must choose atleast 4 from the first five questions. The number of choices available to him is
(a) 140 (b) 196
(c) 280 (d) 346

26. A box contains 2 white balls. 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box. if atleast one black ball is to be included in the draw?
(a) 64 (b) 24
(c) 3 (d) 12

27. Four speakers will address a meeting, where speaker Q will always speak after speaker P. Then, the number of ways in which the order of speakers can be prepared is
(a) 256 (b) 128
(c) 24 (d) 12

28. Sum of digits in the unit's place formed by the digits 1,2, 3 and 4 taken all at a time is
(a) 30 (b) 30
(c) 59 (d) 61

29. A teacher takes 3 children from her class to the zoo at time as often as she can, but she does not take same three children to the zoo more than once finds that she goes to the zoo 84 times more that particular child goes to the zoo. The number children in her class is
(a) 12 (b) b
(c) 60 (d) None of the above

30. A student is allowed to select almost n books her collection of (2n + I) books. If the total number ways in which he can select at least one book is 225, then the value of n is
(a) 6 (b) 5
(c) 4 (d) 3

31. In how many ways can 5 prizes be distributed among four students, when every student can take one more prizes?
(a) 1024 (b) 625
(c) 120 (d) 600

32. The number of ways in which the digits 1, 2, 3, 4, 3,2,1 can be arranged so that the odd digits always occupy the odd places, is
(a) 6 (b) 12
(c) 18 (d) 24

33. ${}^{m}C_{r+1} + \sum_{k=m}^{n} {}^{k}C_r$ is equal to a
(a) ${}^{n}C_{r+1}$ (b) ${}^{n+1}C_{r+1}$
(c) ${}^{n}C_r$ (d) None of these

34. $\frac{C_1}{C_0} + 2\frac{C_2}{C_1} + 3\frac{C_3}{C_2} + ... + n\frac{C_n}{C_{n-1}}$ is equal to
(a) $\frac{n(n-1)}{2}$ (b) $\frac{n(n+1)}{2}$
(c) $\frac{n(n+1)(n+2)}{2}$ (d) Noneof these

35. If n is an integer with $0 \leq n \leq 11$, then the minimum value of n! (11 – n)! is attained, when a value of n is
(a) 11 (b) 5
(c) 7 (d) 9

36. How many ways are there to arrange the letters in the word 'GARDEN' with the vowels in alphabetical order?
(a) 120 (b) 240
(c) 360 (d) 480

37. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is
(a) {1, 2, 3} (b) {1, 2, 3, 4, 5, 6}
(c) {1, 2, 3, 4} (d) {1, 2, 3, 4, 5}

MULTIONMIAL, REPEATED ARRANGEMENT & SELECTION

38. The numbers of integers between 1 and 10^6 have the sum of their digit equal to K : (where $1 < K < 18$)
(a) ${}^{(K+6)}C_6 - {}^{(K-4)}C_6$ (b) ${}^{K}C_6 - 6.\,{}^{K}C_4$
(c) ${}^{K}C_6 - 6.\,{}^{K-4}C_6$ (d) None of these

39. The number of selections of four letters from the letters of the word ASSASSINATION is
(a) 72 (b) 71
(c) 66 (d) 52

40. Words are formed from the letters *AAAABBBCCDEF*. The number of words in which the letters *C* are separated, is given by

(a) 2772000 (b) 1386000
(c) 4158000 (d) None of these

41. The number of words formed by any 4 letters of word MEERUT is equal to
(a) 180 (b) 120
(c) 72 (d) 192

42. If the permutations of a, b, c, d, e taken all together be written down in alphabetical order as in dictionary and numbered, then the rank of the permutation debac is
(a) 90 (b) 91
(c) 92 (d) 93

43. The number of positive integral solutions $15 < x_1 < x_2 < x_3 \leq 20$, is equal to
(a) 1025 (b) 685
(c) 1245 (d) None of these

44. Total number of positive integral solution of $15 < x_1 + x_2 + x_3 \leq 20$, is equal to
(a) 1025 (b) 685
(c) 1245 (d) None of these

45. The number of subsets of {1, 2, ..., n} having least element m and greatest element k, $1 \leq m < k \leq n$, is
(a) $2^{n-(k+m)}$ (b) 2^{k-m-2}
(c) 2^{k-m-1} (d) 2^{k-m+1}

46. Total number of polynomials of the form $x^3 + ax^2 + bx + c$ that are divisible by $x^2 + 1$, whre $a, b, c \in$ {1, 2, 3, ..., 9, 10} is equal to
(a) 90 (b) 45
(c) 5 (d) 10

47. From 12 books. the difference between numbs ways of selection of 5 books when one specifed book is always excluded and one specified boot always included, is
(a) 64 (b) 118
(c) 132 (d) 330

48. The number of integers greater than 6000 that can formed with 3, 5, 6, 7 and 8, where no digit repeated, is
(a) 120 (b) 192
(c) 216 (d) 72 (e) 202

49. If ${}^{56}P_{r+1} : {}^{54}P_{r+1} = 30800 : 1$, then the value of r is
(a) 40 (b) 51
(c) 41 (d) 510

50. The number of permutations by taking all letters keeping the vowels of the word 'COMBINE' in the places. is
(a) 96 (b) 144
(c) 512 (d) 576

51. From 12 books. the difference between numbers ways of selection of 5 books when one specifed book is always excluded and one specified boot always included, is
(a) 64 (b) 118
(c) 132 (d) 330

DISTRIBUTION OF OBJECT INTO GROUP DE ARRANGEMENT AND GEOMETRY IN PERMUTATION

52. The number of positive integral solution of $2x_1 + 3x_2 + 4x_3 + 5x_4 = 25$ is
(a) 20 (b) 22
(c) 23 (d) 7

53. The total number of six digit numbers $x_1x_2x_3x_4x_5x_6$ having the property that $x_1 < x_2 \leq x_3 < x_4 < x_5 \leq x_6$ is equal to
(a) ${}^{10}C_6$ (b) ${}^{12}C_6$
(c) ${}^{11}C_6$ (d) None of these

54. Total number of positive integral solutions of the equation $x_1.x_2.x_3 = 60$ is equal to
(a) 27 (b) 108
(c) 54 (d) 64

55. A number is said to be a nice number if it has exactly four factors, (including one and number itself). Let $n = 2^3 \times 5^5 \times 3^2 \times 11^2 \times 7$, then the number of factors, which are nice number, is
(a) 10 (b) 12
(c) 36 (d) 143

56. 10 different toys are to be distributed among 10 children. Total number of ways of distributing those toys, so that exactly two children do not get any toy, is

(a) $\frac{10!}{2!3!7!}$ (b) $\frac{10!}{(2!)^4 \times 6!}$

(c) $(10!)^2\left[\frac{1}{(2!)\times 6!} + \frac{1}{2!\times 3!}\right]$

(d) $\frac{10! \times 10!}{(2!)^2 \times 6!}\left[\frac{25}{84}\right]$

57. Out of 7 consonants and 4 vowels, the number of words (not necessarily meaningful) that can be made each consisting of 3 consonants and 2 vowels. is
(a) 24800 (b) 25100
(c) 25200 (d) 25400

58. In how many ways can a student choose a program of 5 courses, if 9 courses are available and 2 specific course are compulsory for every student?
(a) 34 (b) 36
(c) 35 (d) 37

59. S = {1, 2, 3, ..., 20} is to be partitioned into four sets A, B, C and D of equal size. The number of ways it can be done, is

(a) $\frac{20!}{4!\times 5!}$ (b) $\frac{20!}{4^5}$

(c) $\frac{20!}{(5!)^4}$ (d) $\frac{20!}{(4!)^5}$

60. A student is allowed to select atmost n books from a collection of (2n + 1) books. If the number of ways in which he can do this, is 64, then the value of n is
(a) 6 (b) n
(c) 3 (d) None of these

61. Out of thirty points in a plane, eight of them are collinear. The number of straight lines that can be formed by joining these points, is
(a) 296 (b) 540
(c) 408 (d) 348

SECTION-B
(Assertion-Reason)

1. Consider the word 'SMALL'
Statement–1 : Total number of 3 letter words from the letters of the given word is 13.
Statement–2 : Number of words having all the letters distinct = 4 and number of words having two are alike and third different = 9
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

2. **Statement–1:** Number of non integral solution of the equation $x_1 + x_2 + x_3 = 10$ is equal to 34.
Statement–2 : Number of non integral solution of the equation $x_1 + x_2 + x_3 + \ldots x_n = r$ is equal to ${}^{n+r-1}C_r$
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

3. **Statement–1 :** The number of ways of selecting 5 students from 12 students (of which six are boys and six are girls), such that in the selection there are at least three girls is ${}^6C_3 \ {}^9C_2$.
Statement–2 : If a work has two independent parts, of which first part can be done in m way and for each choice of first part, the second part can be done in n ways, then the work can be completed in m × n ways.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

4. **Statement–1:** The number of ways of writing 1400 as a product of two positive integers is 12.
Statement–2 : 1400 is divisible by exactly three prime numbers.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

5. **Statement–1 :** A polygon has 44 diagonals and number of sides are 11.
Statement–2 : From *n* distinct object *r* object can be selected in ${}^nC_r$ ways.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

6. **Statement-1:** The number of positive integral solutions of the equation $x_1x_2x_3x_4x_5 = 1050$ is 1875.
Statement-2: The total number of divisor of 1050 is 25.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

7. **Statement-1:** $\left(\sum_{r=0}^{100} {}^{500-r}C_3\right) + {}^{400}C_4 = {}^{501}C_4$
Statement-2 : ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

8. **Statement-1 :** $\frac{(n^2)!}{(n!)^n}$ is a natural number for all $n \in N$

Statement-2 : The number of ways of distributing mn things in m groups each containing n things is $\frac{(mn)!}{(n!)^m}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

9. **Statement-1:** The number of selections of four letters taken from the word PARALLEL must be 22.

Statement-2: Coefficient of x^4 in the expansion of $(1-x)^3$ is 10.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

10. **Statement-1:** The total number of words with letters of the word civilization (all taken at a time) is 19958393.

Statement-2: The number of permutations of n distinct objects (r taken at a time) is ${}^{n}p_{r+1}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

SECTION-C
(Previous Years Questions)

1. The number of integers greater than 6000 that can be formed using the digits 3, 5, 6, 7 and 8 without repetition is

(a) 216 (b) 192

(c) 120 (d) 72

2. The number of seven-digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only, is

(a) 55 (b) 66

(c) 77 (d) 88

3. How many different nine-digit numbers can be formed from the number 22 33 55 888 by rearranging its digits so that the odd digits occupy even positions?

(a) 16 (b) 36

(c) 60 (d) 180

4. An n-digit number is a positive number with exactly n digits. Nine hundred distinct n-digit numbers are to be formed using only the three digits 2, 5 and 7. The smallest value of n for which this is possible, is

(a) 6 (b) 7

(c) 8 (d) 9

5. In a collage of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is

(a) atleast 30 (b) atmost 20

(c) exactly 25 (d) none of these

6. A five digits number divisible by 3 is to be formed using the numbers 0, 1, 2, 3, 4 and 5, without repetition. The total number of ways this can be done, is

(a) 216 (b) 240

(c) 600 (d) 3125

7. Eight chairs are numbered 1 to 8. Two women and three men wish to occupy one chair each. First the women choose the chairs from amongst the chairs marked 1 to 4 and then the men select the chairs from amongst the remaining. The number of possible arrangements is

(a) ${}^{8}C_3 \times {}^{4}C_2$ (b) ${}^{4}P_2 \times {}^{4}P_3$

(c) ${}^{4}C_2 + {}^{4}P_2$ (d) None of these

8. The different letters of an alphabet are given. Words with five letters are formed from these given letters. Then, the number of words which have at least one letter repeated, is

(a) 69760 (b) 30240

(c) 99748 (d) None of these

9. Let T_n be the number of all possible triangles formed by joining vertices of an n-sided regular polygon. If $T_{n+1} - T_n = 10$, then the value of n is

(a) 7 (b) 5

(c) 10 (d) 8

10. If r, s, t are prime numbers and p, q are the positive integers such that LCM of p, q is $r^2s^4t^2$, then the number of ordered pairs (p, q) is

(a) 252 (b) 254

(c) 225 (d) 224

11. The value of the expression ${}^{47}C_4 + \sum_{j=1}^{5} {}^{52-j}C_3$ is

(a) ${}^{47}C_5$ (b) ${}^{52}C_5$

(c) ${}^{52}C_4$ (d) None of these

12. The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN, is

(a) 360 (b) 192
(c) 96 (d) 48

13. The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball, is

(a) 75 (b) 150
(c) 210 (d) 243

14. The number of arrangements of the letters of the word BANANA in which the two N's do not appear adjacently, is

(a) 40 (b) 60
(c) 80 (d) 100

SECTION-D
(Board pattern)

1. There are 20 books of which 4 are single volume and the other are books of 8, 5 and 3 volumes respectively. In how many ways can all these books be arranged on a shelf so that volumes of the same book are not separated.

2. In a class of 10 students, there are 3 girls. In how many different ways can they be arranged in a row such that no two of the three girls are consecutive.

3. In how many ways can the letters of the word "civilization" be rearranged?

4. How many words can be formed with the letters of the word 'Pataliputra' without changing the relative order of the vowels and consonants?

5. How many numbers greater than a million can be formed with the digits 2, 3, 0, 3, 4, 2 and 3?

6. How many different words can be formed with five given letters of which three are vowels and two are consonants, no two vowels being together in any word?

7. A man invites a party of (m + n) friends to dinner and places m at one round table and *n* at another. Find the number of ways of arranging the guests.

8. The streets of a city are arranged like the lines of a chessboard. There are m streets running North and South and n East and West. Find the number of ways in which a man can travel from the N. W. To the S.E. corner, going shortest possible distance.

9. There are m points in a plane which are joined by straight lines in all possible ways and of these no two are coincident or parallel and no three of them are concurrent except at the points. Show that the number of points of intersection, other than the given points, of the lines so formed $\frac{\lfloor m}{8\lfloor m-4}$.

10. Prove that (n!)! is divisible by $(n!)^{(n-1)!}$.

11. If a and b are positiveintegers, show that $\frac{\lfloor ab}{\lfloor a(\lfloor b)^a}$

12. Show that the total number of permutations of n different things taken not more than r at a time, when each thing may be repeated any number of times is $\frac{n(n^r-1)}{(n-1)}$.

13. How many different car licence plates can be constructed if the licences contain three letters of the English alphabet followed by a three digit number

(i) If repetitions are allowed?

(ii) If repetition are not allowed?

14. Prove that that the product of *r* consecutive integers is divisible by r!.

15. A man has 8 children to take them to a zoo. He takes three of them at a time to the zoo as often as he can without taking the same 3 children together more than once. How many times will he have to go to zoo? How many times a particular child will go?

16. On a new year day every student of a class sends a card to every other student. The postman delivers 600 cards. How many students are there in the class.

17. Out of 6 gentlemen and 4 ladies a committee of 5 is to be formed. In how many ways can this be done so as to include at least one lady in each committee?

18. There are ten points in a plane. Of these ten points four points are in a straight line and with the exception of these four points, no other three points are in the same straight line. Find the number of quadrilaterals formed, by joining these ten points.

19. A committee of 12 is to be formed from 9 women and 8 men. In how many ways this can be done if at least five women have to be included in a committee? In how many of these committees.

(i) The women are in majority?

(ii) The men are in majority?

20. Eighteen guests have to be seated half on each side of a long table. Four particular guests desire to sit on one particular side and three other on the other side. Determine the number of ways in which the sitting arrangements can be made.

21. Let $n \geq 2$ be an integer. Take *n* distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of *n* is

22. A student is allowed to select atmost *n* books from *n* collection of (2n + 1) books. If the total number of ways in which he can select at least one books is 63, find the value of *n*.

23. *m* men and *n* women are to be seated in a row so that no two women sit together. If m > n, then show that the number of ways in which they can be seated, is $\frac{m!(m+1)!}{(m-n+1)!}$.

24. Let $n_1 < n_2 < n_3 < n_4 < n_5$ be positive integers such that $n_1 + n_2 + n_3 + n_4 + n_5 = 20$. The number of such distinct arrangements $(n_1, n_2, n_3, n_4, n_5)$ is

25. Let n and k be positive integers such that $n \geq \frac{k(k+1)}{2}$. The number of solution $(x_1, x_2,....,x_3)$.

$x_1 \geq 1, x_2 \geq 2,...., x_k \geq k$ for all integers satisfying $x_1 + x_2 +....+ x_3 = n$ is....

26. Five balls of different colours are to be placed in three boxes of different sizes. Each box can hold all five. In how many different ways can we place the balls so that no box remains empty?

27. There are four balls different colours and four boxes of colours, same as those of the balls. The number of ways in which the balls, one each in a box, could be placed such that a ball does not go to a box of its own colour is..... .



ANSWER KEY

CHAPTER-01 (QUADRATIC EQUATION)

SECTION-A (Straight objective type questions)

1. (d)	**2.** (c)	**3.** (c)	**4.** (d)	**5.** (a)	**6.** (a)	**7.** (a)
8. (a)	**9.** (a)	**10.** (c)	**11.** (b)	**12.** (c)	**13.** (b)	**14.** (b)
15. (a)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (a)	**20.** (b)	**21.** (a)
22. (d)	**23.** (d)	**24.** (a)	**25.** (b)	**26.** (b)	**27.** (a)	**28.** (b)
29. (a)	**30.** (b)	**31.** (d)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (b)
36. (a)						

SECTION-B (Assertion-Reason type questions)

1. (a)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (a)

SECTION-C (Previous Years Questions)

1. (c)	**2.** (d)	**3.** (c)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (a)
8. (b)	**9.** (b)	**10.** (c)	**11.** (b)	**12.** (a)	**13.** (b)	**14.** (a)
15. (a)	**16.** (a)	**17.** (c)	**18.** (b)	**19.** (a)	**20.** (b)	**21.** (c)
22. (c)	**23.** (a)	**24.** (b)	**25.** (a)	**26.** (d)	**27.** (d)	

SECTION-D (School/Board level type questions)

1. (b)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (d)	**6.** (c)	**7.** (a)
8. (a)	**9.** (d)	**10.** (d)	**11.** (c)	**12.** (c)	**13.** (a)	**14.** (a)
15. (c)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (d)	**20.** (b)	

CHAPTER-02 (COMPLEX NUMBER)

SECTION-A (Straight objective type questions)

1. (c)	**2.** (d)	**3.** (d)	**4.** (a)	**5.** (d)	**6.** (a)	**7.** (d)
8. (a)	**9.** (a)	**10.** (a)	**11.** (d)	**12.** (c)	**13.** (c)	**14.** (a)
15. (d)	**16.** (b)	**17.** (c)	**18.** (d)	**19.** (b)	**20.** (b)	**21.** (d)
22. (d)	**23.** (a)	**24.** (b)	**25.** (c)	**26.** (a)	**27.** (a)	**28.** (b)
29. (a)	**30.** (d)	**31.** (b)	**32.** (d)	**33.** (c)	**34.** (a)	**35.** (a)
36. (d)	**37.** (b)	**38.** (a)	**39.** (d)	**40.** (b)	**41.** (c)	**42.** (c)
43. (b)	**44.** (b)	**45.** (b)	**46.** (a)	**47.** (d)	**48.** (a)	**49.** (a)
50. (a)	**51.** (a)	**52.** (d)	**53.** (a)	**54.** (a)	**55.** (d)	**56.** (a)
57. (a)	**58.** (b)	**59.** (a)	**60.** (b)	**61.** (a)	**62.** (a)	**63.** (a)
64. (a)	**65.** (d)	**66.** (a)	**67.** (c)	**68.** (b)	**69.** (a)	**70.** (c)
71. (a)	**72.** (b)	**73.** (d)	**74.** (c)	**75.** (a)	**76.** (a)	**77.** (c)
78. (b)	**79.** (a)	**80.** (a)	**81.** (b)	**82.** (a)	**83.** (a)	**84.** (d)

SECTION-B (Assertion-Reason type questions)

1. (a)	**2.** (d)	**3.** (a)	**4.** (a)	**5.** (c)

SECTION-C (PREVIOUS YEARS QUESTIONS)

1. (c)	**2.** (b)	**3.** (d)	**4.** (a)	**5.** (d)	**6.** (b)	**7.** (a)
8. (b)	**9.** (a)	**10.** (d)	**11.** (d)	**12.** (b)	**13.** (b)	**14.** (d)
15. (b)	**16.** (a)	**17.** (a,d)	**18.** (a,b,c)	**19.** (c)	**20.** (a)	**21.** (d)
22. (c)	**23.** (a, c, d)	**24.** (d)	**25.** (d)	**26.** (a)	**27.** (c)	**28.** (d)
29. (c)	**30.** (b)	**31.** (c)	**32.** (d)	**33.** (d)		

CHAPTER-03 (PERMUTATION AND COMBINATION)

SECTION-A (Straight objective type questions)

1. (a)	**2.** (b)	**3.** (c)	**4.** (d)	**5.** (b)	**6.** (b)	**7.** (c)
8. (d)	**9.** (c)	**10.** (d)	**11.** (b)	**12.** (b)	**13.** (b)	**14.** (a)
15. (b)	**16.** (c)	**17.** (a)	**18.** (d)	**19.** (a)	**20.** (d)	**21.** (a)
22. (a)	**23.** (d)	**24.** (b)	**25.** (b)	**26.** (a)	**27.** (d)	**28.** (b)
29. (d)	**30.** (c)	**31.** (a)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (b)
36. (c)	**37.** (a)	**38.** (d)	**39.** (a)	**40.** (b)	**41.** (d)	**42.** (d)
43. (b)	**44.** (b)	**45.** (c)	**46.** (d)	**47.** (c)	**48.** (b)	**49.** (c)
50. (d)	**51.** (c)	**52.** (d)	**53.** (c)	**54.** (c)	**55.** (b)	**56.** (d)
57. (c)	**58.** (c)	**59.** (c)	**60.** (c)	**61.** (c)		

SECTION-B (Assertion-reason type questions)

1. (a)	**2.** (d)	**3**. (d)	4. (b)	**5.** (a)	**6.** (c)	**7.** (a)
8. (a)	**9.** (c)	**10.** (c)				

SECTION-C (PREVIOUS YEARS QUESTIONS)

1. (b)	**2.** (c)	**3.** (c)	**4.** (b)	**5.** (c)	**6.** (a)	**7.** (d)
8. (a)	**9.** (b)	10. (c)	**11.** (c)	**12.** (c)	**13.** (b)	**14.** (a)

SECTION-D (School/Board Level type questions)

1. $\lfloor 7 \lfloor 8 \lfloor 5 \lfloor 3$ **2.** ${}^{8}P_{3} \times 7!$ **3.** 19958392 **4.** 3600

5. 360 **6.** 12 **7.** $\dfrac{\lfloor m+n}{mn}$ **8.** $\dfrac{\lfloor m+n-2}{\lfloor m-1 \lfloor n-1}$

13. (i) $(26)^3(999)$; (ii) $({}^{26}P_3)({}^{10}P_3)$ **15.** 56, 21 **16.** 25 **17.** 246 **18.** 185

19. (i) ${}^{9}C_5.{}^{8}C_7 + {}^{9}C_6.{}^{8}C_6 + {}^{9}C_7.{}^{8}C_5 + {}^{9}C_8.{}^{8}C_4 + {}^{9}C_9.{}^{8}C_3$; (ii) ${}^{9}C_5.{}^{8}C_7$ 20. ${}^{9}P_4 \times {}^{9}P_3\ (11)!$

21. (n = 5) **22.** n = 3 **24.** (7) **25.** $\frac{1}{2}(2n - k^2 + k - 2)$ **26.** (300) **27.** (9)